ANNEX 1

Horizon Europe (HORIZON)

Description of the action (DoA)

Part A

Part B

DESCRIPTION OF THE ACTION (PART A)

Associated with document Ref. Ares(2025)2863795 - 08/04/2025

COVER PAGE

Part A of the Description of the Action (DoA) must be completed directly on the Portal Grant Preparation screens.

PROJECT *Grant Preparation (General Information screen) — Enter the info.*	
Project number:	101181658
Project name:	Effective management strategies to tackle Clavibacter sepedonicus and Ralstonia solanacearum outbreaks on POtato and toMATO crops
Project acronym:	POMATO
Call:	HORIZON-CL6-2024-FARM2FORK-02
Topic:	HORIZON-CL6-2024-FARM2FORK-02-4-two-stage
Type of action:	HORIZON-RIA
Service:	REA/B/02
Project starting date:	fixed date: 1 May 2025
Project duration:	48 months

TABLE OF CONTENTS

Project summary 3
List of participants 3
List of work packages 5
Staff effort 20
List of deliverables 21
List of milestones (outputs/outcomes) 36
List of critical risks 38
Project reviews 42

PROJECT SUMMARY

Project summary

Grant Preparation (General Information screen) — Provide an overall description of your project (including context and overall objectives, planned activities and main achievements, and expected results and impacts (on target groups, change procedures, capacities, innovation etc)). This summary should give readers a clear idea of what your project is about.

Use the project summary from your proposal.

The Solanaceae family, encompassing economically significant genera like Solanum, plays a vital role in global agriculture, particularly through crops. The Solanaceae family, encompassing economically significant genera like Solanum, plays a vital role in global agriculture, particularly through crops like potatoes and tomatoes. These crops, crucial for human consumption and nutrition, face significant threats from pest diseases, with annual losses amounting to billions of euros. Notably, the bacteria Clavibacter sepedonicus – Cs (ring rot in potato) and Ralstonia solanacearum Rs (bacterial wilt in potato and tomato) pose severe economic and environmental risks, warranting stringent regulatory measures in EU and globally being included in Part B Annex of the Regulation 2019/2072 and are classified as EPPO A2 quarantine pests. In response, the POMATO project aims to safeguard potato and tomato health by focusing on four key pillars: isolation and molecular characterization of resistance genes of potato and tomato native and wild varieties against Cs and Rs, early detection using advanced technologies like AI and digital predictive platforms, development of natural bio-control solutions, and field validation at TRL 5 of the Integrated Pest Management (IPM) strategies. This collaborative 48-month initiative involves a multi-actor approach of the potato/tomato value chain, including academic research institutions, agrochemical companies, farmers, and international partners from affected regions like Latin America. By leveraging expertise and resources, POMATO seeks to mitigate the spread of these quarantine pests and enhance food security sustainably by aiming to decrease between 40-60% the incidence of Cs/Rs. Sharing IPM POMATO's strategies among relevant stakeholders in the policy and decision-making cycle will ensure economic sustainability of EU potato/tomato production, increase farm competitiveness as well as replicate the outcomes of the project to other crops.

LIST OF PARTICIPANTS

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
1	COO	UBU	UNIVERSIDAD DE BURGOS	ES	998959642
2	BEN	IHAR	INSTYTUT HODOWLI I AKLIMATYZACJI ROSLIN - PANSTWOWY INSTYTUT BADAWCZY	PL	988289933
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4	BEN	USFQ	UNIVERSIDAD SAN FRANCISCO DE QUITO	EC	938699847
5	BEN	IRIS	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	ES	998993786
6	BEN	IDEN	IDENER RESEARCH & DEVELOPMENT AGRUPACION DE INTERES ECONOMICO	ES	901322643
7	BEN	WU	WAGENINGEN UNIVERSITY	NL	999981634
8	BEN	FUND	FUNDACION TECNOLOGICA ADVANTX	ES	995825960
9	BEN	CBGP	UNIVERSIDAD POLITECNICA DE MADRID	ES	999974844

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
10	BEN	JSI	INSTITUT JOZEF STEFAN	SI	999971837

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13	BEN	INO	INOSENS DOO NOVI SAD	RS	933205185
14	BEN	PIN	POLSKA IZBA NASIENNA	PL	907718532
15	BEN	EUPAT	UNION EUROPEENNE DU COMMERCE DE LA POMME DE TERRE	BE	922651003
16	BEN	F4S	FOOD4SUSTAINABILITY - ASSOCIACAO PARA A INOVACAO NO ALIMENTO SUSTENTAVEL	PT	895673654

LIST OF WORK PACKAGES

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
WP1	Project Management	1 - UBU		1	48	D1.1 – Data Management Plan
WP2	Characterisation of pathogens, resistance sources and predisposing factors	3 - AGRO		1	42	D2.1 – I-matrix for improved isolation of Rs D2.2 – Cs/Rs completely characterised at molecular level D2.3 – Resistant tomato/potato genotypes against Cs/Rs and epigenomic markers associated to quanti-qualitative disease resistance traits D2.4 – Information on population diversity and structure of microbiota associated with environmental, soil factors and native soils that are suppressive/conducive to Cs/Rs D2.5 – Common platform with molecular data suitable for AI/ML

WP3	Early Detection and Surveillance Tools	5 - IRIS		1	42	D3.1 – Mechanistic models of establishment and spread of Cs/Rs D3.2 – Infection signatures including omics markers associated to infection onset and progression D3.3 – AI-based Disease Detection and Risk Forecasting Models D3.4 – Sensory polymeric material D3.5 – Final version of the multiparametric disease detection device D3.6 – Functional AI-powered Cloud data management platform D3.7 – User guides, and documentation

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						to support the deployment and use of the platform
WP4	Natural and Bio-Control solutions	8 - FUND		1	40	D4.1 – Formulation of biocontrol coating solutions D4.2 – Plant derived organic substances products D4.3 – Report on PGPM consortia D4.4 – Solution based on disruption of Cs/Rs chemoperception D4.5 – Report of the natural and bio-control solutions manufacturing for pilot plant

WP5	Integrated Pest Management Strategies validated in field trials, SSbD Framework and Policy Recommendations	1 - UBU		1	48	D5.1 – Pilot validation for Rs in potato fields in LATAM (Colombia and Ecuador) (CS#1) D5.2 – Pilot validation for Rs in tomato fields in LATAM (Colombia) (CS#2) D5.3 – Pilot validation for Rs in tomato fields in Portugal (CS#3) D5.4 – Pilot validation for Cs in potato fields in Poland (CS#4) D5.5 – Safety pre-assessment of the materials and processes D5.6 – Safety of natural and bio-control solutions D5.7 – Final safety and sustainability assessment of the materials and processes D5.8 – IPM guidelines for farmers D5.9 – Final IPM guidelines for farmers D5.10 – Policy recommendations

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables

WP6	Dissemination, Communication, Exploitation and Multi-Actor Engagement	13 - INO		1	48	D6.1 – Dissemination, Exploitation and Communication Plan D6.2 – Multi-Actor Participatory activities and platform D6.3 – Update on Multi-Actor Participatory activities and platform D6.4 – Report on DC, networking and clustering activities D6.5 – Final report on DC, networking and clustering activities D6.6 – Practice Abstracts - batch 1 D6.7 – Practice Abstracts - batch 2 D6.8 – Practice Abstracts - batch 3 D6.9 – POMATO Exploitation Plan D6.10 – Final POMATO Exploitation Plan

Work package WP1 – Project Management

Work Package Number	WP1	Lead Beneficiary	1 - UBU
Work Package Name	Project Management		
Start Month	1	End Month	48

Objectives

- Coordinate and manage the project and the efficient, smooth and timely execution of all activities.
- Establish a liaison with the EC for reporting the project progress and financial administration.
- Develop and update the Data Management Plan.
- Regularly update the risk assessment to ensure success in the execution of the work plan.

Description

T1.1 Scientific and technical coordination (Leader: UBU; Partners: all) (M1-M48).
As project coordinator (PC), UBU will monitor the proper implementation and progress of the project activities in compliance with the Grant and Consortium Agreements. UBU will act as intermediary between the project partners and the EC officer and will collect, review and submit technical and financial reports, deliverables and documents specifically requested by the EC. UBU will organise the consortium meetings including General Assembly (GA) and Executive Management Board (EMB) meetings. Each partner will be represented in the GA, which will take place every 6 months (1 in-person meeting per year and 1 online meeting in between) and will be duly authorised to deliberate, negotiate and decide on all project matters. The EMB, formed by the PC and WP leaders, will coordinate among and within WPs, ensuring proper execution and implementation of the decisions of the GA. WP meetings between WP leaders and members will be held every 6 months combined with the GA and as needed in between. In addition, an External Experts Advisory Board (EEAB) will be created, composed of invited experts in different scientific areas with influence on European and international level, an ethics advisor and a policymaker representative. We envisage to also include a JRC representative on the EEAB. They will review the project progress, provide assistance and external recommendations providing quality assurance of POMATO's results through peer review, integrating recommendations and effectively communicating on project results and outcomes to relevant stakeholders.
The responsibilities of the various management bodies, the decision-making process, conflict resolution, etc. will be described in the CA that will be signed by all partners before starting the project. By M3, UBU will prepare a Project Management Handbook (D1.1) with the procedures for optimal project management and quality assurance, including an internal review procedure for deliverables. If necessary, quality issues will be on the agenda of EMB meetings, possible resulting in preventive or corrective actions.
T1.2 Administrative and financial management (Leader: UBU; Partners: all) (M1-M48).
A dedicated Administrative and Financial Office within UBU will oversee the financial monitoring of the project and the delivery and follow-up of all the administrative documents required by the EC. UBU will establish a direct communication channel with each partner's administrative responsible to monitor the financial progress of the project, to support the partners in administrative, financial, legal and other aspects of the project and to fulfil the payments to each partner of the received financial contributions of the EC according to the Grant and Consortium Agreements. The procedures for administrative and financial management will be described in the PMH (D1.1).
T1.3. Research data management (Leader: UBU; Partners: all) (M1-M48).
UBU will create a Data Management Plan (DMP) by M6 (D1.2a) following the FAIR data principles to describe the data management lifecycle and balance openness and protection of scientific information, commercialisation and IPR, privacy concerns, security and data management. The aim of the DMP is to define what data will be collected, processed or generated and what methodology and standards will be followed, and it will cover the data handling during and after the project, including how data will be shared, curated and preserved. The DMP will be updated whenever major changes arise and at least in M24 (D1.2b) and M46 (D1.2c).
T1.4. Risk management (Leader: UBU; Partners: all) (M1-M48).
UBU will coordinate the management of the risks on the implementation of the project activities. The identified risks will be tracked and managed according to the risk level (low, medium, high). The risk table included in the proposal (Table 3.1e) will be reviewed at least during each GA meeting, unforeseen risks will be included, the level of likelihood and severity assessed, and mitigation measures proposed and implemented. The procedures for risk management will be described in the D1.3 at M6 and the corresponding versions at M24 and M46.

Work package WP2 – Characterisation of pathogens, resistance sources and predisposing factors

Work Package Number	WP2	**Lead Beneficiary**	3 - AGRO
Work Package Name	Characterisation of pathogens, resistance sources and predisposing factors		
Start Month	1	**End Month**	42
Objectives			
-To develop of I-matrix for the specific isolation of Rs bacterial cells from contaminated environmental samples. - To characterise Cs/Rs at genomic and transcriptomic level to understand the genetic diversity and pathogen virulence signatures. - To identify resistance sources against Cs/Rs using tomato and potato broad germplasm collections and to understandthe genetic basis of disease resistance. - To identify the key predisposing physical and biological factors that underlie the susceptibility of tomato/potato cropsto Cs/Rs. - To integrate genomic and transcriptomic data in a digital platform for POMATO project.			
Description			

T2.1 Pathogen characterisation (Lead: AGRO; Partners: IHAR, USFQ, IDEN, WUR, CBGP) (M1-M30) Subtask 2.1.1 I-matrix approach for Rs isolation

This task consists of developing a tool and procedure for isolating Rs bacteria from highly contaminated environmental samples (plant extracts, soil, water). These solutions will allow for specific isolation and concentration of live Rs bacteria cells while simultaneously removing contaminants in the tested samples. For this purpose, a matrix for isolating Rs cells will be developed, using: i) Selection and verification of biological material for research (Rs strains, potato and tomato varieties); ii) Development of optimal conditions for the isolation of Rs bacteria through the selection of appropriate microbiological media, buffers and isolation time of Rs; iii) Development of receptor molecules enabling a specific reaction with bacterial cells; iv) Development of matrix bases for bacterial isolation (modification of material, shape and surface to allow covalent attachment of receptor particles) and v) Development of prototypes of an i-matrix based on the template and receptor particles.

The developed prototype of the tool (I-matrix) and the methodology for isolating Rs bacteria based on it will first be tested on nominated Rs assays under controlled laboratory conditions before being introduced on a wider scale. The subject of the next stage of work will be the validation of the developed solution on samples obtained under environmental conditions. The results of this task will allow the development of a protocol for rapid and effective isolation of live Rs cells directly from contaminated environmental samples and therefore will improve and make the Rs diagnostics more efficient.

Subtask 2.1.2 Determining genetic diversity and virulence factors for Cs/Rs

At least 100 isolated strains from potato/tomato infected and germplasm collections (25 isolates of Rs from AGRO, 25 isolates of Rs from USFQ and 50 isolates of Cs from IHAR) will be obtained. Genomes from 12 representative isolates (6 from Cs and 6 from Rs) from different areas and registered levels of disease will be completely sequenced, annotated and compared to understand the genetic diversity of field samples. Moreover, these strains will be assayed “in planta” to perform transcriptomics (CBGP) and describe the virulence signatures of field samples of Cs/Rs.

Methods developed at WUR to ensure consistent and uniform genome reannotation will be deployed and newly generated datasets will be queried together with the machine-actionable semantic database of >100.000 consistently de novo annotated microbial genomes. The selection of representative isolates will lead to a pangenome reconstruction for the pathogens.

T2.2 Compilation of resistance sources against Cs/Rs (Lead: AGRO; Partners: IHAR, USFQ, IDEN) (M1-M36)

The primary objective of this task is to identify and characterise genes associated with both qualitative and quantitative disease resistance in order to link the genetic variants to the different degrees of Cs/Rs susceptibility. This includes a thorough examination of genes conferring resistance (R genes) and susceptibility (S genes) within the context of their molecular and functional roles. Additionally, special attention will be given to plant metabolic pathways and regulatory networks that interact with these R and S genes, providing a comprehensive understanding of their contributions to plant defence mechanisms. Further transcriptomic data will provide which of those mechanisms are actively working for each studied plant under each Cs/Rs strain. A total of 200 samples of potato and 200 of tomato will be analysed and characterised for their tolerance level to Cs/Rs using broad germplasm working collections of IHAR and AGRO:

i) Potato: 50 varieties and 20 wild varieties from Europe; 130 native and landraces varieties from LATAM; ii) Tomato: 50 native varieties, 130 wild varieties and 20 tomato improved varieties from both LATAM and Europe. Whole-genome transcriptome and genome-wide association studies (GWAS) analysis of incompatible interactions between Cs/Rs and potato/tomato to identify candidate RGs against Cs/Rs will be evaluated.

T2.3 Linking pathogen–microbiome–host interactions (Lead: JSI; Partners: IHAR, AGRO, USFQ, IDEN, WUR) (M1-M36)

The task will assess and identify the environmental and soil factors (temperature, humidity, soil, pH, texture and organic matter content) associated to potato/tomato susceptibility against Cs/Rs. It will also determine 10 different plantassociated consortia exhibiting pathogen-promoting (positive interaction) and pathogen-tolerating (neutral interaction) properties. We will utilise 5 pathovars each of Cs and Rs, selected based on their highest and lowest pathogenicity characteristics identified in T2.1 (St 2.1.2). Individual pathogenic strains will be stained with viable fluorescent dyes, co-aggregated with extracted plant-associated microbiota, and encapsulated in 40 µm-sized droplets of alginate or agarose using a microfluidic approach. Plant-associated microbiota will be extracted at following disease stages: i) before infection, ii) before symptom onset, iii) during symptoms, and iv) during the final disease stage. These droplets containing co-aggregated bacteria will then be cultured in microwell plates, with continuous monitoring of pathogen and associated co-aggregated strain growth separately using two different fluorescence excitation/emission wavelengths. The results of this task will elucidate the involvement of microbiota in promoting and spreading pathogens and determine their contribution to plant susceptibility and diseas progression. These outcomes will feed the tasks carried out in WP4. T2.4 Compilation of molecular data into a common platform (Lead: IDEN; Partners: IHAR, AGRO, USFQ, WUR, CBGP, F4S) (M13-M42)

FAIR by design pipelines for data and metadata capture and analysis will be deployed to ensure the collected data and processed data remains machine actionable and is suitable for the deployment of AI/ML in WP3. Additionally, all molecular data obtained will be available in a public database such as GenBank-NCBI.

Work package WP3 – Early Detection and Surveillance Tools

Work Package Number	WP3	**Lead Beneficiary**	5 - IRIS
Work Package Name	Early Detection and Surveillance Tools		
Start Month	1	**End Month**	42

Objectives

- To scale up and extend an AI-driven cloud data management platform that collects data from the field devices andother information sources and runs the predictive models for disease prediction.
- To develop polymeric sensors for the detection of key metabolites generated in the early stages of the infection allowingearly detection of Cs/Rs in potato and tomato plants.
- To develop a self-contained compact multiparametric analyser for on-site real-time and non-destructive early detectionof diseases in potato and tomato plants.

Description

T3.1 Identification of biotic and abiotic factors affecting bacterial disease establishment and spread (Lead: IDEN; Partners: AGRO, WUR) (M1-M36)

Subtask 3.1.1: Mechanistic models: MAGS and GEMs

The available GEM for target pathogen Cs/Rs will be extensively tested and refined, when needed with data from WP2. Analysis of publicly available data and the metagenomic data generated in WP2 will uncover the taxonomic and functional composition of the microbial communities associated to disease progression or suppression. Metagenomics datasets will be mined to reconstruct MAGs. Keystone species will be inferred from co-occurrence networks. Their GEMs will be automatically reconstructed and will be employed to simulate their metabolic interactions, simulate their spatio-temporal dynamics, and their interaction with environmental factors In this way, this approach permits the quantitative inference of key metabolites during bacterial infection, which will be statistically validated in subtask 3.1.2 and utilised by T3.3. Existing models of plant metabolism will be combined with the GEMs of the pathogen and their ecological opponents to study host pathogen interaction and to further characterise the determinants of disease progression and attenuation.

Subtask 3.1.2: Definition of omics signatures associated with infection through multiscale data integration

Molecular signatures associated with bacterial infection and progression in tomato and potato plants through the integration of plant and microbial omics and phenotypic data will be defined. For this, a combination of multivariate

modelling and network inference will be used. Prior to integration and analysis, raw omics data will undergo a process of annotation, filtering of low-quality data, and normalisation. Generalised linear models will be used to account for potential confounding factors and batch effects. Data integration will be performed at both top and bottom levels: At the top level, we will combine results from standard analyses, including univariate, multivariate or machine-learning-based analyses. At the bottom level, we will concatenate data sets from various measurements. Bottom level integration will account for data error structure by modelling of additive and error components and explicit modelling using or adapting the existing component methods. Procedures to avoid overfitting will be implemented (e.g. cross-validation, resampling and LASSO). Extraction of underlying common and distinctive processes will be achieved by specialised methods for analysing complex multi-set data (e.g., GSVD, DISCO-SCA and MOFA). To complement the statistical approach, construction of multilayer networks will provide the relationships between molecular features, and disease. We will utilise network inference approaches to handle various data types, including Mixed Gaussian Graphical Models (GMM), probabilistic GMM and Random Forest. These techniques will enable the construction of conditional independence/ dependence graphs. Lastly, the topology of the estimated networks will be analysed by focusing on network hubs and motifs.

We will also work in tandem with the activity in T3.2 to integrate, with similar strategies, biomolecular data and disease signatures with image data acquired in T3.2 to assess the possibility of using images as proxies of molecular markers.

T3.2 Advanced AI-based models for robust disease detection and climate-driven risk assessment (Lead: FUND; Partners: WUR AGRO, IDEN, F4S) (M6-M40)

To develop and implement advanced AI-based models for comprehensive disease detection and severity assessment in potatoes and tomatoes crops, extending beyond traditional leaf-focused diagnostics to include various plant parts using advanced imaging techniques. The task also aims to incorporate environmental and climatic data to assess disease progression and integrate explainability techniques to ensure trustworthy AI.

To achieve this ambitious impactful objective, the task is divided into three subtasks:

Subtask 3.2.1: Acquisition of infected tomato and potato crops using NDVI Drone

This sub-task focuses on the acquisition of NDVI images from healthy and infected tomato and potato crops. Early detection and surveillance of Cs/Rs will be performed using high-resolution images obtained by drone equipped with multispectral camera. This equipment will be used to efficiently detect, differentiate, and quantify the severity of the symptoms induced by Cs/Rs infections under field conditions. Two aerial surveys per infected/healthy field will capture multispectral images with spatial resolution of 5-20 cm, which will be analysed using vegetation indices like NDVI and NDRE. This data will be also integrated with remote sensing data obtained from Sentinel- 2's, in which the range of multispectral data allows other indices (CCI, EVI or MSAVI) to be evaluated. The data collected will be integrated with AI-based predictive models to forecast potential outbreaks and suggest targeted interventions. The algorithms developed includes a processing phase that converts the read data into a collection of features, which are then analysed. An orthomosaic multispectral raster will be loaded into the algorithm to calculate spectral indexes and improve the detection rates of disease in infected and not infected fields. All five bands in the multispectral raster, as well as in the estimated vegetation spectral indexes, are denominated as input features. The labelled regions from the groundbased assessments are exported from QGIS and loaded into an array. This work is performed through different machine learning regression methods (e.g., RF, DT, KNN and XGB) to detect the Cs/Rs from multispectral images. Finally, the fitted model is validated using k-fold crossvalidation and the prediction stage, unlabelled pixels are processed in the optimised classifier, and their values are displayed in the same 2D spatial image from the orthorectified multispectral raster. Two campaigns of images acquisition, one in Portugal from tomato and potato crops and one in Poland from potato crops will be carried out.

Subtask 3.2.2: Robust Trustworthy Early Detection Image Analysis based on symptoms and biomolecular metadata This subtask focuses on developing AI models for early detection of diseases in potato and tomato crops using image analysis combined with biomolecular data. The models will analyse various plant parts using advanced imaging techniques, and symptoms will be correlated with biomolecular patterns to enhance detection accuracy and reliability. AI models like Visual Transformers and Vision-Language Models (VLMs) will be explored, incorporating lightweight guided visual search algorithms to optimise image analysis. The novelty and challenge reside in the enhanced disease detection accuracy through the integration of symptom and biomolecular correlations with reasoning capabilities and certainty awareness of detection. The performance of the AI detection models will be measured using reliable detection accuracy rates, like Matthews Correlation Coefficient or Kappa Coefficient, and false positive/negative rates. To address the interpretability, robustness, and trustworthiness of the models, we will compare AI predictions with human annotations to ensure a high correlation. Additionally, we will measure the consistency of explanations provided by the AI through feedback from human experts.

Subtask 3.2.3: Climate Data Driven forecasting model of disease spread and risk

This subtask aims to develop AI models that integrate environmental and climatic data to predict the spread and risk of the study of diseases in potato and tomato crops. The models will consider abiotic factors (e.g., temperature, humidity) that will be enriched with biotic factors specific to the pathogens under study (e.g., pathogen populations, reproduction cycle, transmission routes) to predict disease progression. The goal is to use climate forecasting to understand how

changing environmental conditions affect disease dynamics and provide timely risk assessments. This task pursuits shortterm (weeks to months) and long-term (seasons to years) disease risk forecasts capable of upgrading and adapting based on incoming environmental changeable data to make dynamic active predictions including interval of confidence and certainty of the prediction.

T3.3 Smart polymers for straightforward and early disease detection (Lead: UBU; Partners: CBGP, IRIS) (M6-M40)

This task aims to develop polymeric sensors for the detection of key metabolites generated in the early stages of the infection (identified in T3.1), considering them as specific biomarkers for Cs/Rs and allowing the early detection of them in potato and tomato crops in a simple and economically feasible way. The chemical structure of the proposed polymers is based on vinyl copolymers, which allow for a flexible design of hydrophilic polymers for the preparation of films with gel-like behaviour (meaning they can swell in water) that respond to the targets with a visual alert (change in colour or fluorescence). A simple yet efficient methodology will be followed for this purpose. Well-established receptors for the target molecules (key metabolites) will be chosen after a bibliographic search to produce highly robust sensory polymers from a chemical standpoint. These receptors will be slightly modified to introduce polymerizable groups and then they will be copolymerised with commercial comonomers (>99 mol%). The selection of proven receptor molecules ensures the success of the sensor materials, and commercial comonomers provide simplicity to the material and the ability to tailor properties based on prior knowledge, such as hydrophilicity, allowing the receptors to be used in aqueous media.

Subtask 3.3.1: Design of the receptor/sensory units

The viability of the selective and sensitive detection of the key metabolites identified in T3.1 will be studied to select the most appropriate targets for the early detection of the Cs/Rs in potato and tomato plants by a thorough examination of the literature, taking into account the possible interferences. Should the selected key metabolites be simple organic molecules, we will design equally simple organic or organometallic receptor/sensory units based on well-established examples. If the key metabolites prove to be complex biomolecules such as proteolytic enzymes (proteases), we will use receptors based on chemically anchored fluorogenic peptide substrates.

Subtask 3.3.2: Synthesis, characterisation and response evaluation of the sensory polymers

In this subtask the final smart sensory materials will be prepared, characterised and evaluated. Firstly, the acrylic/vinylic monomers based on the receptor/sensory units designed in 3.3.1 will be synthesised using traditional methods in organic synthesis and characterised (NMR, FTIR, CHN, HPLC, and GM). After obtaining high-purity sensory monomers, the polymeric materials will be prepared through their copolymerisation with commercial comonomers. In these materials, the sensory monomers will be present in a molar proportion generally lower than 1%, while the commercial comonomers that make up the structural matrix, without active groups, will be present in a molar ratio greater than or equal to 99%. These acrylic polymers will be prepared by conventional free radical polymerisation both in solution (linear polymers) and by bulk polymerisation (cross-linked polymers as films) thermally or photochemically initiated and the characterisation of both non-crosslinked and cross-linked materials (FTIR-ATR, UV/Vis spectroscopy, fluorescence spectroscopy, SEM, XPS, AFM-RAMAN, WAXS, and more) will be conducted. Finally, to develop tools for the early behaviour and effectiveness of the chromogenic/fluorogenic sensory polymers in terms of selectivity and sensitivity will be primarily analysed in aqueous media. Their performance under real conditions will be tested in growth chambers by CBGP and this technology will be finally validated in larger scale in WP5.

T3.4 Multiparametric Disease Detection (MDD) sensing device (Lead: IRIS; Partners: FUND, UBU) (M6- M40)

In this task, IRIS will design and develop a compact self-contained multiparametric device that will be suited for in-field (portable or handheld) and on-drone use and which will be powered by machine learning models for enabling the early detection of diseases in potato and tomato plants. The work is divided into two subtasks:

Subtask 3.4.1: Analysis and selection of monitoring technologies

Firstly, a series of multivariate spectroscopic analysers, imaging technology and univariate sensors will be screened to evaluate their suitability for monitoring the parameters as candidates for integration into a portable multiparametric device. The proposed monitoring technologies are, among others: (1) Temperature sensors and soil moisture sensors (sensor-based or NIR-based)- complemented with weather information; (2) NIR miniaturised spectroscopy sensors for plant disease identification; (3) Thermal imaging using InfraRed cameras to detect changes in plant temperature during latent infection periods; (4) Machine Vision using RGB cameras to detect visible symptoms, wounds, and other signs of plant diseases.

The performance, accuracy and reliability of each technology in detecting and identifying plant pathogens and diseases with a focus on early detection during latent infection periods will be assessed.

Based on the results obtained, the optimal combination of these technologies that can be effectively integrated into portable multiparametric device will be determined.

Subtask 3.4.2: Design and development and testing of the Portable Disease Detection Multiparametric Device

Based on the results from the previous subtask, the first step in developing the portable device will be to define the initial specifications and protocols. These definitions will be determined in collaboration with the WP3 partners. IRIS will then conduct a thorough study to map out and design the device architecture. The hardware will be designed and

engineered to deliver a compact, lightweight device, and a highly durable casing to house all the sensing, imaging and processing units. Software will be developed for on-device data processing the sensing and imaging data with machine-

learning algorithms for providing users with clear information for the early detection and classification plant diseases. A user-friendly interface on the device will also allow users to easily operate the device and to obtain easy to understand information. Connectivity with the Cloud platform developed in T3.5 will also be ensured.
Once the device and its algorithms are fully integrated and functional, its performance will be validated in all the four Case Studies of WP5, assessing its ability to accurately detect and identify plant bacteria and diseases.
Feedback from end-users will also be gathered to identify potential improvements of the detection device.
T3.5 AI powered Cloud data management for disease prediction (Lead: IRIS; Partners: IDEN, WUR, FUND, UBU) (M13-M42)
Connectivity with polymeric sensors developed in T3.3 and with the Portable Disease Detection Multiparametric Device from T3.4 will be established for data collection from these field devices, as well as from other relevant information sources (climatological data (temperature, humidity, precipitation) from sources such as, European Space Agency (ESA) and NASA satellites, geographic and temporal data on past disease outbreaks in potato and tomato crops, among others). The identified soil factors and climate conditions affecting bacterial disease establishment and spread from T3.1 and the predictive model for bacterial disease progression developed in T3.2 will be integrated into the platform and will run based on all the processed data. The cloud platform will be interfaced by a web and mobile application with easy to understand and clear visuals.

Work package WP4 – Natural and Bio-Control solutions

Work Package Number	WP4	**Lead Beneficiary**	8 - FUND
Work Package Name	Natural and Bio-Control solutions		
Start Month	1	**End Month**	40
Objectives			
- To study biocontrol agents actions as antagonist bacteria and specific enzymes incorporated into solutions to test onplants against Cs/Rs. - To identify the most effective plant derived-organic substance against Cs/Rs. - To obtain PGPM consortia active against Cs/Rs. - To scale up the natural and bio-control solutions at pilot plant (TRL 5).			
Description			

T4.1 Biocontrol coating solutions (Lead: FUND; Partners: CBGP, IDEN) (M1-M36)
Improved coating formulations based on effective microbial strain combinations and enzyme systems will be tested against pathogen candidates on seed and plant crops. Screening of a wide range of microbial stains (mainly Streptomyces, Pseudomonas and Bacillus spp.) will be carried out and selected according to their antibacterial effects against Cs/Rs. Further antimicrobial testing will be conducted with candidate bacteria isolated from the rhizosphere of Cs/Rs diseased plants. For this, initial identification will be carried out using matrix-assisted laser desorption/ionisation time-of-flight mass spectrometry (MALDI-TOF MS) and 16S rRNA sequencing. These results will be correlated with the supporting data on the identification of Cs/Rs resistance candidate microbes provided in T3.1. Different fermentation processes conditions will be evaluated in order to optimise the metabolite production as antibiotics, lipoproteins, and antimicrobial peptides from the antagonistic bacteria selected. The antagonistic ability of bacteria candidates will be determined at different concentrations against Cs/Rs through the disc diffusion method. For lactoperoxidase system (LPS) production, first step will be to define and prepare several substrate ratios and make defined dilution. Metabolites concentrations will be analytically determined to reach highest system yield. For lysozyme obtained from eggs white origin (purity 95%) several concentrations will be prepared in water sterile to be tested as antimicrobial solutions. Turbidity test based on optical densities determinations at different incubation time will be carried out against Cs/Rs. Besides, in in vitro challenge tests (inhibition area) will be also carried out in specific agar medium. Regarding the coating formulations, different polymeric materials will be evaluated for the colloidal delivery of microbial strain combinations, LPS and lysozyme. Encapsulations techniques as coacervation complex, emulsion solvent evaporation or spray dried will be considered. Size, dispersion, and Z potential of particles will be determined by dynamic light scattering (DLS).
Besides, encapsulation efficiency and active compounds release rate will be characterised in in vitro assays for both microbial strains and enzyme-based solutions. Phytotoxicity effect as burnings, spots and stains generated by biocontrol coating solutions will be evaluated on tomato and potato leaves. Finally, biocontrol coating solution prototypes, that showed best results in in vitro trials, will be tested in in vivo assays on potato and tomato crops at growth chamber. Seed

and plants will be previously infected, afterwards enzyme-based and microbial strain-based solution candidates alone or in combination of them will be tested by spray application. Periodically, presence (incidence and severity) of symptoms provoked by pathogens will be observed in plants.

T4.2 Plant derived organic substances (Lead: FUND; Partners: CBGP) (M1-M36)

A screening in in vitro assays of several plant-derived organic substances will be carried out to identify and test the most effective biocide agents against Cs/Rs. Candidate organic substances as alkaloids, phenolics compounds and terpenoids from different plant extracts and essential oils origin will be tested in in vitro assays. Screening of antibacterial activity of the different substances selected against Cs/Rs will be done by disc diffusion method at different concentrations studied. Substances that showed high antimicrobial effects will be then nanoencapsulated by emulsion and solvent evaporation techniques, using carrier material as cyclodextrine. Besides, similar screening studies will be also carried out with several candidates as plants extracts (e.g. garlic, onion, neem, etc.), essential oils (e.g. thyme, clove, cinnamon, lemongrass etc.) and natural resins as propolis to evaluate their antimicrobial effects against the pathogens selected. Then, best antimicrobial solution will be microencapsulated following the different methodology and encapsulation material already described in the T4.1. Potential phytotoxicity assessments of the compounds selected will be carried out on tomato and potato leaves. In in vivo assays of the plant derived organic substances selected will be also evaluated as described in T4.1.

T4.3 Plant growth-promoting microorganisms (PGPMs) (Lead: JSI; Partners: UBU, FERT) (M1-M36)

The task involves isolating PGPMs from the rhizosphere of healthy plants and plants resistant to pathogens identified in a previous phase (T2.3). UBU will obtain individual isolates that are showing PGP traits and JSI will employ their proprietary iACME method, which includes i) non-cultivation-based extraction of microbial cells, ii) altering cell surface potential to induce aggregation), and iii) encapsulating these aggregates in 40 µm alginate beads. The size of these aggregates will determine the complexity of intercellular interactions as well as enable to test log4 various combinations of bacterial cells. The same iACME approach will be used on isolates obtained by UBU to determine their compatibility and synergistic effects.

The resulting consortia will be tested for various PGP phenotypes, such as phosphorus solubilisation, hydrogen cyanide (HCN) production, siderophore secretion, and others. JSI will develop spectroscopic methods for largescale testing to detect these phenotypes inside the microbeads. Additionally, microdroplet-based assays will be developed to evaluate the interaction of consortia against Cs/Rs, aiding in the selection of effective PGPM consortia. Furthermore, JSI will introduce setups using microdroplet methods to evaluate i) contact inhibition, ii) water-soluble excretions, and iii) the production of volatile organic compounds by the consortia. Based on spectroscopic parameters and anti-pathogenic activity, JSI aims to identify 10 effective consortia for each plant species. Metagenomic analysis will characterise the PGP genomic information, particularly the genes involved in interspecies interactions within consortia. The synthetic biofilms made of the most efficient consortia will be attached on inanimate and plant roots and further succession of species composition and spatial distribution will be studied. The study will explore synergistic interactions within aggregates and assess potential antagonistic activities against pathogens. The most active consortia against Cs/Rs will be used to prepare the formulations and methods for efficient large-scale coating of plant root system using electrostatic approach. Five developed formulations will be tested separately on 30 plants in growth chambers for determining resistance to infection and the progress of severity of the progress of the disease signs.

T4.4 Chemoperception disruptive products (Lead: CBGP) (M1-M36)

Identification of genes involved in the chemoperception function in Cs/Rs will be performed by computational analysis based on genome sequences (WP2) and phylogenomic comparison of the chemoperception genes. The annotation of chemoreceptors using a de novo methodology, implemented for the identification of those genes relevant for the Cs/Rs plant interaction, will be carried out. The functional study of chemosensing proteins in the selected Cs/Rs strains (see T2.1) will be developed following several analyses: i) construction of mutant strains in the identified chemoreceptors and analysis of their chemotactic behaviour to identify their putative ligands; ii) determination of the chemotactic profile of these mutants respect to that of the wild-type strains using a commercial (BIOLOGTM) and a custom compound collection created in our lab containing potato/tomato plant-derived compounds. Increasing amounts of tested compounds will be used to determine the saturating concentrations that inhibit bacterial motility. The chemicals inducing a differential response between wild type and mutants will be considered putative ligands of the analysed chemoreceptors. The biochemical characterisation of selected chemoreceptors will be achieved through the expression and purification of recombinant ligand binding domains (LBDs). To determine if there is a direct ligand binding between ligands and the selected LBDs, we will carry out the following approaches: i) Fluorescence-based thermal shift assays (FTS); ii) Isothermal Titration Calorimetry (ITC). Phenotypic characterisation of the Cs/Rs chemoreceptor mutants to evaluate the role of each chemoreceptor in the context of plant infection will be performed through plant infection assays in tomato and potato plants.

The previously identified ligands of the most relevant chemoreceptors (based on functional data) will be assayed as infection disruptive agents. We will perform infection assays in the presence of different saturating compound concentrations.

T4.5 Scale up process of the solutions at pilot plant (TRL5) (Lead: CHEM; Partners: FERT, FUND, CBGP, JSI) (M12M40) CHEM and FERT will be in charge of manufacturing, at pre-industrial (pilot plant) scale (TRL5), the solutions developed in T4.2 and T4.4; and T4.1 and T4.3, respectively. The task will include several optimisation process steps and the definition of protocols for pilot plant scale production (up to 200L) for each final product. Functional activity of each solution manufactured will be evaluated during a defined storage shelf life (maintaining antimicrobial activity, e.g.). Besides, pre-industrial manufacturing will be carried out according to standard of quality, safety, and respect for the environment (toxicity assessments, in T5.4). Lower amounts (up to 10 L) of each solution will be produced to carry out trials on potato and tomato plants in growth chamber by FUND, JSI and CBGP and evaluate the antimicrobial efficacy against Cs/Rs. The best two solutions, that will show the most promising results in the growth chamber trials, will be manufactured to be tested in field validation trials during the WP5 execution.

Work package WP5 – Integrated Pest Management Strategies validated in field trials, SSbD Framework and Policy Recommendations

Work Package Number	WP5	**Lead Beneficiary**	1 - UBU
Work Package Name	Integrated Pest Management Strategies validated in field trials, SSbD Framework and Policy Recommendations		
Start Month	1	**End Month**	48
Objectives			
- To assess the effectiveness of all mentioned early detection and bio-control strategies at TRL5. - To implement 4 case studies to validate POMATO's strategies in LATAM and Europe: CS#1-3 addressing Rs in potato/tomato; CS#4 addressing Cs in potato. - To provide assessment on the developed bio-control solutions regarding their safety, environmental and economicperformance. - To develop of an Integrated Pest Management (IPM) Plan to integrate the results of the validated project strategies.			
Description			

T5.1 Validation field trials against Rs infection under real conditions in Colombia and Ecuador in potato crops: Early Detection & Bio-control solutions (Lead: USFQ; Partners: AGRO, UBU, IRIS, CBGP, JSI, FUND, CHEM, FERT) (M25-M48)

The developed early detection (smart sensory polymers and MDD device) and two selected bio-control solutions will be validated against Rs for potato crops in a two-phase fashion: first in greenhouse conditions (Colombia) and then at larger scale in field conditions (Ecuador). The bio-control validation studies will be planned according to the results gathered from growth chamber tests in T4.5, selecting the two best-performing bio-control solutions. CS #1: A total of 750 potato samples (susceptible varieties Solanum tuberosum subsp. andigenum, S. tuberosum subsp. Tuberosum) will be at disposal for the validations in greenhouse conditions in Colombia. The performance of the smart sensory polymers and the MDD device will be evaluated for the pre-symptomatic and symptomatic Rs disease detection and the two selected bio-control solutions will also be tested in the greenhouse (each one for a total of 375 potato samples). Regarding the validations in field conditions with local landowners in Ecuador, the MDD device will be tested in 15 different locations (5 locations per province) of 1-2 ha of potato plant fields at different phenological development stages. The two bio-control solutions will be validated in fields under two crop management strategies (conventional and organic) in 3 different locations (1 per province) of 0.5-1 ha each. The field validations will be performed twice (in the dry season and in the rain season) for at least one commercial potato variety.

T5.2 Validation field trials against Rs pest disease under real conditions in Colombia in tomato crops: Early Detection & Bio-control solutions (Lead: AGRO; Partners: UBU, IRIS, CBGP, JSI, FUND, CHEM, FERT) (M25-M48)

The developed early detection (smart sensory polymers and MDD device) and two selected bio-control solutions will be validated against Rs for tomato crops in Colombia both in greenhouse and field conditions. The bio-control validation studies will be planned according to the results gathered from growth chamber tests in T4.5, selecting the two best-performing bio-control solutions. CS #2: A total of 750 tomato samples (susceptible tomato cultivar Santa Cruz Kada) will be at disposal for the validations in greenhouse conditions in Colombia. The performance of the smart sensory polymers and the multiparametric device will be evaluated for the pre-symptomatic and symptomatic Rs disease detection and the two selected bio-control solutions will also be tested. Field trials with local landowners will take place

in commercial tomato fields in Boyacá Department (Colombia) with an extension of 0.5 ha. The smart sensory polymers, the multiparametric device and the two selected bio-control solutions will be evaluated twice (in the dry season and in the rain season) for at least one commercial tomato variety.

T5.3 Validation field trials against Cs/Rs infection under real conditions in European potato and tomato crops: Early Detection and Bio-control solutions (Lead: F4S; Partners: IHAR, UBU, IRIS, CBGP, JSI, FUND, CHEM, FERT) (M25-M48)

Strict regulations are to apply in the EU regarding protective measures against pests of plants. Particularly, EU Regulations 2022/1193 and 2022/1194 establish measures to eradicate and prevent the spread of Rs and Cs, respectively, which hinders the possibility to carry out validation field trials in the European territory. Due to these constrains, CS #3 and #4 cannot take place in a two-phase fashion (greenhouse and field) and will face some limitations to comply with the Union legislation and ensure protection of the Union territory. CS #3 (F4S): The performance of the MDD device will be evaluated for Rs in tomato fields in Portugal in a location yet to be determined. The strategy to follow to select the most appropriate field, will be to work in close contact with the Direcao-Geral da Alimentacao e Veterinária (DGAV) to know the areas where there is a suspicion of the presence of Rs and immediately test the multiparametric device prior to the official confirmation of the infection and the application of the eradication measures. CS #4 (IHAR): The developed early detection (smart sensory polymers and MDD device) and the two best-performing bio-control solutions (from T4.5) will be validated against Cs for potato crops in Poland in an experimental field with an extension of 400 m2. This field is located in one of IHARPIB's divisions at Bydgoszcz, is guarded and inaccessible to unauthorised persons and has a permit from the State Plant Health and Seed Inspection Service (SPHSIS) to perform field tests involving quarantine Cs bacteria. Potato varieties of Solanum tuberosum selected during the POMATO project as most susceptible and most resistant to Cs will be employed. Validation tests of the selected solutions will be carried out over two growing seasons from the turn of April/May to the turn of September/October. Each solution will be evaluated on a research sample of 200 plants.

T5.4 SSbD Framework and safety hazard application to natural and bio-control solutions (Lead: UBU; Partners: All) (M1-M48)

In POMATO, the safety and the sustainability of all the materials and compounds used in the project for the development of natural and bio-control solutions for tackling outbreaks of plant pests will be evaluated. Hence, subtasks have been implemented to cover these aspects: Subtask 5.4.1 Safety Hazard assessment and Safe by Design framework: (Step 1-3) For materials and chemicals that can be assessed using the SSbD framework, their specifications will be gathered. To evaluate safety, the first three steps of the SSbD framework will be applied to understand the safety of substances used at various stages—from material selection (Step 1) to production (Step 2) and use phase (Step 3), in compliance with REACH and EFSA standards. For compounds where the SSbD framework does not apply, a comprehensive safety assessment will be conducted using various bioassays adapted from OECD guidelines for biological pesticides. We will ensure that the natural and bio-control solutions (WP4) comply with the EU regulatory framework 2019/1009 and 528/2012, which govern the regulation of fertilising products and biocidal products in the EU market. The safety of solutions developed in WP4, will be evaluated through several in vitro and/or in vivo assays. Initially, an environmental toxicity assessment will be performed on non-target microorganisms using soil at a microcosm scale, with the most efficient treatment potentially tested in a lab mesocosm. The safety of natural and bio-control solutions in soil will be assessed by soil characterisation and analysing the soil microbial community. Assays will include microbial soil respiration test (ISO-16072), determination of soil microflora abundance and activity using respiration curves (ISO-17155), determination of microbial community abundance from soil DNA extracts (ISO-17601), nitrogen transformation test (OECD 216), carbon transformation test (OECD 217), soil enzyme activity test (ISO-20130), and microbial metabolite residue analysis. Changes in microbial communities will be evaluated through metabarcoding analyses following direct extraction of soil DNA (ISO-11063). Additional assays for nontarget organism risk assessment may include the in vitro eco-toxicity assay OECD 249 to test acute toxicity using the RT-gill W1 trout cell line, microtoxicity assay on Aliivibrio fischeri, genetic toxicity assay using the Ames test for mutagenicity on bacteria, and the Algal Growth Inhibition Test (OECD 201) to evaluate effects on algae. Some in vivo tests will be subcontracted to certified labs, including the aquatic toxicity test Acute Toxicity to Fish (OECD 203) to determine the LC50 for fish and the Honey Bee Toxicity Test (OECD 213) to determine contact and oral toxicity to honey bees. Phytotoxicity assays to study germination and oxidative stress (photosynthetic pigment determination, soluble proteins, catalase and ascorbate peroxidase activities in leaves) in crops or model plants such as Arabidopsis or Brassica may also be performed in a plant growth chamber with controlled conditions. These will include the Terrestrial Plant Test for Vegetative Vigour (OECD 227) and the Seedling Emergence and Seedling Growth Test (OECD 208). Additionally, human hazard will be assessed through cytotoxicity testing on cell lines/tissues representative of relevant human exposure pathways (inhalation, oral, or dermal) using 2D and advanced 3D in vitro models to study viability, oxidative stress/DNA damage, membrane integrity, and transcriptomic analysis to reveal toxicity mechanisms. Skin irritation will be evaluated using the OECD 439 assay. This approach will estimate the local toxicity of the product and evaluate major risks associated with the usual preparation operations of the microbial pesticide or in the event of an accident.

Subtask 5.4.2 Environmental and Economic Assessment: (Step 4 and 5): The evaluation of the environmental and

economic impacts of the production of the natural and bio-control solutions will be performed using Life Cycle Assessment (LCA) and Techno-Economic Analysis (TEA) methodologies. The ISO 14040 and 14044 standards for LCA will be followed, determining the goal and scope of the study in the first step. Collaboration from the partners involved in the development and validation of the solutions will be needed to gather all the foreground data necessary to complete the Life Cycle Inventory (LCI), which will also be used to determine the capital and operational costs of the processes. The LCI will be modelled in LCA software (e.g., SimaPro) using databases (e.g., ecoinvent), and the impact assessment will be calculated using the Environmental Footprint methods developed by the European Commission. For the TEA, different financial indicators will be calculated (e.g., Net Present Value, Internal Rate of Return). The interpretation stage will identify the main hotspots and compare the solutions developed within the project with the main benchmarks, both in environmental and economic terms. Special attention will be given to the consideration of different TRLs of the alternatives. Therefore, prospective scenarios for upscaling the production of the solutions can be evaluated and complemented with sensitivity and uncertainty analysis.

T5.5 IPM Guidelines for farmers and policy recommendations (Lead: EUPAT; Partners: UBU, PIN, F4S) (M19-M48)

Subtask 5.5.1 Delivering IPM guidelines to farmers: UBU will deliver the Integrated Pest Management reports, covering the technical, economic, environmental, and social aspects of the proposed POMATO solutions.

The reports will include comprehensive recommendations, future directions, and guidelines, aiming to serve as support material for farmers and other key target groups and to align with policy recommendations from next subtask. To ensure the efficacy of the strategies outlined in POMATO's IPM Plan, the 1st IPM guidelines, based on the molecular characterisation of the bacteria and the resistance genes of tomato and potato (WP2), will be released in M30. A 2nd report, which will incorporate the final outcomes of the early detection and bio-control solutions, will be delivered in M48. All of this will be in accordance with policy directives, while also developing new material intended for farmers to serve as practical guide.

Subtask 5.5.2 Delivering Policy recommendations: EUPAT and PIN, with the help of POMATO partners, will prepare a policy paper which will assist EU and Member States' policy makers in designing, monitoring, reviewing, and adjusting (if needed) current policies, rules and programs regarding Cs and Rs, as well as shaping and supporting the implementation of new policy initiatives and decisions on these pathogens. To achieve this, existing EU policies will be identified and assessed if any further advice is needed to cover the issues such as detection, monitoring, prevention, etc. In addition, EUPAT and PIN will prepare a list of recommendations advising on further research building on the results of the POMATO project. This will be prepared at the end of the project with the help of all project partners and will be presented to and discussed with the EC in a clustering event. If needed, all partners will translate these recommendations into their respective national languages and extensively distribute them among relevant stakeholders involved in policy and decision-making. Most partners have extensive experience in lobbying and advocacy and can leverage their existing networks and contacts for this purpose. The same approach will be applied to the EU recommendations, which will be submitted to the EC, the European Parliament, the Committee of the Regions, various EU networks, and the Commission Expert Group on Plant Health Legislation.

Work package WP6 – Dissemination, Communication, Exploitation and Multi-Actor Engagement

Work Package Number	WP6	**Lead Beneficiary**	13 - INO
Work Package Name	Dissemination, Communication, Exploitation and Multi-Actor Engagement		
Start Month	1	**End Month**	48
Objectives			
- To set up and execute a clear and strategic Dissemination, Exploitation and Communication (DEC) Plan. - To launch the POMATO Multi-Actor Participatory Platform. - To set up Multi-Actor Approach Framework and Activities. - To actively engage with a wide range of stakeholders including farmers, industry experts, policymakers, and widersociety, fostering strong synergies that enhance the adoption and impact of POMATO technologies. - To build synergies with sister projects and related initiatives, EU networks and associations to amplify the projectimpact. - To craft IPR Strategy and a roadmap for sustainability of key exploitable results.			

Description

T6.1 Dissemination, Exploitation and Communication Plan (Lead: INO; Partners: all) (M1-M6)
T6.1 INO and EUPAT will tailor a multi-channel plan for dissemination and exploitation of results, including communication activities (D6.1). The Plan will: 1) Lay out the narrative and messages in a dissemination methodology; 2) outline the key benefits POMATO's novel technology, and its unique value proposition compared to existing solutions; 3) map the interplay between stakeholders groups to deliver a robust engagement strategy across the language technology value chain; 4) set up and maintain the tools and channels to promote POMATO and its solutions across digital, offline, and social media platforms; 5) design mechanisms for cooperation with relevant partnerships; 6) coordinate dissemination of research outputs at conferences, and seminars; 7) draft the overall exploitation strategy for our KERs, describe the steps that will be taken to exploit the commercial potential and roadmap. Potential regulatory hurdles associated with the novel biocontrol products will be identified and assessed as an integral part of our exploitation activities. Strategies to address these challenges will be developed with active participation from POMATO product providers and the multi-actor community. The Plan will be reviewed against engagement metrics, adjusting resources and channels (T6.2 and T6.3) accordingly.
T6.2 Public outreach and case study specific promotional campaigns (Lead: F4S; Partners: INO, UBU, EUPAT) (M1-M48)
This task will focus on execution of the DEC Plan activities, which will target all stakeholder groups on the European and global levels and include: 1) development of brand material (logo, templates, toolkits and identity guidelines); 2) facilitation of (online) discussions and debates, via YouTube, podcasts, social media platforms, and press outreach; 3) submission of high quality papers at high impact, peer-reviewed journals and magazines; 4) cooperation with various media outlets targeting a broader audience (press release, blogs, explanatory videos, podcast, radio and TV outreach. In addition, this task will focus on set of outreach activities and close interaction with POMATO stakeholders, including:
a) engagement of the farmers and wider society (including youth and general publics) through 6 dedicated activities (2 workshops in LATAM (AGRO, USFQ) and 4 workshops or plenary sessions in target congress in Europe (UBU, EUPAT, PIN and F4S) and four "POMATO Pilot Open Days" (UBU, FUND, together with CBGP, F4S, and PIN) in order to raise awareness about environmental and economic problems related to pest outbreaks and inform them about POMATO solutions and cutting-edge technologies to address these challenge; b) organise consultations and one-on-one meetings with key industry and policy stakeholders at relevant events; c) presenting POMATO results and achievements at relevant events (e.g. conferences, agricultural fairs, farmer days, etc.) (See Section 2.2 for more details about event-related KPIs). T6.3 POMATO Multi-Actor Participatory Platform (Lead: INO; Partners: all) (M6-M48)
As the principal amplifier of the POMATO results and outputs, we will launch the POMATO Multi-Actor Participatory Platform. This platform will be an extension of the project website, providing access to information and resources, realtime reporting, data visualisation, and insights. Specifically, the platform will offer valuable multimedia information for: i) conventional and organic potato and tomato growers on natural and bio-control solutions for tackling pests (inputs from WP4), including information about the POMATO's IPM Plan to reduce reliance on chemical pesticides, minimise pest damage, and promote crop health (input from WP5); ii) retailers, wholesalers, and distributors on POMATO's Early Detection and Surveillance Tools (input from WP3); iii) Researchers and innovation sectors on the characterisation of pathogens, resistance sources, and predisposing factors (inputs from WP2); iv) Environmental organisations and policymakers on the environmental and economic benefits of POMATO solutions, as well as policy recommendations streamed directly from the project pilot activities (inputs from WP5). INO will be responsible for the technical development and management of the platform, while all partners will contribute to providing the multimedia content. In addition, UBU will orchestrate the delivery of 25 practice abstracts to the European CAP Network.
T6.4 Multi-Actor approach framework and activities (Lead: F4S Partners: ALL, M1-M48)
This task aims to facilitate the co-creation and implementation of POMATO solutions for tackling potato and tomato pests by establishing a Multi-Actor Approach (MAA) Framework to engage a variety of (local) stakeholders, identify their specific needs, and encourage collaborative input. As such, this task will include the following activities: (1) Identification of key stakeholders interested in the project outcomes at both the pilot-site level and the international level. (2) For each identified stakeholder group, this task will assess their real-life needs, problems, and opportunities to engage in the co-creation process of POMATO's solution development and evaluation, as well as knowledge sharing throughout the project's lifespan. (3) Establishment of the MAA Governance Model (D 6.1) at the beginning of the project to define responsibilities among POMATO partners and MAA stakeholders, guide the partners on effective communication flow with the identified multi-actor community, and provide partners with reporting and periodic assessment procedures on the predefined goals that the established network of stakeholders accomplished. Different partners will be chosen to facilitate discussion, provide input, co-create, co-decide, and disseminate project results during a set of different (local) knowledge exchange activities (workshops, Open Days, Practice abstracts) with the MAA community. The Governance Model will be included in D6.1, while the activities will be reported in each D6.2 (M18 and M48).
T6.5 Clustering, synergies, and interlinkages with the European R&D community (Lead: UBU; Partners: All) (M1-M48) This task focuses on the clustering with other related projects and international initiatives across Europe and other

world regions, including selected operational groups of the EU CAP Network. POMATO will assess collaboration opportunities against qualitative/quantitative indicators, such as relevance, feasibility, and terms for collaboration. A

range of actions will be pursued, including alignment based on high-level interactions (e.g., identification of overlaps and mapping of incentives), joint cross-project task forces (contribution to standardisation activities, capacity building, joint dissemination and communication activities) outcome utilisation (share, reuse, or exchange results).

T6.6 Exploitation and IPR Management (Lead: INO; Partners: All) (M1-M48)

This task will set up the framework, plan and implemented all necessary activities for the exploitation and sustainability of POMATO results after the end of the project. The handling of the exploitation activities will be consolidated into an 'Exploitation Plan and IPR Management', which will be integrated into D6.1 (M6) and updated in M24 and M48. The Plan will include: i) assessment and categorisation of the key exploitable results ii) assessment of exploitation strategies for POMATO results (scientific, policy, societal and potential commercial) and roadmap for their implementation; iii) the joint and individual exploitation plans; iv) IPR assessment and strategy; v) exploration of business models to promote adoption of POMATO 's key exploitable results as well as to develop better awareness of the benefits among different end-user groups; vi) assessment on how different multiactor governance models have been implemented in the project pilots, and how these models could be utilised for the promotion of POMATO's Integrated Pest Management Plan.

Finally, the feasibility of an operational and organisational model for a post-project partnership that will serve to advance and support the joint exploitation of the POMATO results and ensure its sustainability beyond the duration of the project will also be evaluated. Finally, the consortium will utilise all available "Horizon" tools to boost uptake of results, such as Horizon Results Booster and Horizon Results Platform.

LIST OF DELIVERABLES

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

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Public — fully open (⚠ automatically posted online)

Sensitive — limited under the conditions of the Grant Agreement

EU classified —RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision 2015/444

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D1.1	Data Management Plan	WP1	1 - UBU	DMP — Data Management Plan	PU - Public	6
D2.1	I-matrix for improved isolation of Rs	WP2	2 - IHAR	R — Document, report	SEN - Sensitive	30
D2.2	Cs/Rs completely characterised at molecular level	WP2	3 - AGRO	R — Document, report	SEN - Sensitive	30
D2.3	Resistant tomato/potato genotypes against Cs/Rs and epigenomic markers associated to quanti-qualitative disease resistance traits	WP2	3 - AGRO	R — Document, report	SEN - Sensitive	36
D2.4	Information on population diversity and structure of microbiota associated with environmental, soil factors and native soils that are suppressive/conducive to Cs/Rs	WP2	10 - JSI	R — Document, report	SEN - Sensitive	36
D2.5	Common platform with molecular data suitable for AI/ML	WP2	6 - IDEN	R — Document, report	SEN - Sensitive	32
D3.1	Mechanistic models of establishment and spread of Cs/Rs	WP3	6 - IDEN	R — Document, report	SEN - Sensitive	36

D3.2	Infection signatures including omics markers associated to infection onset and progression	WP3	7 - WU	R — Document, report	PU - Public	36

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D3.3	AI-based Disease Detection and Risk Forecasting Models	WP3	8 - FUND	OTHER	SEN - Sensitive	40
D3.4	Sensory polymeric material	WP3	1 - UBU	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	40
D3.5	Final version of the multiparametric disease detection device	WP3	5 - IRIS	OTHER	SEN - Sensitive	40
D3.6	Functional AI-powered Cloud data management platform	WP3	5 - IRIS	OTHER	SEN - Sensitive	42
D3.7	User guides, and documentation to support the deployment and use of the platform	WP3	5 - IRIS	R — Document, report	SEN - Sensitive	42
D4.1	Formulation of biocontrol coating solutions	WP4	8 - FUND	R — Document, report	SEN - Sensitive	36
D4.2	Plant derived organic substances products	WP4	8 - FUND	R — Document, report	SEN - Sensitive	36

D4.3	Report on PGPM consortia	WP4	10 - JSI	R — Document, report	SEN - Sensitive	36
D4.4	Solution based on disruption of Cs/Rs chemoperception	WP4	9 - CBGP	R — Document, report	SEN - Sensitive	36
D4.5	Report of the natural and bio-control solutions manufacturing for pilot plant	WP4	11 - CHEM	R — Document, report	SEN - Sensitive	40
D5.1	Pilot validation for Rs in potato fields in LATAM (Colombia and Ecuador) (CS#1)	WP5	3 - AGRO	R — Document, report	PU - Public	48

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

The labels used mean:

Public — fully open (⚠ automatically posted online)

Sensitive — limited under the conditions of the Grant Agreement

EU classified —RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision 2015/444

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D5.2	Pilot validation for Rs in tomato fields in LATAM (Colombia) (CS#2)	WP5	3 - AGRO	R — Document, report	PU - Public	48
D5.3	Pilot validation for Rs in tomato fields in Portugal (CS#3)	WP5	16 - F4S	R — Document, report	PU - Public	48
D5.4	Pilot validation for Cs in potato fields in Poland (CS#4)	WP5	2 - IHAR	R — Document, report	PU - Public	48
D5.5	Safety pre-assessment of the materials and processes	WP5	1 - UBU	R — Document, report	SEN - Sensitive	24

D5.6	Safety of natural and bio-control solutions	WP5	1 - UBU	R — Document, report	SEN - Sensitive	48
D5.7	Final safety and sustainability assessment of the materials and processes	WP5	1 - UBU	R — Document, report	SEN - Sensitive	48
D5.8	IPM guidelines for farmers	WP5	1 - UBU	R — Document, report	PU - Public	30
D5.9	Final IPM guidelines for farmers	WP5	1 - UBU	R — Document, report	PU - Public	48
D5.10	Policy recommendations	WP5	15 - EUPAT	R — Document, report	PU - Public	40
D6.1	Dissemination, Exploitation and Communication Plan	WP6	13 - INO	R — Document, report	SEN - Sensitive	6
D6.2	Multi-Actor Participatory activities and platform	WP6	13 - INO	DEC —Websites, patent filings, videos, etc	PU - Public	18

Deliverables

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D6.3	Update on Multi-Actor Participatory activities and platform	WP6	13 - INO	DEC —Websites, patent filings, videos, etc	PU - Public	48
D6.4	Report on DC, networking and clustering activities	WP6	13 - INO	R — Document, report	PU - Public	18

D6.5	Final report on DC, networking and clustering activities	WP6	13 - INO	R — Document, report	PU - Public	48
D6.6	Practice Abstracts - batch 1	WP6	1 - UBU	R — Document, report	PU - Public	12
D6.7	Practice Abstracts - batch 2	WP6	1 - UBU	R — Document, report	PU - Public	36
D6.8	Practice Abstracts - batch 3	WP6	1 - UBU	R — Document, report	PU - Public	48
D6.9	POMATO Exploitation Plan	WP6	13 - INO	R — Document, report	SEN - Sensitive	24
D6.10	Final POMATO Exploitation Plan	WP6	13 - INO	R — Document, report	SEN - Sensitive	48

Deliverable D1.1 – Data Management Plan

Deliverable Number	D1.1	**Lead Beneficiary**	1 - UBU
Deliverable Name	Data Management Plan		
Type	DMP — Data Management Plan	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP1
Description			
Data Management Plan (DMP) to describe the data management lifecycle and balance openness and protection of scientific information, commercialisation and IPR, privacy concerns, security and data management.			

Deliverable D2.1 – I-matrix for improved isolation of Rs

Deliverable Number	D2.1	**Lead Beneficiary**	2 - IHAR
Deliverable Name	I-matrix for improved isolation of Rs		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	30	**Work Package No**	WP2
Description			
A tool and procedure for isolating Rs bacteria from contaminated environmental samples will be developed to allow the isolation and concentration of live Rs bacteria cells while simultaneously removing contaminants in the tested samples therefore improving and making the Rs diagnostics more efficient.			

Deliverable D2.2 – Cs/Rs completely characterised at molecular level

Deliverable Number	D2.2	**Lead Beneficiary**	3 - AGRO
Deliverable Name	Cs/Rs completely characterised at molecular level		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	30	**Work Package No**	WP2
Description			
Cs/Rs completely characterised at molecular level leading to a pangenome reconstruction for the pathogens.			

Deliverable D2.3 – Resistant tomato/potato genotypes against Cs/Rs and epigenomic markers associated to quanti-qualitative disease resistance traits

Deliverable Number	D2.3	**Lead Beneficiary**	3 - AGRO
Deliverable Name	Resistant tomato/potato genotypes against Cs/Rs and epigenomic markers associated to quanti-qualitative disease resistance traits		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP2
Description			

Genes associated with both qualitative and quantitative disease resistance will be identified and characterised in order to link the genetic variants to the different degrees of Cs/Rs susceptibility.

Deliverable D2.4 – Information on population diversity and structure of microbiota associated with environmental, soil factors and native soils that are suppressive/conducive to Cs/Rs

Deliverable Number	D2.4	**Lead Beneficiary**	10 - JSI
Deliverable Name	Information on population diversity and structure of microbiota associated with environmental, soil factors and native soils that are suppressive/conducive to Cs/Rs		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP2
Description			
The environmental and soil factors (temperature, humidity, soil, pH, texture and organic matter content) associated to potato/tomato susceptibility against Cs/Rs will be assessed and identified. As a result, the involvement of microbiota in promoting and spreading pathogens will be elucidated and their contribution to plant susceptibility and disease progression will be determined.			

Deliverable D2.5 – Common platform with molecular data suitable for AI/ML

Deliverable Number	D2.5	**Lead Beneficiary**	6 - IDEN
Deliverable Name	Common platform with molecular data suitable for AI/ML		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	32	**Work Package No**	WP2
Description			
FAIR by design pipelines for data and metadata capture and analysis will be deployed to ensure the collected data and processed data remains machine actionable and is suitable for the deployment of AI/ML in WP3. Additionally, all molecular data obtained will be available in a public database such as GenBank-NCBI.			

Deliverable D3.1 – Mechanistic models of establishment and spread of Cs/Rs

Deliverable Number	D3.1	**Lead Beneficiary**	6 - IDEN
Deliverable Name	Mechanistic models of establishment and spread of Cs/Rs		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP3
Description			
Mechanistic models of establishment and spread of Cs/Rs to study host pathogen interaction and to further characterise the determinants of disease progression and attenuation.			

Deliverable D3.2 – Infection signatures including omics markers associated to infection onset and progression

Deliverable Number	D3.2	**Lead Beneficiary**	7 - WU
Deliverable Name	Infection signatures including omics markers associated to infection onset and progression		

Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP3
Description			
Molecular signatures associated with bacterial infection and progression in tomato and potato plants through the integration of plant and microbial omics and phenotypic data will be defined.			

Deliverable D3.3 – AI-based Disease Detection and Risk Forecasting Models

Deliverable Number	D3.3	**Lead Beneficiary**	8 - FUND
Deliverable Name	AI-based Disease Detection and Risk Forecasting Models		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	40	**Work Package No**	WP3
Description			
Advanced AI-based models will be developed and implemented for comprehensive disease detection and severity assessment in potatoes and tomatoes crops, extending beyond traditional leaf-focused diagnostics to include various plant parts using advanced imaging techniques. Environmental and climatic data will also be incorporated to assess disease progression and integrate explainability techniques to ensure trustworthy AI.			

Deliverable D3.4 – Sensory polymeric material

Deliverable Number	D3.4	**Lead Beneficiary**	1 - UBU
Deliverable Name	Sensory polymeric material		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	40	**Work Package No**	WP3
Description			
Polymeric sensors will be developed for the detection of key metabolites generated in the early stages of the infection, considering them as specific biomarkers for Cs/Rs and allowing the early detection of them in potato and tomato crops in a simple and economically feasible way.			

Deliverable D3.5 – Final version of the multiparametric disease detection device

Deliverable Number	D3.5	**Lead Beneficiary**	5 - IRIS
Deliverable Name	Final version of the multiparametric disease detection device		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	40	**Work Package No**	WP3
Description			
A compact self-contained multiparametric device will be designed and developed, suited for in-field (portable or handheld) and on-drone use and which will be powered by machine learning models for enabling the early detection of diseases in potato and tomato plants.			

Deliverable D3.6 – Functional AI-powered Cloud data management platform

Deliverable Number	D3.6	**Lead Beneficiary**	5 - IRIS
Deliverable Name	Functional AI-powered Cloud data management platform		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	42	**Work Package No**	WP3
Description			
Functional AI-powered Cloud data management platform interfaced by a web and mobile application with easy to understand and clear visuals.			

Deliverable D3.7 – User guides, and documentation to support the deployment and use of the platform

Deliverable Number	D3.7	**Lead Beneficiary**	5 - IRIS
Deliverable Name	User guides, and documentation to support the deployment and use of the platform		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	42	**Work Package No**	WP3
Description			
Training and support materials related to the usage of the Cloud data management platform will be developed such as user guides, instructional videos and documentation on platform features and functionalities.			

Deliverable D4.1 – Formulation of biocontrol coating solutions

Deliverable Number	D4.1	**Lead Beneficiary**	8 - FUND
Deliverable Name	Formulation of biocontrol coating solutions		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP4
Description			
Improved coating formulations based on effective microbial strain combinations and enzyme systems against Cs/Rs.			

Deliverable D4.2 – Plant derived organic substances products

Deliverable Number	D4.2	**Lead Beneficiary**	8 - FUND
Deliverable Name	Plant derived organic substances products		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP4
Description			
Identified and encapsulated plant-derived organic substances as biocide agents against Cs/Rs.			

Deliverable D4.3 – Report on PGPM consortia

Deliverable Number	D4.3	**Lead Beneficiary**	10 - JSI

Deliverable Name	Report on PGPM consortia		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP4
Description			
Report on the most active identified consortia pf PGPMs against Cs/Rs.			

Deliverable D4.4 – Solution based on disruption of Cs/Rs chemoperception

Deliverable Number	D4.4	**Lead Beneficiary**	9 - CBGP
Deliverable Name	Solution based on disruption of Cs/Rs chemoperception		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP4
Description			
Identified ligands of the most relevant chemoreceptors as infection disruptive agents for Cs/Rs.			

Deliverable D4.5 – Report of the natural and bio-control solutions manufacturing for pilot plant

Deliverable Number	D4.5	**Lead Beneficiary**	11 - CHEM
Deliverable Name	Report of the natural and bio-control solutions manufacturing for pilot plant		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	40	**Work Package No**	WP4
Description			
Report on the protocols for manufacturing and evaluation of the natural and bio-control solutions at pre-industrial scale.			

Deliverable D5.1 – Pilot validation for Rs in potato fields in LATAM (Colombia and Ecuador) (CS#1)

Deliverable Number	D5.1	**Lead Beneficiary**	3 - AGRO
Deliverable Name	Pilot validation for Rs in potato fields in LATAM (Colombia and Ecuador) (CS#1)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP5
Description			
Validation field trials of the early detection and bio-control solutions against Rs infection under real conditions in greenhouse (Colombia) and in field conditions (Ecuador) in potato crops.			

Deliverable D5.2 – Pilot validation for Rs in tomato fields in LATAM (Colombia) (CS#2)

Deliverable Number	D5.2	**Lead Beneficiary**	3 - AGRO
Deliverable Name	Pilot validation for Rs in tomato fields in LATAM (Colombia) (CS#2)		

Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP5
Description			
Validation field trials of early detection and bio-control solutions against Rs infection under real conditions both in greenhouse and field conditions in Colombia in tomato crops.			

Deliverable D5.3 – Pilot validation for Rs in tomato fields in Portugal (CS#3)

Deliverable Number	D5.3	**Lead Beneficiary**	16 - F4S
Deliverable Name	Pilot validation for Rs in tomato fields in Portugal (CS#3)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP5
Description			
Validation field trials of early detection solutions (MDD device) and against Rs infection under real conditions in Portugal in tomato crops.			

Deliverable D5.4 – Pilot validation for Cs in potato fields in Poland (CS#4)

Deliverable Number	D5.4	**Lead Beneficiary**	2 - IHAR
Deliverable Name	Pilot validation for Cs in potato fields in Poland (CS#4)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP5
Description			
Validation field trials of early detection and bio-control solutions against Cs infection under real conditions in Poland in potato crops.			

Deliverable D5.5 – Safety pre-assessment of the materials and processes

Deliverable Number	D5.5	**Lead Beneficiary**	1 - UBU
Deliverable Name	Safety pre-assessment of the materials and processes		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP5
Description			
Safety pre-assessment of the materials and processes			

Deliverable D5.6 – Safety of natural and bio-control solutions

Deliverable Number	D5.6	**Lead Beneficiary**	1 - UBU
Deliverable Name	Safety of natural and bio-control solutions		

Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP5
Description			
Safety Hazard assessment of natural and bio-control solutions.			

Deliverable D5.7 – Final safety and sustainability assessment of the materials and processes

Deliverable Number	D5.7	**Lead Beneficiary**	1 - UBU
Deliverable Name	Final safety and sustainability assessment of the materials and processes		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP5
Description			
Final safety and sustainability assessment of the materials and processes.			

Deliverable D5.8 – IPM guidelines for farmers

Deliverable Number	D5.8	**Lead Beneficiary**	1 - UBU
Deliverable Name	IPM guidelines for farmers		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	30	**Work Package No**	WP5
Description			
Integrated Pest Management report based on the molecular characterisation of the bacteria and the resistance genes of tomato and potato.			

Deliverable D5.9 – Final IPM guidelines for farmers

Deliverable Number	D5.9	**Lead Beneficiary**	1 - UBU
Deliverable Name	Final IPM guidelines for farmers		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP5
Description			
Integrated Pest Management report incorporating the final outcomes of the early detection and bio-control solutions covering the technical, economic, environmental, and social aspects.			

Deliverable D5.10 – Policy recommendations

Deliverable Number	D5.10	**Lead Beneficiary**	15 - EUPAT
Deliverable Name	Policy recommendations		
Type	R — Document, report	**Dissemination Level**	PU - Public

Due Date (month)	40	**Work Package No**	WP5
Description			
Policy paper which will assist EU and Member States' policy makers in designing, monitoring, reviewing, and adjusting (if needed) current policies, rules and programs regarding Cs and Rs, as well as shaping and supporting the implementation of new policy initiatives and decisions on these pathogens and list of recommendations advising on further research building on the results of the POMATO project.			

Deliverable D6.1 – Dissemination, Exploitation and Communication Plan

Deliverable Number	D6.1	**Lead Beneficiary**	13 - INO
Deliverable Name	Dissemination, Exploitation and Communication Plan		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP6
Description			
Multi-channel plan for dissemination and exploitation of results, including communication activities.			

Deliverable D6.2 – Multi-Actor Participatory activities and platform

Deliverable Number	D6.2	**Lead Beneficiary**	13 - INO
Deliverable Name	Multi-Actor Participatory activities and platform		
Type	DEC —Websites, patent filings, videos, etc	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP6
Description			
Initial report on the Multi-Actor Participatory activities and platform.			

Deliverable D6.3 – Update on Multi-Actor Participatory activities and platform

Deliverable Number	D6.3	**Lead Beneficiary**	13 - INO
Deliverable Name	Update on Multi-Actor Participatory activities and platform		
Type	DEC —Websites, patent filings, videos, etc	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP6
Description			
Final report on Multi-Actor Participatory activities and platform.			

Deliverable D6.4 – Report on DC, networking and clustering activities

Deliverable Number	D6.4	**Lead Beneficiary**	13 - INO

Deliverable Name	Report on DC, networking and clustering activities		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP6
Description			
Initial report on DC, networking and clustering activities.			

Deliverable D6.5 – Final report on DC, networking and clustering activities

Deliverable Number	D6.5	**Lead Beneficiary**	13 - INO
Deliverable Name	Final report on DC, networking and clustering activities		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP6
Description			
Final report on DC, networking and clustering activities.			

Deliverable D6.6 – Practice Abstracts - batch 1

Deliverable Number	D6.6	**Lead Beneficiary**	1 - UBU
Deliverable Name	Practice Abstracts - batch 1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	12	**Work Package No**	WP6
Description			
This deliverable contains 5 practice abstracts.			

Deliverable D6.7 – Practice Abstracts - batch 2

Deliverable Number	D6.7	**Lead Beneficiary**	1 - UBU
Deliverable Name	Practice Abstracts - batch 2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP6
Description			
This deliverable contains 10 practice abstracts.			

Deliverable D6.8 – Practice Abstracts - batch 3

Deliverable Number	D6.8	**Lead Beneficiary**	1 - UBU

Deliverable Name	Practice Abstracts - batch 3		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP6
Description			
This deliverable contains 10 practice abstracts.			

Deliverable D6.9 – POMATO Exploitation Plan

Deliverable Number	D6.9	**Lead Beneficiary**	13 - INO
Deliverable Name	POMATO Exploitation Plan		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP6
Description			
POMATO Exploitation Plan with all necessary activities for the exploitation and sustainability of POMATO results after the end of the project.			

Deliverable D6.10 – Final POMATO Exploitation Plan

Deliverable Number	D6.10	**Lead Beneficiary**	13 - INO
Deliverable Name	Final POMATO Exploitation Plan		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP6
Description			
Final update of POMATO Exploitation Plan with all necessary activities for the exploitation and sustainability of POMATO results after the end of the project.			

LIST OF MILESTONES

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
1	Data Management Plan completed	WP1	1 - UBU	Document approved by the relevant consortium members	5
2	Development of I-matrix for improved isolation of Rs	WP2	2 - IHAR	I-matrix verification completed on Rs bacteria strains and environmental tissue samples.	21
3	Cs/Rs genome sequence and transcriptome data are obtained and uploaded successfully to NCBI database	WP2	6 - IDEN	Data is registered in NCBI database with their respective accession numbers.	30
4	Identifying a group of resistant/tolerant tomato/potato genotypes against Cs/Rs	WP2	3 - AGRO	Screening for resistance completed and data quality validated.	30
5	Developed microfluidic method for testing Cs/Rs neutral and positive interactions with the plant microbiome	WP2	10 - JSI	Obtaining positive growth of Cs/Rs separately in droplets and within aggregates containing neutrally interacting bacteria.	30
6	1st keystone microbial taxa related to healthy crops identified for its study	WP3	6 - IDEN	Initial identification of microbial strains with metabolic and/or ecological potential for pathogen attenuation.	12
7	Alpha version of AI-based Disease Detection and Risk Forecasting Models	WP3	8 - FUND	Development of a testable alpha version ready for preliminary testing.	24
8	Developed and characterised smart polymers for early detection	WP3	1 - UBU	Laboratory prototype exhibits 85% detection efficiency tested under predefined conditions. with 85% of feasibility in detection.	30

9	MDD Disease sensing device defined	WP3	5 - IRIS	Monitoring sensing technologies selected based on the results (performance, accuracy, reliability) of the experimental work for screening the most	20

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
				suitable techniques(s) for measuring the defined parameters.	
10	AI powered Cloud data management platform defined	WP3	5 - IRIS	Collection of all the data to feed the platform is ready.	36
11	Effectiveness of plant biocontrol coating solutions, natural compounds products and chemoperception disruptive products application to control Cs/Rs	WP4	8 - FUND	Plant application of both solutions will decrease at least 40% the incidence of Rs and Cs.	30
12	Developed method for large scale testing for PGP activity of consortia grown in the microdroplets	WP4	10 - JSI	Methods will be verified using known PGPM for various PGP traits.	30
13	Natural and biocontrol solutions manufacturing at pilot plant	WP4	11 - CHEM	Natural and biocontrol solutions prototypes will be manufactured producing at least 10L per solution to perform the trials in growth chamber and 100 L for the field trials.	36
14	Validation field trials addressing Rs in potato and tomato in LATAM and Europe	WP5	1 - UBU	Conducted field trials in selected regions of LATAM and Europe, achieving 90% operational reliability in the performance of MDD sensor device, smart polymers, and biocontrol treatments across various climatic and agricultural conditions.	36

15	Validation field trials addressing Cs in potato in Europe	WP5	2 - IHAR	Conducted field trials in Poland, achieving 90% operational reliability in the performance of MDD sensors device, smart polymers, and biocontrol treatments across various climatic and agricultural conditions.	36
16	Required samples for toxicological assessments in place	WP5	1 - UBU	Reception of the materials to be analysed at UBU.	18

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
17	Life Cycle Inventory Assessment for sustainability fulfilled	WP5	1 - UBU	Inventory data sheets completed.	24
18	POMATO project website	WP6	13 - INO	POMATO domain and website published.	3
19	Multi-Actor Participatory Platform	WP6	13 - INO	Engagement with other successful projects under the addressed HE call and other future HE projects.	36
20	Deploying dissemination activities	WP6	1 - UBU	At least 25 Practise abstract published; a total of 15 (national and international) events participated, 4 Open Days and 6 engagement workshops/plenary sessions.	40

LIST OF CRITICAL RISKS

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures

1	Delays in reporting and quality of the provision of deliverables	WP1	The proposed management structure that includes a continuous follow-up on the WP developments, including the deliverables. Also, the quality plan will be used to track progress. Therefore, early indication of delay will be present to all participating partners, which in turn intensifies the pressure on the delaying partner (reputation at risk). Project management procedures and regular control of the work to detect problems in advance.
2	Consortium problems (e.g. underperforming partner; a key partner or expert leaves the project;	WP1	Consortium has a strong R&D - delivery capacity and some partners overlapping skills that can fully cover this risk. The CA will fully cover conflict resolution and decision making.

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
	disagreements between partners; need to reallocate budget)		Effective management structure and procedures have been established to timely intercept problems, remove, or reallocate resources, redistribute budget renegotiating the value-formoney with WP leaders/partners to achieve a better-balanced output against payment, prior approval by the EC.
3	Aggregation protocol not suitable for Cs/Rs and might cause toxicity	WP2	The polyelectrolytes used in the aggregation will be modified to decrease density of positive charges and decrease accessibilities of primary amines.
4	Insufficient Quality of Input Data	WP3	Collaborate with agricultural institutions and data providers to ensure access to highquality, comprehensive datasets. Implement rigorous data preprocessing and augmentation techniques to enhance data quality.
5	Ineffective sensors or not selective to Cs/Rs	WP3	Redesign the material (monomer nature and molar ratio, format, etc.) to solve the problems.
6	Challenges in obtaining sensor materials incorporating specific metabolite in monomer structure	WP3	Utilise polymer matrices that have already been studied (including dense films and foamed materials).
7	Microdroplet approach cannot be used for characterisation all PGP traits	WP4	The selected traits that cannot be used in microdroplets approach will be then tested in classical microbiological way using consortia.

8	Low effectiveness of biocontrol solutions, plant derived organic substance product and chemoperception disruptive agents of compounds	WP4	Extend the screening and improve the concentrations of active antimicrobial agents in order to enhance the effectiveness of the solutions prototype against Cs/Rs.
9	Low performance properties of the sustainable solutions manufactured at pilot plant	WP4	Manufacturing processing steps will be reviewed and adjusted at pilot plant level productions.
10	Along the project, the phytosanitary restrictions can be modified, tightening the use of quarantine organisms	WP5	Adapting project activities to new conditions and establish contact with national phytosanitary authorities of the countries where field trials are performed (Ecuador, Colombia, Portugal and Poland).
11	Technical issues with the smart sensory polymers and MDD sensors device may result in inconsistent data, complicating the validation of early detection tools	WP5	Implement advanced data filtering and analysis techniques to isolate anomalous data points caused by device malfunction or environmental interference and conduct iterative improvements on real-time data from initial trials, providing remote technical support during field trials to troubleshoot issues in real-time.

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
12	Unpredictable weather conditions and climate variations in Ecuador, Colombia, Poland and Portugal could affect the consistency and reliability of field trial outcomes	WP5	Have alternative fields located in different geographic regions in LATAM and Europe to carry out the experiments.
13	Coordinating and managing field trials across multiple locations (Ecuador, Colombia, Poland, Portugal) may face logistical issues, including transportation of samples, timely data collection, and resource allocation	WP5	Implement a centralised digital project management platform to coordinate activities, track progress, and facilitate communication between teams in Ecuador, Colombia, Poland, and Portugal, assigning dedicated local field coordinators in each region to oversee logistics.
14	The effectiveness of plant-derived organic substances, biocontrol coatings, and other natural solutions may vary based on local soil conditions	WP5	Delivery system strategy of the solutions developed will be adapted according to the soil condition variation.

15	Information and/or materials sampling missing or delayed developing the safety and sustainability assessment	WP5	Commercial materials will be used for testing in case those from the project are not available. Missing information will be completed if needed with: professional databases – scientific literature – scientific based assumptions.
16	Lack or poor participation from stakeholders	WP6	Provision of information, insights & accurate explanation for follow; involvement of wider groups; increase variety & tailoring of communication and dissemination measures.
17	Loss, death and/or destruction of plant material (in vitro plants, tubers, bacteria) shipped from the European Union to Latin America and vice versa, caused by problems at the border, customs and/or phytosanitary services.	WP2	The possibility of shipment between both continents and individual countries will be well studied and the shipment will be arranged with the phytosanitary services from the source and destination countries and, after sending the material, a continuous monitoring of the shipment will be performed.
18	Environmental variability across countries affecting the accuracy and robustness of distribution models	WP3	The project will implement a robust calibration process by integrating region-specific environmental data during model training and validation phases. The models will consider abiotic factors (e.g., temperature, humidity) that will be enriched with biotic factors specific to the pathogens under study to predict disease progression. The disease risk forecasts will be capable of upgrading and adapting based on incoming environmental changeable data

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
			to make dynamic active predictions including interval of confidence and certainty of the prediction.

PROJECT REVIEWS

Project Reviews

Grant Preparation (Reviews screen) — Enter the info.

Review No	**Timing** (month)	**Location**	**Comments**
RV1	20	to be agreed with consortium	
RV2	38	to be agreed with consortium	
RV3	48	to be agreed with consortium	

Associated with document Ref. Ares(2025)2863795 - 08/04/2025

Effective management strategies to tackle *Clavibacter sepedonicus* and *Ralstonia solanacearum* outbreaks on POtato and toMATO crops (POMATO)

Topic: HORIZON-CL6-2024-FARM2FORK-02-04 two stages: Tackling outbreaks of plant pests Type of Action: Research and Innovation Action
Duration: 48 months

HISTORY OF CHANGES Part A		
VERSION	DATE	CHANGE
1.0	22.01.2025	Update of WP1 (T1.1) description to add information about the constitution of an External Experts Advisory Board (EEAB) in response to shortcoming 3.
2.0	22.01.2025	Update of WP5 (T5.5.2) description in response to shortcoming 4.
3.0	22.01.2025	Update of WP6 (T6.1) description in response to shortcoming 6.
4.0	22.01.2025	Update of Critical Risks for Implementation in response to shortcoming 7 (two risks added).
5.0	17.02.2025	Update of Deliverables: Project Management Handbook and Risk Management Plan have been removed. For the Data Management Plan, the type of deliverable has been changed from R to DMP.
6.0	17.02.2025	Update of IRIS budget (equipment costs have been changed to the OGWS category in response to FO's comments).
7.0	27.02.2025	Subcontracting costs have been changed to the OGWS category according to AFO's comments.

HISTORY OF CHANGES Part B		
VERSION	DATE	CHANGE
1.0	16.09.2024	Initial version.
2.0	16.01.2025	Update to delete header, remove the list of participants and remove tables 3.1a, 3.1b, 3.1c, 3.1d, 3.1e and 3.1f from section 3.1 to comply with DoA format.
3.0	17.01.2025	Update to add footer and section 4 for "ethics self-assessment" to comply with DoA format.
4.0	22.01.2025	Update of the "3.2 Consortium as a whole section" and the "Acronyms" section to add information about the External Experts Advisory Board (EEAB) in response to shortcoming 3.
5.0	22.01.2025	Update of "Means of Verification" of EO3 in the Impact Section in response to shortcoming 4.
6.0	22.01.2025	Update of the "2.1.2 Requirements and potential barriers" section to include the assessment of potential economic barriers in POMATO's strategy in response to shortcoming 5.
7.0	22.01.2025	Update to add disclaimer text pointed out in point 17 of the POMATO Guidance on Action document.
8.0	05.02.2025	Number of practice abstracts added in the disclaimer below (page 1).
9.0	17.02.2025	Tables 3.1g, 3.1h and 3.1i have been updated following AFO's comments.
10.0	20.02.2025	Table 3.1h has been updated following AFO's comments.
11.0	27.02.2025	Table 3.1g has been removed and the subcontracting costs have been changed to the OGWS category according to AFO's comments.

"New knowledge and innovative solutions generated by the projects will feed into the EIP-AGRI project database of the EU CAP network website for broad dissemination to practitioners. End-user material will be produced in the form of a number of summaries of practice-oriented solutions for practitioners in the EIP-AGRI common format ("practice abstracts"). The project profile will also be uploaded to the EU CAP network website at project start. Guidance about how to complete and submit project profile and practice abstracts through the online EIP-AGRI common format is available on the EU CAP network website.
A total target number of 25 practice abstracts is foreseen for the project."

Acronyms Associated with document Ref. Ares(2025)2863795 - 08/04/202

AI: Artificial Intelligence; **CNN**: Convolutional Neural Networks; ***Cs***: Clavibacter sepedonicus; **DSS**: Decision Support Systems; **EEAB:** External Experts Advisory Board; **EPPO**: European and Mediterranean Plant Protection Organization; **FAIR**: Findable Accessible Interoperable Reusable; **FTS**: Fluorescence-based Thermal Shift Assays; **GEMs**: Genome Scale Metabolic Models; **GWAS**: Genome-Wide Association Studies; **IPM**: Integrated Pest Management; **IPR**: Intellectual Property Rights; **I-matrix:** Isolation Matrix**; ITC**: Isothermal Titration Calorimetry; **JRC**: Joint Research Centre; **LATAM**: Latin America; **LCA**: Life Cycle; Assessment; LCI: Life Cycle Inventory; **MAGs**: Metagenomically Assembled Genomes; **MDD**: Multiparametric Disease Detection; **OECD**: Organisation for Economic Co-operation and Development; **PCR**: Polymerase Chain Reaction; **PGPM**: Plant Growth-Promoting Microorganisms; **RGs**: Resistance Genes; ***Rs***: Ralstonia solanacearum; **SCA**: Single Cell Analysis; **SSbD**: Safe and Sustainable by Design; **T/Y**: T-type/Ytype junctions; **TEA**: Techno-Economic Analysis; **VLM**: Visual Language Models.

Table of contents

1. Excellence 2
1.1. OBJECTIVES AND AMBITION 4
1.2. METHODOLOGY 10
2. Impact 17
2.1. PROJECT'S PATHWAY TOWARDS IMPACT 17
2.2. MEASURES TO MAXIMISE THE IMPACT 21
2.3. IMPACT SUMMARY 25
3. Quality and efficiency of the implementation 26
3.1. WORK PLAN AND RESOURCES 26
3.2 CAPACITY OF PARTICIPANTS AND CONSORTIUM AS A WHOLE 28
4. Ethics self-assessment 32

1. Excellence

Project overview

The Solanaceae family is one of the largest plant families in the world being present on all continents except Antarctica, and its economic importance lies in many different fields such as food, medicines, cosmetics, etc. The most economically important genus of the family is *Solanum*, which includes the **potato** *(Solanum tuberosum)* and the **tomato** (*Solanum lycopersicum*). Potato and tomato crops are integral sources of carbohydrates, vitamins, and minerals essential for a balanced and healthy diet, making them staples in human consumption for centuries and being cultivated globally. Indeed, within the EU, these two crops stand out as the most significant among fresh vegetables, underscoring their importance in both regional and global agriculture. In 2022, the **harvested area of potatoes across** the EU was 1.4 M Ha, of which **47.5 MT** of crops were produced. Germany (22.5%, 10.7 MT), France (17.0%), the Netherlands (14.6%) and Poland (12.7%) together accounted for around two thirds of the EU's potato harvest.[1] In Latin America (LATAM), cultivated potatoes are widely distributed in the

Figure 1. 1 Symptoms on potato tubers affected by Cs and on tomato by Rs.

[1] Key figures on the European Food Chain (2023). Eurostat. Available here.

Andean region with over 4,500 varieties and more than 100 wild species.[2] Regarding **tomato**, the **total production** in 2022 was **15.4 MT** of harvested production, with Italy (39.8%) and Spain (23.7%) at the top of EU, followed by Portugal (9.1%). Both crops are vulnerable to several pest diseases, a threat that is greater than ever because of the considerable increase in the volumes, commodity types and origins of trade in plant material from third countries, the

introduction of new crops, and the impact of climate change, which is affecting the boundaries of pests and their vectors. Based on range of annual production, output values and estimated percentage of damage, annual losses caused by pests range from **2.4-3.1 B€ for potatoes** and **1.5-2.3 B€ for tomatoes**, corresponding to 20-25% of total production damage.[3] Among the quarantine and priority pests that threaten crops and that have caused severe losses in recent years, the bacteria ***Clavibacter sepedonicus (Cs)*** (ring rot in potato) and the bacteria ***Ralstonia solanacearum (Rs)*** (Smith, phylotype II) [RALSS] (bacterial wilt in potato/tomato) are considered the most economically and environmentally relevant diseases and constitute a menace for seed production in particular (Fig 1).[4] Both bacteria are included in the **Part B Annex of the Regulation 2019/2072** and are classified as **EPPO A2 quarantine pests**.[5,6] Furthermore, *Cs/Rs* are regulated by their respective EU policy establishing measures for

eradicating and preventing their spread in EU.[7,8,9,10] In this context, and in line with the regulation Associated with document Ref. Ares(2025)2863795 - 08/04/2025**2016/2031 on**

protective measures against pests of plants, effective front-line detection methods and innovative monitoring technologies suitable for surveillance and prevention are critical for recognizing infected crops and finding incursions of quarantine plant pathogens or pests in an early stage.[11] In addition, fossil-base disease management practices are still applied to prevent new outbreaks and further spread of currently infected areas. More sustainable practices, aiming for the development of rapid response pest eradication capabilities, that are non-dependant on fossil-based pesticides (in line with the **EU Green Deal target of 50% pesticides reduction**) are necessary to safeguard ecosystems and biodiversity while improving food safety.

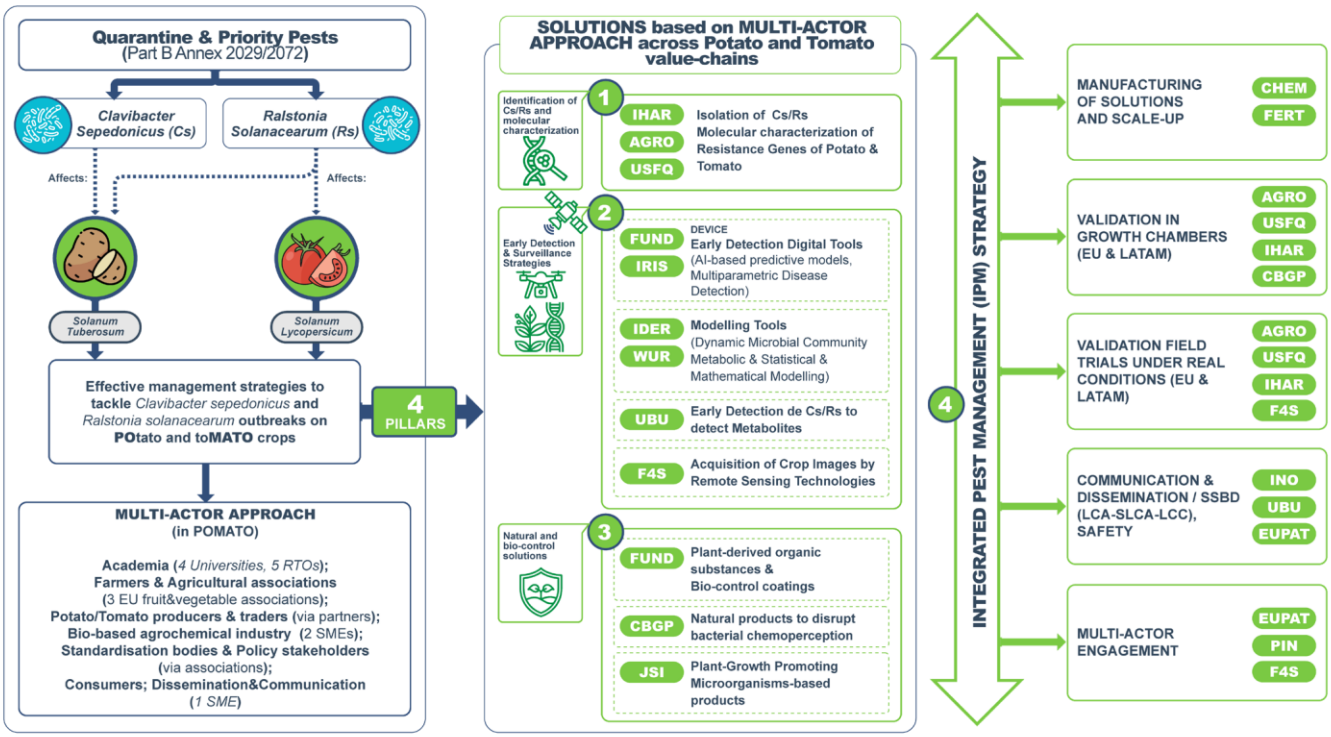

Figure 1.2. POMATO overview of the multi-actor approach towards the IPM strategy to tackle *Clavibacter sepedonicus* (*Cs*) and *Ralstonia solanacearum* (*Rs*) outbreaks.

In view of the current scenario, **POMATO's** primary goal is to safeguard the health of potato and tomato crops, as well as the quality of the final product, by effectively combating outbreaks of *Cs* and *Rs*. This mission is anchored in

[2] Rondon, S.I.(2022). In: Alyokhin, A. et al.Insect Pests & Potato 317-330.
[3] Damage is estimated by manufacturer of plant protection products.
[4] Nouioui I, et. al. (2018). Front. Microbiol. 9:2007. Available here.
[5] Regulation (EU) 2019/2072 protective measures against pests of plants. Available here.
[6] PM1/002(32) EPPO A1 & A2 Lists of pests recommended for regulation as quarantine pests (2023). Available here.

four key pillars, guiding the project's strategic initiatives and activities (Fig 1.2): ***1)*** to **isolate and identify *Cs/Rs*** and to **characterise at molecular level the resistance genes of potato/tomato**, ***2)*** to provide **effective early detection and surveillance strategies** (Artificial Intelligence (AI) predictive models, Multiparametric Disease Detection (MDD) sensing device, mathematical modelling tools, polymeric sensors, AI-driven Cloud data management platform and adquisition of infected field with drones), ***3)*** to develop **natural and bio-control solutions** applied to seed, plant and soil (plant-derived organic substances, biocontrol coatings, plant growth-promoting microorganisms (PGPMs) and chemoperception disruptors), ***4)*** to evaluate and validate the performance of all the above mentioned strategies at TRL5 in **four case studies (CS#1-#4)** at field trials and to prepare the **POMATO**'s **Integrated Pest Management (IPM) Plan with the aim to reduce the incidence of the pests by 40-60%**. **POMATO** is the first of this class of project that brings together a **multi-actor approach** to prevent and eradicate the spread of quarantine pests known to occur in the EU. **POMATO** is a **48-month** project that will involve strategic actors of the potato/tomato value chain. The actors include academic research organizations (**4 universities (UBU, USFQ, WUR, CBGP) and 5 RTOs (IHAR, AGRO, IDER, FUND, JSI**) with high expertise on molecular biology of plant's resistance characterization of *Cs/Rs*, on data analysis, modelling, AI-based models, polymeric sensors development and crop protection by natural bio-control solutions and **2 agrochemical companies (CHEM, FERT)** to scale-up and validate the solutions. Moreover, **POMATO** will involve **farmers** (as field validators) through its **3 agriculture EU associations (EUPAT, PIN, F4S), 1 SME (IRIS)** to develop the multiparametric disease detection

[7] Regulation (EU) 2022/1194 on Clavibacter sepedonicus (Spieckermann & Kotthoff 1914) Nouioui et al. 2018. Available here.
[8] Regulation (EU) 2022/1193 on Ralstonia solanacearum (Smith 1896) Yabuuchi et al. 1996 emend. Safni et al. 2014. Available here.
[9] *Cs* present in: Belarus, Bulgaria, Czech Republic, Estonia, Finland, Georgia, Germany, Greece, Hungary, Latvia, Lithuania, Norway, Poland, Romania, European part of Russia, Slovakia, Spain, Sweden, Turkey, Ukraine.
[10] *Rs* present in: Georgia, Moldova, Russia, Serbia and Ukraine.
[11] Regulation (EU) 2016/2031 on protective measures against pests of plants. Available here.

device**, and 1 SME** responsible for **dissemination, communication and exploitation (INO)** of the project's results.

Associated with document Ref. Ares(2025)2863795 - 08/04/2025

Finally, **POMATO**'s international partners from Colombia (**AGRO**) and Ecuador (**USFQ**), currently heavily affected by *Cs/Rs,* will ensure the exchange of good practices while at the same time providing access to a wide germplasm collection of potato/tomato and to infected fields to evaluate the developed strategies.

1.1. Objectives and ambition

1.1.1. Objectives

POMATO strategies are aligned with the topic "HORIZON-CL6-2024-FARM2FORK-02-04 two stages: Tackling outbreaks of plant pests". The project proposes **5 objectives** (**O**) with their respective **5 outcomes** (**OC**), articulated under **21 results** (**R**).

O1. To develop molecular approaches for Cs/Rs disease Resistance Genes (RGs) identification in potato and tomato varieties. *Linked to WP 2*

R1	Tool (I-matrix) and procedure for specific isolation of bacterial cells directly from contaminated environmental samples to reach at least **80% effectiveness**.
R2	At least **100 isolated bacteria** from potato/tomato infected areas and germplasm culture collections from LATAM and Poland, **12 completely sequenced** bacterial genomes, and at least **12 transcriptomic analyses** will be performed to understand the genetic diversity and virulence signatures of field samples of *Cs/Rs*.
R3	**Identification**, **compilation**, and **mapping** of all **annotated potato** and **tomato resistance (R genes), susceptibility (S genes)** and plant **metabolic pathways** and **regulatory networks** that interact with these R and S genes.
R4	Identification of potential **resistance genes/regions (RGs) against *Cs/Rs*** through whole-genome transcriptome and genome-wide association studies (**GWAS**) analysis of incompatible interactions between *Cs/Rs* and potato/tomato using extensive germplasm working collections from Europe and LATAM: **200 potato and 200 tomato samples.**
R5	Isolating and characterizing **10 different non-suppressive consortia** that are involved in or are allowing the establishment and spread of pathogen infection. These consortia will allow the identification of the interactions between at least **6 genetic, environmental and soil factors** (temperature, humidity, soil, pH, texture, organic matter content).

R6	Implement FAIR by design pipelines to prepare **data and metadata for interoperability**, ensuring alignment with open-science principles and readiness for subsequent data analysis models and machine learning.
OC1	The identification of RGs and the development of new molecular tools will enable breeding programs to produce *Cs/Rs*-resistant potato and tomato varieties. This will reduce disease incidence, enhance crop resilience, and decrease pesticide use.

O1's pertinence to topic: *one or more plant pests: Union quarantine plant pests present in the EU or priority pests in the EU. Effective implementation of the principles of IPM. Innovative biological and other non-chemical control and resistance breeding. Drivers of plant pest introduction, spread and establishment and its interaction with host plants and antagonists.*

O2. To develop tools for the early detection and surveillance of Cs/Rs in potato and tomato. *Linked to WP 3*

R7	**Metagenomically Assembled Genomes** (**MAGs**) will be used to build **Genome Scale Metabolic models (GEMs)**, understand the plant's immune response against *Cs/Rs* and identify key metabolites generated in the early stages of the infection (**at least 10 metabolites**) that will be detected some of them using **polymeric sensors** allowing the early detection of both pests.
R8	Machine learning and deep learning approaches will be combined with mechanistic modelling to identify dysregulated pathways related to disease progression, their regulators and molecular signatures. The pangenome of the characterized pathogens and **at least 10% of the most relevant genetic signatures identified** will be subjected to extensive comparative genomic analysis.
R9	Acquisition and analysis of **>1000 high-resolution multispectral images** from drones and satellites to detect, differentiate, and quantify the severity of *Cs/Rs* infections in tomato and potato crops, integrating this data with AI-based predictive models for early detection and targeted interventions.
R10	Analysis of **>1,000 images with AI based models**, to establish a relationship between plant phenotype and severity of the bacterial infection to reach at least **80% of weighted accuracy**.
R11	An advanced **AI-driven cloud platform** that integrates diverse data sources and employs machine learning algorithms, interfaced through a digital application, to deliver tailored insights based on specific environmental conditions and crop health metrics, thereby enhancing disease prediction for potato and tomato crops.
OC2	The development of advanced tools for early detection and surveillance of *Cs/Rs* will enable rapid identification of infections in potato and tomato crops. AI-driven models, along with sensor-based detection and image analysis, will significantly enhance disease prediction accuracy.

Associated with document Ref. Ares(2025)2863795 - 08/04/2025

O2's pertinence to topic: *Influence of climate change, ecosystem degradation, and globalisation. Efficient surveillance methods and strategies for early-detection.*

O3. Natural and bio-control solutions against Cs/Rs applied on potato and tomato seeds, plants and soil beneficial for both conventional and organic farming. *Linked to WP 4*

R12	**Plant-derived organic substances, biocontrol coatings** and **chemoperception disruptive products** to **decrease at least 40%** the incidence of *Cs/Rs* with respect to untreated plants.
	~~Development of sustainable practices using the treatment with **PGPMs** obtained from the healthy plants to~~ **decrease at least 60%** the incidence of *Cs/Rs* with respect to untreated plants.
	~~The adoption of sustainable solutions to improve seed, plant and soil health and pest control for the reduction of chemical pesticide and fungicide use in potato/tomato crops.~~

O3's pertinence topic: *target to reduce by 50% the overall use and risk of chemical pesticides and reduce the use by 50% of the more hazardous pesticides. Develop efficient (bio)control of the pest(s).*

O4. To validate the Integrated Pest Management (IPM) strategies at TRL 5 in 4 Case Studies (CS#) in greenhouses and real infected field in LATAM and EU. *Linked to WP 5*

R14	**4 Case Studies (CS#)** to validate **POMATO**'s IPM strategies at both greenhouse and infected fields in LATAM and Europe: 3 related to *Rs* in potato/tomato (CS#1 CS#2 and CS#3) and 1 to *Cs* in potato (CS#4) at several locations in Colombia, Ecuador, Portugal and Poland.

R15	**The MDD sensing device (90% of reliability)** and portable **polymeric sensors (>90% accuracy)** will be validated in field trials in the CS# for in-situ diagnostic detection of *Cs/Rs* in potato/tomato crops.
R16	**Plant-derived organic substances, biocontrol coatings, PGPMs and chemoperception disruptive products** will be formulated at pre-industrial level (TRL5) and validated in field trials in the CS# as biocontrol strategies for *Cs/Rs* in potato/tomato fields.
R17	The generation of recommendations to optimize processes and products using the **Safe and Sustainable by Design (SSbD) Framework**.
OC4	The field trials will lead to the deployment of reliable diagnostic tools and effective biocontrol solutions for *Cs/Rs* in potato/tomato plants (at TRL5) and validate the effectiveness of the IPM strategies to control outbreaks of these pests with potential to be replicable in other Solanaceae crops.
O4's pertinence to topic: *tools and technologies of economically and environmentally sound solutions, taking into account the use of non-chemical or biological control methods; analyse the ecological impact of plant pest(s) spread and establishment; International cooperation with countries affected by the same pest(s).*	

O5. To disseminate, communicate and exploit the main achievements of POMATO project. *Linked to WP 5 and WP 6 (detailed KPIs in Section 2.2)*

R18	Creation of **POMATO**'s multi-actor participatory platform (MAPP) and activities to share knowledge, expertise, and background across the interdisciplinary consortium and the target actors. To generate at least **25 "practice abstracts"** to contribute to European CAP Network repository and insights for farmers and potato/tomato traders; at least 10 peer reviewed publications, 10 publications in specialised agriculture media outlets.
R19	Project **dissemination** results will be **shared** through **relevant channels** to support the **impact** and **exploitation outcomes** including websites, social media, training materials, project related videos, social events (including both physical and online), etc.
R20	**4 engagement activities in Europe** and **2 in LATAM** to increase the project's visibility across the target groups, especially focused on farmers (see section 2.1). **4 Thematic POMATO's Open Days** addressing the 4 different pillars.
R21	**POMATO IPM Plan** (translated into **at least 3 languages**) among relevant stakeholders, including those related in the **policy and decision-making cycle**. EU recommendations will be sent to the EC, the European Parliament, the Committee of the Regions, the Commission Expert Group on Plant Health Legislation (2 policy briefs, 1 position paper, at least 2 policy roundtables and briefings).
OC5	Providing the foundations and drivers for future development and implementation of IPM strategies by farmers and other key players in the event that these potato/tomato pests outbreak in EU.
O5's pertinence to topic: *social and economic implications for farmers, foresters and other economic operators affected by the outbreaks of the plant pest(s); implement the multi-actor approach; cooperation with other proposals.*	

1.1.2. Ambition and progress beyond the state of the art (BsoA) Associated with document Ref. Ares(2025)2863795 - 08/04/2025

POMATO project aims to provide a powerful European initiative for a sustainable IPM strategy to control and treat the presence of *Cs/Rs* in potato and tomato crops. Beyond the state of art, several innovative strategies will be implemented within the project which are articulated in 5 ambitions:

Ambition #1: Isolation, genomic and transcriptomic characterization of *Cs/Rs* and molecular characterization of potato/tomato RGs (WP 2, T2.1-2.2, 2.4): ***SoA***: *Cs/Rs* pests possess distinct characteristics that complicate both their diagnosis and management of their spread. These pathogens may exist in potato plants and tubers as a latent infection, which may be caused by low concentration of bacteria, different virulence of strains, and/or different tolerance of potato varieties to the presence of pathogens. On the other hand, their symptomatic form is often confused with the symptoms of wilting and other potato diseases. In tomato, *Rs* typically gains access to host root systems through natural wounds and then bacteria move to the vascular system, where they multiply in the xylem. As tomato and potato affected by bacterial wilt (*Rs*) present latent infections, and the contaminants present in the tested samples (extracts from plant tissue, soil and water) make diagnostics difficult and false, disease control continues to be a

challenge ***BSoA:*** **IHAR,** based on previous patent to identify *Cs*,[7] will develop an effective protocol based on **isolation matrix (I-matrix)** for the specific isolation of *Rs* cells from plant tissues and tubers, soil and water with the simultaneous possibility of concentrating the tested cells, maintaining their viability and eliminating burdensome contaminants. This will allow to enhance field monitoring by improving detection of *Rs* in latent and/or symptomatic infected environmental samples.

POMATO will also conduct **genomic and gene expression analyses** of *Cs/Rs* to understand the process of bacterial pathogenesis and virulence during vascular colonization and plant immunity, and to gain new insights into the genetic diversity of bacteria populations. This will allow the early detection of genes and effectors involved in pathogenicity and strengthen biosecurity measures and field monitoring of *Cs/Rs* in latent infection and/or symptomatic plants. The risk of *Cs/Rs* intrusions and putative dissemination of pathogens represents a challenge for Europe and optimized and validated methodologies for the detection of *Cs/Rs* are required. Commercial kits are available for the detection of *Cs/Rs* (Agdia®); however, these diagnostic kits present some relevant constraints, validations are limited to few reference isolates and kits are expensive, showcasing the need for the development of new approaches for *Cs/Rs* diagnosis. The phenotypic and genotypic characterization of *Cs/Rs* will contribute to the advance of ***SoA*** by providing new insights into the genetic diversity of pathogen populations which is still highly unexplored in both LATAM and Europe. Additionally, much remains to be learned about the genes involved in the pathogenicity and virulence of *Cs/Rs* during vascular colonization. Identification of such genes could improve understanding of how pathogens cause disease.

POMATO will complete a broad screening for the **identification of new sources of potato and tomato RGs** to *Cs/Rs* including wild and native varieties from European and LATAM germplasm collections. A prerequisite for finding new sources of disease resistance is the characterization of genes involved in plant resistance. Sources of resistance to bacterial wilt originating from tomato wild relatives, *S. pimpinellifolium*, *S. pennellii* and *S. lycopersicum* var. *cerasiforme* have been already reported.[8] Published results suggested the presence of polygenic and strain-specific resistance in tomato wild relatives.[9] Considering the centre of origin of potato/tomato, the Andean region in LATAM provides a high diversity of wild potato/tomato germplasm and a potential source of variability and genetic resistance against plant pathogens, such as *Cs/Rs*. Part of the diversity of potato/tomato genetic resources (> 3.000 accessions) is maintained in different potato/tomato Germplasm Banks (PGB and TGB) in LATAM and Poland, which has mostly not yet been characterized for their resistance/susceptibility to *Cs/Rs*. This TGB is constituted by an important number of tomato wild relative accessions, such as, *Solanum peruvianum*, *S. lycopersicum* var *ceraciforme*, *S. arcanum S. pimpinellifolium*, *S. pennellii* and *S. hirsutum,* among others. A screening for resistant plants to *Cs/Rs* including wild potato/tomato accessions will advance the ***SoA*** through the identification of new sources of resistance that could be used as resistant parentals in breeding programs. Additionally, deciphering the molecular mechanisms of potato/tomato wild genotypes during interaction with effector genes from the pathogens will contribute to the understanding of the processes underlying plant immune responses.

Ambition #2: Factors affecting the establishment and spread of *Cs/Rs* (WP 2 T2.3, 2.4 & WP 3 T3.1): ***SoA***: Climate, soil, and agronomic management practices play a crucial role in the susceptibility of potato/tomato to *Cs/Rs*.[10] For instance, temperature and humidity levels can impact pathogen growth and spread, while soil characteristics such as pH, texture, and organic matter content can affect the plant's ability to uptake nutrients and resist infection.[11] Agronomic management practices, such as the use of irrigation systems, fertilizers, and pesticides,

[7] Patent PL 427988. Link.
[8] Barchenger, Derek W., et al. (2022) Scientific Reports 12.1, 8374.
[9] Shin, Il Sheob, et al. (2020) Euphytica 216, 1-20.
[10] Singh, D., et al. (2014). Arch. Phytopathol. Plant Prot, 47(13), 1574-1583.
[11] Jiang, G., et.al (2021). Soil Ecol. Lett. Vol 3, 356-366.

can also influence disease incidence and severity. For instance, improper use of pesticides can lead to the development of pesticide-resistant variants of *Cs/Rs*. ***BSoA:*** Understanding the influence of climate, soil, and agronomic management practices on *Cs/Rs* susceptibility is critical to developing effective disease management strategies. In **POMATO**, activities will be carried out to map and determine the impact of environmental factors such as soil type and climate on the predisposition of potato/tomato fields to bacterial wilt disease, as a base study to avoid and control the spread of *Cs/Rs.* This activity will provide for the first time a map of factors affecting the susceptibility of potato/tomato to *Cs/Rs* such as precipitation, temperature, soil characteristics, and farming conditions and how these factors can influence the disease dynamics. Furthermore, **POMATO** will identify the keystone microbial species and molecular features from both plants and plant-associated bacterial microbiome that prevent the progression of *Cs/Rs* strains. To do so, two modelling strategies will be deployed: *i)* **MAGs and their GEMs** will be obtained from publicly available data. If not available, automated tools for GEM reconstruction will be used, and *ii)* **Uni- and multivariate statistical modelling** in combination with predictive modelling using machine and deep learning (like Random forest and Neural networks), network inference and analysis, and reverse engineering approaches will be used to integrate and analyse data and metadata. Mechanistic models will be simulated using ***SoA*** approaches that consider spatio- temporal dynamics that will be expanded to include host pathogen. Statistical approaches and machine learning methods[12] will be expanded to identify the genetic makeup of the observed virulence. Identification and characterization of the key microbes in the complex network of plant pathogen interactions will enable development of artificial microbial communities with plant protective effects. Based on the results obtained from the modelling approach, the rhizosphere microbial sub-communities will be constructed using an innovative microfluidic approach. This construction will serve as the foundation for determining and comparing interactome signatures with the modelling data under exposure of plants together with their rhizosphere microbiome to *Cs/Rs*.

Ambition #3: Early detection digital tools (WP 3 T3.1-3.2, T3.4-3.5): ***SoA*:** Current AI-based decision support systems (DSS) utilize advanced machine learning models such as Convolutional Neural Networks (CNN) and Transfer Learning to identify plant diseases from images, particularly focusing on plant leaves.[13,14,15] These models have been developed beyond traditional machine learning models like Random Forest and Naïve Bayes. However, their application is limited to leaf diagnostics and is heavily reliant on public datasets like PlantVillage, which have limitations such as regional specificity, imbalanced class distributions, and a focus on common diseases. This results in high false detection rates in different environmental conditions and challenges in early disease detection and surveillance due to a lack of temporal analysis and detailed disease progression insights.[21,16] ***BSoA:*** **POMATO** aims to transcend current limitations by extending AI applications to include various plant parts for a more holistic disease detection approach. This involves utilizing advanced imaging and deep learning techniques, such as new computer vision architectures and Visual Language Models (VLMs), to correlate molecular patterns with visual symptoms. Key innovations include assessing disease severity and progression over time, integrating environmental and climatic data for dynamic disease management, and enhancing model transparency and trustworthiness through explainable AI (XAI) techniques. These advancements are designed to improve early detection, provide comprehensive risk management, and adapt to changing environmental conditions, thereby significantly advancing the current state of plant disease diagnostics and management. Genomic databases will be inspected using bioprospecting approaches to identify genomic markers of microbial actors involved in each situation.

Furthermore, **POMATO** will develop **mechanistic models for severity and progression assessment** by analysing changes of microbial community composition over time. To that purpose, MAGs and their constructed GEMs will facilitate the elucidation of the metabolic interactions that drives ecological relationships. In this way, dynamic models of the microbial community are meant to provide insights into how pathogenic bacteria evolve and spread within a crop microbiome, facilitating the identification of strategic interventions at critical stages of disease development. In addition, **POMATO** will go beyond traditional detection methods[17] to develop advanced and automated contactless, non-destructive detection methods tailored for in-field (portable or handheld) and on-drone applications, enabling real-time and accurate disease information for potato/tomato farmers through a **MDD sensing device**. The integration of multivariate spectroscopic analysers, imaging technology and univariate sensors into a single, innovative sensing device will significantly enhance the accuracy of detecting and identifying plant pests and diseases, empowering early detection during latent infection periods. Additionally, the device will have a userfriendly

[12] Poncheewin et al (2022) Scientific Reports 12, no. 1 (June 27, 2022): 10857.
[13] Latif, G., et al. (2022) Plants. 11 (17), p. 2230.
[14] Arafath, M. et al (2023) Advances in Science and Technology, 124, 236-245.
[15] N. N. Moon, et al (2021) International Conference on Smart Generation Computing, Communication and Networking pp. 1-8
[21] Rashid, J., et al (2021) Electronics, 10 (17) p. 2064.
[16] Jafar, A., et al (2024) Frontiers in Plant Science, 15, p. 1356260.
[17] Zhou, C. et al (2023). Appl. Spectrosc. 77(5), 491-499.

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interface for easy operation. Finally, **POMATO** will develop an **AI powered Cloud data management platform**, interfaced by both web and mobile applications. The cloud platform will integrate data collected from polymeric sensors, multispectral images at high spatial resolutions (5-20 cm) from both healthy and infected potato
Associated with document Ref. Ares(2025)2863795 - 08/04/2025
and tomato crops, the MDD sensing device and other relevant information sources. In addition, it will also integrate the AI models developed within the project framework to run based on the available data for disease prediction.

Ambition #4: Monitoring strategy based on polymeric sensors to detect plant metabolites (WP 3 T3.3): ***SoA***: Smart polymers react sharply and present a selective intelligent response to physical and chemical stimuli (chemical species, biomolecules, pH) and so, the interest in such materials has recently increased exponentially, finding applications in environmental, biomedical and food safety sectors, among others[18]. Particularly, **polymeric sensors** exhibit physical measurable responses like a change in colour or fluorescence visible to the naked eye, enabling any inexperienced user to employ them for qualitative and quantitative analysis. ***BSoA:*** **POMATO** will design innovative solutions for the rapid, simple, inexpensive, and on-site detection and quantification of key metabolites, generated by potato/tomato plants in the early stages of infection, through tailor-made sensory polymers that will allow the identification and alert of the presence of both *Cs/Rs* by non-specialized personnel in an early stage.

Ambition #5: Natural and bio-control solutions (WP 4 T4.1 – T4.5):
Biocontrol coating solutions*: **SoA:*** The decrease in the number of authorized synthetic plant protection products coupled with societal demands for a more oriented organic farming involves searching for bio-based solutions to protect crops[19]. In the case of *Cs/Rs*, nowadays there are not commercial sustainable chemical-free solutions to control these pathogen diseases in field crops. Biocontrol agents as antagonist bacteria and specific enzymes are recognised as effective treatments against diverse phytopathogens due to their capacity to metabolise biocide compounds. However, the effectiveness of biocontrol agents depends on application strategies, formulation method, and types of strains used. Because of the genetic diversity of pathogens, no unique biocontrol agent has been universally effective[20]. Moreover, the field application effectiveness of biocontrol agents is reduced by environmental factors. Therefore, the use of coating formulations and encapsulation techniques plays a vital role in prolonging the viability and enhancing their survival against adverse environmental conditions[21]. ***BSoA***: **POMATO** will study the mode of the combinations action of microbial strains which some of them will be isolated from the rhizosphere of diseased plants and/or enzymes with antagonistic antibacterial effect and their improved plant application feasibility by using coating formulation including encapsulation techniques to provide significant biocide effectiveness against *Cs/Rs*.
Plant-derived organic substances: ***SoA:*** Detection of hazardous chemical pesticide residues in foods and increased consumer awareness on food safety has resulted to ban of certain pesticides in agricultural production and natural plant protection products are gaining popularity as an environmentally friendly tool for the crop protection[22]. Plant pathogens virulence can be suppressed by the application of plant-derived organic substances, such as extracts and essential oils and their active principles which have been studied for their antimicrobic and antioxidant properties[23]. However, several bioactive molecules are highly volatile, chemically unstable and susceptible to the adverse external environment limiting their direct utilization.[24] ***BSoA***: In **POMATO**, the identification of the most effective plantderived organic substances against *Cs/Rs* will be carried out. Moreover, promising delivery strategies for application and release of the bioactive substances in potato and tomato crops will be evaluate and developed.
Plant growth-promoting microorganisms (PGPM): ***SoA***: PGPM have proved to be a green strategy to manage plant disease through production of inhibitory chemicals, inducing immune system, niche exclusion, deprivation of key chemical elements, increasing stress response of plants as well as production of VOC that can completely diminish expression of virulence factors of pathogens.[31] Although not limited to, the most frequently isolated PGPM representatives are various strains from Bacillus and Pseudomonas genera that are showing the inhibitory effects on *Cs/Rs*.[32] However, the microbial networks involving multiple microbes competitively exclude the pathogen, such as

[18] García, J. M. et al. (2022) "Smart Polymers: Principles and Applications", Berlin, Boston: De Gruyter.
[19] Kumar, J. et al. (2021). *Plants*, 10(6), 1185.
[20] Mishra, P. et al.(2016). *Microorganisms for Sustainability,* 11. 259-285.
[21] Riseh, R.S. et al. (2023). *Carbohydrate polymers*, 15, 120384.
[22] Souto, A.L. et al. (2021). *Molecules*, 26(17), 4835.
[23] Arshad, Z. et al. (2014). *Int. J. Chem. Biol. Sci.,* 6, 11-17.
[24] Wang, W. et al. (2026). *Trends Food Sci. Technol.*, 56, 21-38. [31] El-Saadony, M.T. et al. (2022). *Front. Plant Sci*, 13, p.923880 [32] Gautam, S., et al. (2019). *Microb. Pathog*, *130*, pp.196-203.

Rs, through the nutritional interdependence[25]. Microbial community showing low nestedness, high connectance and niche overlap with the pathogen are more resilient to the pathogen invasion, constrained growth of the pathogen, resulting in reduced disease occurrence[26]. However, the random use of combination of PGPM can result in antagonistic effects and can exclude the PGP action of PGPM strains when used separately.[27] Efficient PGPM activity depends on factors like PGPM combination, preparation method, and the delivery system. Recent reports suggest that adding charcoal as a carrier matrix can enhance PGP and pathogen suppression.[28] ***BSoA***: **POMATO** will develop microfluidic/encapsulation platform that will enable to *i)* isolate individuals and micro-community of compatible PGPMs, *ii)* screen their growth-promoting activities on a large scale including several 1000 combinations of PGPMs and *iii)* evaluate and determine their synergistic effects through cell-to-cell interaction assays. On 10 most promising isolated combinations of PGPMs, **POMATO** will determine mechanisms of action against *Rs* and *Cs* using OMIC techniques. **POMATO** will develop electrostatic based carrier system that will have stimulatory effects of PGPMs on PGP activity and having inhibitory effect against *Cs/Rs* when applied on plants.

Chemoperception disruptive products: ***SoA***: Phytopathogenic bacteria actively move to their host via chemotaxis and enter the plant through natural openings. The secretion of natural attractants, such as amino acids and organic acids by the host, triggers the chemoperception systems that enable the control of motility and plant colonization[29,30,31,32]. Considering this, a disrupting chemoperception strategy based on the use of saturating concentrations of natural attractants can impair the bacterial chemoreceptor functions and avoid the infection of the plant by the pathogen.[37] However, information regarding the chemoperception systems of phytopathogenic bacteria *Cs/Rs* is scarce, and the role of the signals perceived in bacterial virulence is not yet defined. ***BSoA*: POMATO** will describe and identify the role of chemoperception in the infection process of *Cs/Rs*, and it will develop natural products solutions to block the infections. The list of the EC of active substances to fight bacterial diseases only include 6 approved substances.[33] The knowledge of the mechanisms used by bacteria to enter the plant tissue and the possibility to interfere it by plant-based compounds open the development of alternative sustainable tools as an efficient alternative to fight these diseases.

1.1.3. Positioning and Technological Readiness Level (TRL)

The activities developed in **POMATO** will start at TRL 2/3. The project will promote the opportunity to evaluate the synergies of the technological solutions within the IPM strategy both for organic and conventional tomato and potato fields at TRL 5 as the following table shows:

Table 1.1. TRL positioning of POMATO innovations (Start- End)

Innovation	TRL	Justify current and final TRL
I-matrix for bacteria isolation	2→5	Contaminants in environmental samples and the lack of effective methods for their removal falsify or prevent correct diagnosis of *Rs*. The development of an I-matrix for the specific isolation of bacteria and its verification on environmental samples will improve and make the diagnostics in these samples more efficient and effective.
Gene Expression of and genome data from *Cs/Rs*	2→4	Transcriptional profiles in *Cs/Rs* are characterized *in planta* using susceptible cultivars and in rich medium. Specific transcriptomic Cs/*Rs* when interacting with wild varieties (more resistant to disease and environmental stresses) will be studied. Whole-genome sequence for *Cs/Rs* representative isolates.
Genetic Resistance of potato/tomato	2→4	Genetic resistance to *Cs/Rs* is mainly polygenic and strain specific. To identify new sources of resistance to *Cs/Rs*, an interesting approach is to look for resistance genes in tomato wild relatives. GWAS combined with transcriptomic analysis will contribute to the identification of new RGs in potato/tomato wild varieties.
AI predictive algorithms and mathematical models	2→5	AI tool advances disease detection with climatological features, evolving from conceptual models to predictive algorithms. It forecasts disease spread in potato/tomato crops, marking a shift from basic research to refined technology tested in controlled environments.

[25] Yu, X., et al. (2022). *Biol. Control*, *170*, p.104937.
[26] Wei, Z., et al. (2015). *Nature communications*, 6(1), p.8413.
[27] Kang, Y., et al. (2014). *Microbiology*, *83*, pp.666-673.
[28] Ahmad, C.A. et al. (2024). *Front. Microbiol*, 15, p.1360703.
[29] Río- Alvarez, I. et al (2015). *Molecular Plant Pathology* 16:685-698
[30] Santamaría-Hernando, S. et al (2022). *Molecular Plant Pathology* 23:1433-1445
[31] Cerna-Vargas JP et al. (2019). mBio. 10(5):e01868-19. doi: 10.1128/mBio.01868-19
[32] Gálvez-Roldán, C et al. (2023). *Phytopathology* 113, 390–399
[33] EU Pesticides database, Active substances, safeners and symergists. Link

MDD sensing device	2→5	Initially, the concept of integrating multivariate spectroscopic analyzers, imaging technology, and univariate sensors will be formulated and investigated. The development will progress through experimental proof of concept, creating a first laboratory-scale prototype. Further advancement will involve developing a functional prototype and validating it in a controlled laboratory environment. At a final stage, the components will be integrated, and the device will be validated in test fields or greenhouses, ensuring accurate disease detection and userfriendly operation.
AI powered Cloud data management for disease prediction	2→5	The AI powered Cloud data management for disease prediction will be first conceptualized undergoing initial development and testing through 50 users in the crop fields for the final deployment.
Smart carriers for PGPMs delivery	3→5	Smart carrier approaches to enhance the delivery and effectiveness of PGPMs acting against pathogens by use of combination of materials that stimulate and on demand release nutrients and stabile soil colonisation through radial encapsulated community expansion.
Polymeric sensors	2→5	The currently existing strategy for the design and development of polymeric sensors will be adapted for the detection of the previously identified key metabolites produced by the plants in the early stages of infection, alerting of the presence of *Cs/Rs*.
Plant-derived organic substances and coatings	3→5	Biocontrol agents and non-pesticide chemicals have been developed as part of integrated management of crops diseases. Improved coating formulations based on microorganisms and enzymes and encapsulated plant-derived substances will be tested against *Cs/Rs*.
Chemoperception disruption products	3→5	From the proof of concept to the development of disease interfering strategies based on the application, in relevant environments, of natural products to inhibit pathogen chemoperception, thereby preventing their entry into crop plants.

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1.2. Methodology

1.2.1. Concept and overall methodology in POMATO

POMATO is a 48-month RIA action project based on the following activities against the presence of *Cs/Rs* pest threats in potato and tomato plants (Fig. 1.3).

Thereby, **POMATO** proposes a work methodology based on 6 work packages (WPs) to be executed in 48 months, where all the molecular studies, early detection and surveillance solutions, as well as the natural and bio-control solutions will be tested in real scenarios in order to validate the Integrated Pest Management strategies to protect potato and tomato against both bacteria. To addressed this, **POMATO**'s methodology is structured in four pillars **(Pillar #1 – Pillar #4)**

The overall methodology of **POMATO** is illustrated in the following figure:

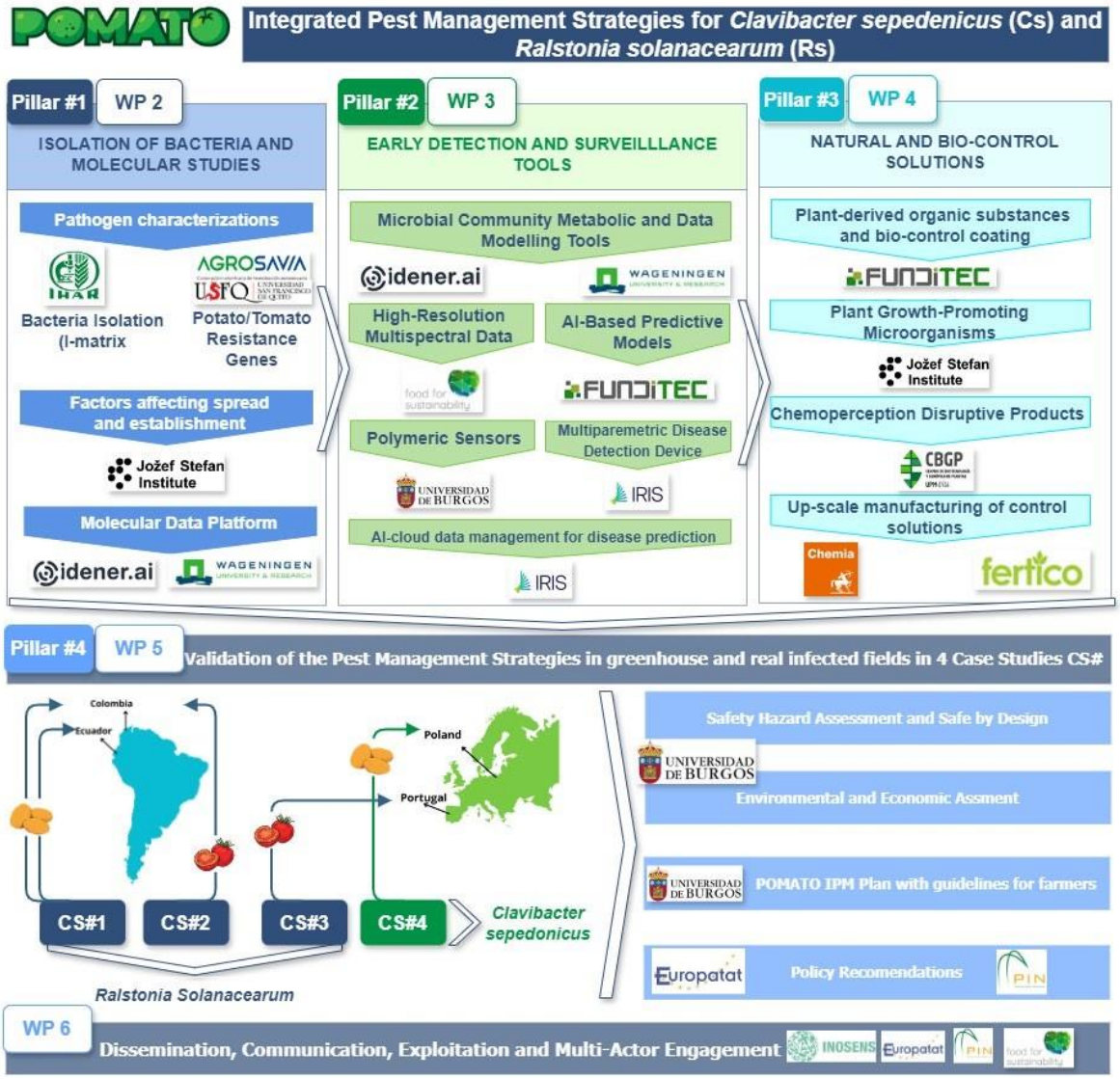

Figure 1.3 POMATO Overall Methodology

Pillar #1: Isolation of bacteria and molecular characterization studies. Linked to WP 2/O1 (R1-R6)

I-matrix for bacteria isolation: The aim of the planned research is to obtain innovative tools for the isolation of *Rs* bacteria that will allow for the concentration and specific isolation of living cells of *Rs* while removing contaminants from the tested environmental samples (plant extracts, soil, water). The methodology of the research will be staged: first, the biological material will be selected (3 strains of *Rs*, 12 varieties of potatoes and tomatoes) and optimal conditions for the isolation of bacterial cells will be developed through the selection of appropriate microbiological media, buffers and *Rs* isolation time. Based on the selected bacterial strains, appropriate receptor molecules will be designed to enable a specific reaction with *Rs* bacterial cells. By choosing suitable materials and shapes, matrix bases with a surface that allow covalent attachment of receptor particles will be created. From the best template bases and the most effective receptor particles will be combined to produce various I-matrix prototypes for *Rs* cell isolation. In the last stage of laboratory tests, the effectiveness and durability of the developed I-matrix will be assessed in the presence of standard strains of *Rs* bacteria and tested samples (extracts from tubers and plants, soil and water) obtained under environmental conditions. The results obtained will allow the development of a protocol using Imatrix to effectively isolate *Rs* cells directly from contaminated environmental samples (the reason of excluding *Cs* is explained in the footnote[34], and due to previous patent by IHAR[12]).

Potato and tomato for resistance against *Cs/Rs*: One of the most significant challenges to address in molecular breeding for bacteria-plant interactions is the strain specificity of the majority of identified Quantitative Trait Locus (QTLs). At least 100 bacterial isolates will be obtained from potato/tomato genotypes collected from 40 field

[34] due to the risk of violating the patent claims of an earlier patent, the development of I-Cs-Matrix on similar principles is too risky for **POMATO**. Trying to circumvent this reservation by finding an alternative solution consisting in replacing the source of receptor molecules from mammals to birds may not be possible due to the constantly increasing requirements and changing regulations of ethical committees for animal experiments.

locations in LATAM, 40 Polish infected potato samples, and 20 (10 *Cs* and 10 *Rs*) from microbial culture collections. First, *Cs/Rs* characterization at phenotypic and genotypic levels will be performed. Then, transcriptome analysis and GWAS studies will be combined for identification and validation of marker genes and RGs in the potato/tomato-*Rs* pathosystem and the potato-*Cs* pathosystem. For this, marker-trait associations (MTAs) previously reported for *Cs/Rs* resistance will be in potato/tomato wild relatives, additionally SNP (Single Nucleotide Polymorphisms) array-based genotyping will be performed to find new QTLs. Finally, a workflow for transcriptome analysis and annotation will be generated for plant wild varieties-bacterial interaction to identify candidate genes related to resistance responses against *Cs/Rs*. Genomes from newly isolated bacterial will be annotated using pipelines designed to maximize data interoperability[35] and to ensure that newly generated datasets can be queried together with the machine-actionable semantic database of >100.000 consistently de novo annotated microbial genomes available at WUR. Features related with gene structure and function, transmembrane domains and protein localization, presence of biosynthetic gene clusters, protein domains and genome properties will be annotated to enable comparative genomic analysis.[36] Classification approaches (Random Forest and support vector machines), followed by feature importance analyses will be followed to identify genetic features associated to pathogenic microbial lifestyle.[37]

The stimulatory **interaction between the plant microbiome and pathogenic *Cs/Rs*** will be studied using microfluidic and microdroplet-based methods. The rhizosphere microbiome will be extracted at various disease stages, with either natural aggregates preserved, or artificial aggregates constructed through microaggregation using polyelectrolytes. Stained aggregates will have electrostatically topically attached cells of pathogenic strains and be further entrapped in microdroplets using T-type microfluidic chips, allowing the formation of 10^4 microbiomepathogen combinations. Pathogen growth will be monitored via fluorescence and analyzed using flow cytometry and microscopy. Microbiomes influencing pathogen growth will be metagenomically characterized with the SqueezeMeta platform, including genome assembly and annotation. Positive microaggregates will be further analyzed by forming 3D structures to test cell distances and microenvironmental conditions using confocal microscopy and microelectrodes. Environmental factors like pH, Fe availability, and plant exudates will be assessed, with transcriptomic studies identifying interspecies responses under stimulatory conditions. These studies will be complemented with insights into the molecular mechanism associated to infection signatures and the underlying molecular networks that may contribute to plant resistance and the development of predictive models of disease progression that include environmental factors and genetic determinants.

Pillar #2: Early detection and surveillance tools. Linked to WP 3/O2 (R7-R11)

POMATO will identify the keystone microbial species and molecular features from both plants and bacteria that prevent the progression of *Cs/Rs* strains. To do so, two complementary modelling strategies will be approached*: i*) **Dynamic microbial community metabolic modelling tools** (including **MAGs and GEMs)** to study the microbial co-occurrence networks of healthy and infected soils and plants Network analysis tools will be applied to explore the associations among genes or microbial species, and to provide insights into the functional potential and interactions within microbial communities. On the other hand, constraint-based stoichiometric models and the Flux balance analysis (FBA)approach[38] for mathematical microbial community modelling will facilitate the identification of their ecological interactions. FBA is the best approximation to infer growth rate and production/consumption of metabolites based on metabolic capabilities of the stoichiometric network and the assumption of the steady state.[39] Additionally, in dynamic FBA,[40] the snapshot the steady-state assumption provided in FBA approach can be replaced

[35] Koehorst, et al. Bioinformatics 34, no. 8 (April 15, 2018): 1401–3.
[36] van, Dam Scientific Data 6, no. 1 (November 4, 2019): 1–9.
[37] Molder, et al. Et BMC Genomics 22, no. 1 (November 23, 2021): 848.
[38] Heirendt et al., Nat. Protoc. 14 (2019): 639–702.
[39] Orth et al., Nat Biotechnol 2010 Mar;28(3):245-8
[40] Mahadevan, R. et al. Biophys J, 83:1331-1340 (2003)

by discretizing time in intervals and assuming quasi-steady-state conditions. Thus, it is possible to update at each time step the metabolite levels based on how much has been consumed or secreted. The ability to predict individual growth rate, metabolite consumption preferences and by-products production, makes dynamic FBA suitable to parametrize the metabolic interactions that drives the dynamical evolution of the individual abundances and the community evolution.[41] *ii*) **Statistical and Mathematical Modelling Data** from various measurements and from metabolic modelling will be integrated to extract underlying common and distinctive processes. This will be achieved using specialized methods for analysing complex multi-set data (e.g., GSVD, DISCO-SCA and MOFA). To complement the statistical approach, construction of (multilayer) networks will provide the relationships between molecular features, phenotypic and disease data. Network inference approaches will be used to handle various data types, including Mixed Gaussian Graphical Models (GMM), probabilistic GMM and Random Forest, which will enable the construction of conditional independence/dependence graphs. This will allow pinpointing and mapping key molecular features from both plant and bacteria to identify and define infection signatures and the underlying molecular networks that may contribute to resistance or be targets for treatment.

Following this, **POMATO** will address the early detection by applying **5** cutting-edge tools (Fig1.4):

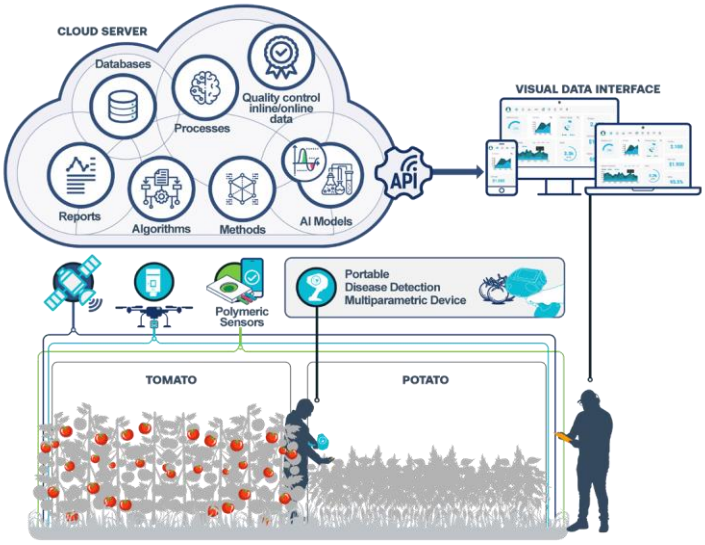

Figure 1.4. Cutting -edge digital technologies applied in POMATO

i) Acquisition of **high-resolution multispectral images using remote sensing technologies** (drone and satellites). These images will be analysed using a range of vegetation indices, including the Normalized Vegetation Index (NDVI), Chlorophyll Index (CI), Leaf Area Index (LAI), Greeness Index, Modified Chlorophyll Absorption in Reflection Index (MCARI), Transformed Chlorophyll Absorption in Reflection Index (TCARI), Triangular Vegetation Index (TVI), and Simple Pigmentation Index (SIPI). By leveraging machine learning algorithms, the data will be processed to detect, differentiate, and quantify the severity of disease symptoms, offering early identification and forecast of potential outbreaks. Satellite imagery from sources like Landsat 8 and Sentinel-2, equipped with visible, near-infrared (NIR), shortwave infrared (SWIR), and thermal bands, will complement drone data to provide a multi-scale view of field conditions. The remote sensing techniques employed will detect plant stress through reduced absorption of visible wavelengths, particularly in the red and blue spectrum, and decreased reflectance in NIR bands. Data collected from both infected and healthy potato/tomato fields in Portugal and Poland will be analysed to classify contamination levels and infer ecosystem health and drivers of contamination. This comprehensive dataset will also provide a training suite for AI models focused on identifying contamination levels and allocating functional measurements of crop diseases across broad geographic areas.

ii) **AI-based predictive models: POMATO** will implement AI-based predictive models using a structured methodology: data acquisition, image and climate data processing, image segmentation, model training, integration of advanced technologies, and iterative testing. The process begins with the collection of comprehensive plant disease data, including high-resolution images and molecular data. FUND will collaborate with consortium partners to optimize image parameters (resolution, brightness, format, colour spaces like HSV, HIS, RGB) and source open weather data from NASA and ECSA databases. Data processing involves standardization, quality control, regularization, colour space adjustment, distortion removal, and application of filters (e.g., Laplacian) and transformations (e.g., Fourier Transform). This ensures data consistency for robust model training. AI model architecture will be designed considering factors like time window size for meteorological data and image resolution. The models will incorporate appropriate neural network blocks for the task. Training will utilize open-source tools like Weights & Biases for tracking and Bayesian hyperparameter optimization. The dataset will be strategically split for training, internal evaluation, and testing to assess real-world performance. Model performance will be evaluated using metrics such as Matthews Correlation Coefficient (MCC), Kappa Coefficient, and false positive/negative rates. To ensure interpretability and trustworthiness, AI predictions will be compared with human expert annotations.

[41] Gomez et al., BMC Bioinformatics 15 (2014): 409.

Finally, the models will undergo validation in real-world scenarios with unseen data, serving as the final optimization phase. This approach aligns with HORIZON-CL6-2023-GOVERNANCE-01-16 projects on digital technologies for plant health early detection. Collaboration on data sharing and model validation with project funded under that call is already envisaged and will enhance the robustness and applicability of the developed models.

iii) **Polymeric sensors:** The development of sensory polymers will be linked to the insights gained through MAGsbased metabolic studies. **POMATO** will firstly identify key metabolites in the *Cs/Rs* pathogenesis process and will study their effects on plant-microbe and microbe-microbe interactions with **chemometrics** and **cheminformatics** together with spatial structuring and selection strategies (consortia isolation, formulation preparation and efficiency testing) of metabolites. Secondly, specific target metabolites associated to *Cs/Rs* in the initial stages of infection of potato and tomato plants will be selected for sensor design. **POMATO** will develop polymeric sensors based on hydrophilic film materials capable of detecting the presence of such key metabolites associated to *Cs/Rs* by triggering a visual alert (a change in colour or fluorescence) upon contact with these target species. To do so, specific receptor molecules will be designed for the selected metabolites and will be co-polymerized to prepare the sensory films. These materials can be used by individuals without specialized training, and even a simple photograph taken with a mobile phone can be used for quantification at ppm or ppb levels of the selected *Cs/Rs*-associated metabolites, providing a timely alert to proactively combat infections in potato/tomato plants.

iv) **Multiparametric Disease Detection (MDD) sensing device**: An innovative multiparametric device able to accurately detect and identify plant pests and diseases in potato and tomato crops using a combination of advanced sensing technologies and machine learning will be developed. The optimal combination of these technologies will be engineered and integrated into the compact device to allow the monitoring of temperature and soil moisture, plant disease identification, changes in plant temperature during latent infection periods, and visible symptoms, wounds, and other signs of plant diseases. The collected data from the different sensors will not only be shown in a userfriendly interface from the device to allow users to operate the device and obtain easy-to-understand information easily, but also to an AI powered Cloud platform.

v) **AI powered Cloud data management** for disease prediction will be a sophisticated multilayer digital tool designed to revolutionize disease detection in agriculture. This Cloud platform, with an Application Programming Interface for data connectivity through a user-friendly web or mobile application, will serve as a comprehensive storage and data processing unit for data obtained through cutting-edge AI algorithms, polymeric sensors, and Portable Disease Detection Multiparametric Devices, among other relevant information sources. The data collected will be seamlessly processed within the platform, utilizing advanced analytics to generate precise and timely responses regarding the detection of agricultural threats.

Pillar #3: Natural and bio-control solutions. Linked to WP 4/O3 (R12-R13)

Plant-derived organic substances and bio-control coating: FUND will formulate biocontrol coating solutions with antimicrobial activity based on the action of biological catalysts (enzymes) and microorganisms (bacterial antagonists) as an alternative solution to the use of chemical pesticides for the protection of potato and tomato crops against *Cs/Rs* diseases. Despite the well demonstrated effectiveness of biocidal agents against a wide range of bacteria, only a limited number of these products are currently available on the market.[42] Therefore, further research efforts are needed to strengthen the feasibility and advancement of promising biocontrol solutions. In **POMATO**, the activity of animal derived enzyme systems, such as lysozyme and lactoperoxidase, will be studied optimising most effective substrate ratios in *in vitro* assays to have high antimicrobial metabolite production. For antagonistic bacteria, a screening will be carried out to identify the best effective microbiological strain candidates. On the one hand, microbial strains from public collection will be studied, evaluating their antimicrobial effects against *Cs/Rs*. On the other hand, in collaboration with CBGP, a pool of microbe candidates will be isolated from the rhizosphere of plants infected by *Cs/Rs*. The identification of selected strains will be performed through Mass Spectrometry and Amplicon Metagenomic Sequencing analyses. Once the microbial candidates are selected as biocidal agents, the fermentation process will be evaluated and optimised studying the effect of different parameters (temperature ranges, agitation, pH, aeration, and different inoculum ratios) to have higher yield of biocidal metabolite production. This biosynthesis from both the biological catalyst and the microorganism will be assessed to evaluate the ability to produce molecules that induce systemic resistance to *Cs/Rs*. The concentration of antimicrobial compounds and the production of lytic enzymes will be determined by chromatographic and enzymatic approaches; the microbiological

[42] Lahlali, R. et al. (2022). *Microorganisms* 10(3), 596.

evaluation will be carried out to assess the antagonistic effects on *Cs/Rs* in agar plates. In parallel, solutions based on plant-derived organic substances will be developed and tested to control the incidence of *Cs/Rs* in potato and tomato crops. The use of natural compounds such as plant extracts and substances and their active principles (alkaloids, phenols, monoterpenes and sesquiterpenes, isoprenoids) is a very promising approach, given their various antimicrobial and antioxidant properties.[43]. In **POMATO**, screening *in vitro* assays of active compounds derived from plant extracts (e.g. garlic, onion, neem, etc.), essential oils (e.g. thyme, clove, cinnamon, lemongrass, etc.) and natural resins such as propolis will be carried out to identify and test the most effective biocidal agents against *Cs/Rs*. Delivery system strategies based on micro- and nanoencapsulation techniques will be implemented to ensure the

feasibility of applying both plant-derived organic substances and biocontrol agents to potato and tomato crops. Monitoring of the biocidal activity of the developed solutions will be evaluated in tomato and potato seeds and plants inoculated with *Cs/Rs* in growth chambers. For biocontrol coating solution developments, several improved coating formulations based on effective microbiological strain combinations (mainly *Streptomyces*, *Pseudomonas* and *Bacillus* spp.) will be tested against pathogen candidates on seed and plant crops. Similar experimental trials will be also carried out testing the antimicrobial effectiveness of animal derived enzyme systems activity as lysozyme and lactoperoxidase.

PGPM to inhibit *Cs/Rs*: JSI will develop a novel approach to screen and select consortia of microorganisms based on the random combination of cells and their electrostatic aggregation. Microbial cells extracted from the rhizosphere, without prior cultivation, will undergo modification through viable staining and coating with polyelectrolytes to alter their surface charge. When oppositely charged cells are mixed, including those of pathogenic strains, aggregates of various sizes are formed. These aggregates will be encapsulated in a polymeric matrix using microfluidic T/Y junctions for droplet generation and encapsulation, or microencapsulation through perlation. JSI will monitor the efficiency of PGPM, determine PGP traits, and assess inhibition of *Cs/Rs* using approaches based on the spectrometry and flow cytometry. Within the **POMATO** project, JSI will develop various matrices for encapsulation of aggregated strains to test the activity of excreted substances, competition for nutrients and microelements, and evaluate effects on microenvironmental niche formation and ecological nesting, among other factors. The most effective consortia against *Cs/Rs*, identified from among thousands of combinations, will undergo characterization using omics tools based on the shotgun sequencing data. For the most effective consortia, we will determine the microspatial distribution of involved microbes using confocal microscopy, which will guide the strategies for encapsulation and rhizosphere inoculation. The efficiency of root attachment, colonisation and biofilm formation will be monitored using molecular tools and microscopy studies.

Disruption of bacterial chemoperception of *Cs/Rs*: Based on available public genome databases and new genomic data generated in WP2, computational analyses of *Cs/Rs* chemoperception genes will be conducted. The classification of chemoreceptors (the proteins responsible for initiating the chemosensory signaling cascade) and their ligand binding domains (LBDs) will be performed using a *de novo* methodology developed by CBGP.[44] A manually curated dataset of 960 representative species of plant-associated bacteria (including 119 phytopathogens) will be employed to classify the chemoreceptor LBDs based on their degree of plant specificity. This will enable the research to focus on candidate chemoreceptors that are potentially more relevant for pathogen-plant interaction as targets for the disruption strategy. Functional analysis of chemosensing proteins in the selected *Cs/Rs* strains will be performed by constructing mutant strains in the selected chemoreceptors. To identify potential ligands for these chemoreceptors, a high-throughput capillary-based chemotaxis assay will be used to detect bacterial chemotactic responses to a BIOLOG™ compound collections. Chemicals that elicit a differential response between wild-type and mutant strains will be considered putative ligands of the disrupted chemoreceptors. This will allow to determine the chemotactic profile of the selected chemoreceptors. To implement a successful chemosensing disruption strategy is crucial to establish whether the attractant compound(s) bind directly to the chemoreceptor LBD, and to determine their binding affinity constants. The selected chemoreceptor LBDs will be cloned, expressed, and purified for biochemical characterization. To determine if there is a direct binding, the following approaches will be followed: *i)* Fluorescencebased Thermal Shift assays (FTS); *ii)* Positive FTS results will be corroborated through Isothermal Titration Calorimetry (ITC). To establish the role of each of the chemoreceptors in the plant infection, the pathogenicity of chemoreceptor mutants will be analysed in tomato and potato plants. For that, plants will be inoculated with bacterial suspensions of the wild-type or mutant strains, and symptoms will be evaluated. Bacterial populations will be analysed during the time-course of infection to have an estimation of the relevance of the functions of these chemoreceptors in the pathogenicity. To assess the effectiveness of plant or natural products that

[43] Simonetti, G. et al. (2019). *Plant Biosyst. Inter. J.* 154, 100–106.
[44] Sanchis-López C., et al. (2021). *Systems* 6:e00951-21.

bind to chemoreceptors in interfering with virulence, the ligands of the most relevant chemoreceptors will be tested as disruptive agents. Infection assays will be conducted with varying saturating concentrations of the compounds to evaluate the extent of interference with disease development.
The solutions developed in the Pillar #3 will be up scaled and manufactured by the companies CHEM and FERT to be used in the validation case studies in the next Pillar #4.

Pillar #4: Integrated Pest Management strategies validated in Case Studies. Linked to WP5/O4 (R14-R17)
In situ pilot validations are planned in 5 case studies (TRL 5), ensuring that the different pest management strategies developed in Pillars #2 and #3 are thoroughly tested in real-world scenarios. The validation studies will be strategically planned based on the best solutions identified in WP3 and 4. The CS will focus on both bacterial pathogens (*Cs/Rs*) and crops (tomato and potato) across LATAM and Europe as depicted in Fig. 1.3. Validation studies for ***Rs*** will be conducted in **3 CS** distributed across **Ecuador**, **Colombia** and **Portugal**. This distribution leverages our experimental fields that have the necessary permits to work with these quarantine bacteria in potato **CS#1** (AGRO, USFQ), and tomato crops **CS#2** (AGRO), **CS#3** (F4S). Validations for ***Cs***, due to stringent

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regulations, can only be conducted in Europe **CS#4** (IHAR). Specifically, these studies will be limited to **Poland**, where the partner in charge, IHAR, has secured the necessary permits to work with this bacterium in **potato** crops. The polymeric sensors and the MDD sensing device developed in WP3, and the solutions identified and scaled up in T4.5 will be implemented according to the tasks described in WP5. The implementation process will take place in the predetermined study locations. These locations have been strategically chosen to maximize the effectiveness of our solutions and ensure compliance with regulatory standards applied by the corresponding local/regional/national authority. The polymeric sensors, the MDD sensing device and the two best scaled solutions will be deployed at the onset of the trials considering the growing season in each country, with subsequent evaluations conducted after 3 months and again after 1 year to assess the long-term impact and efficacy of the solutions. During the validation trials, all necessary permits from relevant agencies will be secured to ensure compliance with local and international regulations. Additionally, throughout the project, efforts will be made to identify potentially contaminated fields, which will aid in the identification and management of infected sites, thereby contributing to the broader goal of disease prevention and control.
CS#1 (AGRO, USFQ) will be conducted in Colombia and Ecuador, focusing on potato crops. Our collaboration with AGRO and USFQ provides access to experimental fields where *Rs* can be studied under real conditions. **CS#2 (AGRO)** will also take place in Colombia and will focus on tomato crops. AGRO's extensive field networks will allow for a thorough examination of *Rs* under real conditions.
CS#3 (F4S) will take place in Portugal, where F4S has secured fields for tomato cultivation, permitting the study of *Rs* under controlled conditions.
CS#4 (IHAR) will be carried out in an experimental field located in one of IHAR's divisions at Bydgoszcz, Poland. IHAR's qualified staff will lead this study focused on potato plants, taking advantage of the permissions obtained from the State Plant Health and Seed Inspection Service (SPHSIS) to conduct field tests involving quarantine *Cs* bacteria.
A **safety and sustainability assessment of POMATO**'s solutions will be carried out since their development phase. The **SSbD Framework** will be applied, and the safety will be assessed to comply with the EU regulatory framework 2019/1009 and 528/2012 providing useful recommendations to optimize the processes and final products, making them safer and more sustainable. **POMATO** will envisage the involvement of the Joint Research Centre (JRC) to develop reference documents in support of a harmonised IPM in the EU legislative acts and policy recommendations. Besides, JRC will be tasked to support UBU on the social life cycle assessment (s-LCA) evaluating the real and potential social and socio-economic impacts of **POMATO** implementation strategies within potato/tomato identified target groups (Table 2.1).

1.2.2. Compliance with the "Do not Significant Harm" principle

All the research activities of **POMATO** will comply with DNSH principle under the Recovery and Resilience Facility Regulation in different ways: *i*) no activity is harming the climate change mitigation as the developed plant health products are fossil-free; *ii*) no activity is harming the circular economy concept. All the contrary, the project will apply circularity principles in the plant health protection products leading to higher resource efficiency, to zero-waste practices, and to sustainable and *iii*) no activity is harming biodiversity and ecosystems as the project enhances potato/tomato genetic varieties.

Link to other R&I activities

POMATO project will capitalize the results of previous and ongoing research and innovation activities, mainly at EU level in H2020 and HEU projects related to crop protection and control where the project's partners have participated. Table 1.2 describes projects relevant for **POMATO** and identifies the connection and valorisation:

Table 1.2. Previous and ongoing initiatives related to POMATO.

Project, partners involved and description	Outcome and valorisation in POMATO
PATAFEST HEU 2023-2027 (**FUND**, **IRIS**, **EUPAT**, **CHEM**): IPM strategies to treat and control the presence of *Candidatus* Liberibacter solanacearum pest in potato plant and address the incidence of soil-borne pathogens during potato postharvest storage.	Biocontrol and digital technologies already developed for potato tubers will be adapted for its application on controlling *Cs/Rs* presence on potato and tomato plants.
Smart polymers as advanced materials, their design and synthesis for agri-food and environmental sustainability applications. AEI 2024-2027 (national project) (**UBU**)	Insights from this project will feed to **POMATO** strategy to detect metabolites of *Cs/Rs* during early stages of infection.
TRIBIOME HEU 2023-2026 (**UBU**): Advanced tools for integration and synergistic interconnection of microbiomes in resilient food systems to reduce the damage to ecosystems and natural biodiversity.	The interconnection strategies will be explored for controlling *Cs/Rs*, enhancing microbiome resilience in **POMATO** crops.
ADAPT H2020 2020-2024 (**EUPAT**): Identify new breeding targets and match potato varieties to specific challenging environmental growth conditions of the future.	The breeding strategies will be aligned with the identification of RGs for *Cs/Rs* to create robust varieties for **POMATO**.

BIOSYSMO HEU 2022-2026 (**UBU**, **IDEN**, **JSI**): Novel approaches for the isolation of whole natural bacterial consortia; computationally assisted framework for designing and optimizing synergistic biosystems mixing bacteria, fungi and plants.	The isolation techniques will support Associated with document Ref. Ares(2025)2863795 - 08/04/2025**POMATO'** aim to characterize the plant microbiome and identify beneficial interactions against *Cs/Rs*.
SECRETed HEU 2021-2025 (**IDEN**): Chemoinformatics, GEM models and other computational tools to unravel the mechanisms for the biosynthesis biosurfactants and siderophores.	The tools will help decipher the chemoperception disruption mechanisms to control the spread of *Cs/Rs*.
CHEMOADAPT 2022-2025 (**CBGP**) Bacterial chemosensing in the adaptation to plant environment and its function in pathogenesis	The insights on the relevance of chemoperception obtained through the analysis in model bacteria/plant interactions using tomato and potato plants as hosts will support their application in the *Cs/Rs* interaction with plants in **POMATO**.
AUSCAR CaribVet 2022-2023 (**AGRO**): Surveillance systems in the Caribbean territories: molecular detection and risk modelling.	Methods and models from AUSCAR will be adapted for developing resilient monitoring systems for early detection of *Cs/Rs* outbreaks in **POMATO**.
An Integrative Approach to Enhance Disease Resistance Against Fusarium Wilt (Foc TR4) in Banana – Phase II, IAEA 2023-2028 (**AGRO**)	The approach for discovering beneficial microbes will be adapted for enhancing potato and tomato resistance *Cs/Rs*.
Plant Microbiome (**USFQ**): In this project, microorganisms related to potatoes and their diseases, such as Pectobacterium and Rizotonia, are being studied.	Data from this project will be leveraged to better understand microbial interactions affecting *Cs/Rs* infections in potato and tomato plants.
PHOTONFOOD H2020 2021-2024 (**IRIS**): Development of the global system control, software and data management and processing platform.	The photonic-based monitoring technology will be explored for real-time detection of *Cs/Rs* in potato and tomato plants.
LILAS4SOILS HEU 2024-2029 (**F4S**): Transformation of the European agricultural sector towards a healthy soil & zeroemission system. Creation of Living Labs in the Mediterranean Region with focus on improving soil health and carbon farming.	Insights from the project will support the acquisition of images in **POMATO** crops.
UNLOCK (**WUR**): Large scale open infrastructure for microbial research focussed on the study of microbial communities and their interactions with their host funded by the Dutch Science Fundation (NWO).	Tools, methods and algorithms to support the analysis of the data generated in **POMATO**.

1.2.3. Digital technologies and Artificial Intelligence applied in Plant Disease Early Detection

As a cross-cutting element, **POMATO** will leverage advanced digital technologies and artificial intelligence (AI) to enhance the detection and management of *Cs* and *Rs*. The AI-based predictive models will be developed with a focus on technical robustness, ensuring accuracy and reproducibility in various environmental contexts. This will be achieved through a systematic methodology encompassing data acquisition, preprocessing, and model training, employing techniques such as Bayesian hyperparameter optimization to fine-tune the models. The performance of these AI models will be rigorously evaluated using reliable metrics like the Matthews Correlation Coefficient, ensuring their ability to handle and inform users about potential inaccuracies and errors, proportionate to the risk posed by these pests. Moreover, to ensure social robustness, the AI models will be developed with a deep understanding of the agricultural context in which they operate, thereby ensuring their reliability and intended functionality. The AI-powered Cloud data management platform will serve as a sophisticated tool for disease prediction, integrating data from polymeric sensors and multiparametric devices to generate precise and timely responses. While providing a detailed explanation of the AI's decision-making processes is complex and beyond the immediate scope, the project will focus on fostering trust and trustworthiness among stakeholders. This will be done by considering objective trustworthiness characteristics and maintaining both "absolute" and "relative" trustworthiness values to reflect the system's reliability over time, thereby supporting human trust decisions.

1.2.4. Gender dimension

Although not specifically required by the topic 2024-F2F-02-04, **POMATO** has a strong multi-actor participatory approach and involvement of citizens, thus it will commit to address the diverse roles, needs, and opportunities, of

citizens regardless their gender and sex. Special attention will be taken to the existing gap between men (68.4%) and women (31.6%) farm managers to foster the engagement of women farmers in the MAPP creation.

1.2.5. Open Science practices

By being compliant with the **European Open Science Policy (EOSP)**, the **POMATO** consortium adheres to achieving responsible research practices to achieve high quality and trustworthy results. The strategy to disseminate or protect the results of **POMATO** is described in Section 2.2. Data associated with results which have potential for commercial or industrial value will not be made accessible for verification and re-use, in general, due to IPR measures. The consortium underlines the importance of OA to scientific publications and research data, which will be managed by INO as lead partner for CDE activities and UBU as task lead Data management, who will ensure adherence to EU guidelines about OA scientific publications and research data under *Horizon Europe*. Furthermore, openness of the data and the results will be addressed throughout the project: *i)* all scientific publications will be made freely accessible as early as possible through OA repositories on the EU innovative OA publishing services such as *Open Research Europe*[53], and shared in *OpenAire- and Google-compliant* repositories, like *Zenodo* and **European Open Science Cloud (EOSC)**, *ii)* the Communication and Dissemination Activities (WP6) have been designed in a OA manner (see section 2.2) aimed at encouraging early sharing of outputs in an open, transparent, and re-usable manner, to a wide audience incl. sector specific target stakeholders and citizens, thereby contributing to increasing the quality of research results, and maximizing the impact for society.

1.2.6. Research data management

The data management process will prioritize how data and results are "Managed," "Protected," "Shared," and "Exploited," using tools, methods, and open repositories to streamline sharing. Research Data Management (RDM) will ensure the data is handled responsibly, adhering to FAIR principles (Findable, Accessible, Interoperable, Reusable). Data will be stored in repositories like Zenodo and Figshare for long-term accessibility, with compliance to Article 17 of the Model Grant Agreement and the IPR policies in the Consortium Agreement. A Data Management Plan (DMP) will be created early in the project (M6) and continuously updated, addressing data security, GDPR, IPRs, and exploitation. The project will generate various data types, including observational, simulation, reference, and experimental, stored in file formats or database connections. UBU will be responsible for curating and preserving all project data, with a dedicated budget for data management and quality assurance. WP2 will implement FAIRcompliant pipelines to ensure the data remains machine-actionable and compatible with AI/ML analysis in WP3, enhancing data accessibility and reuse.

2. Impact

2.1. Project's pathway towards impact

This section covers **POMATO's pathways towards the Expected Outcomes (EOs)** defined in the topic at mediumterm (from project's end to 4 years later) and these two **Expected Impacts (EI)** of the Destination 'Fair, healthy and environment-friendly food systems from primary production to consumption' of the Cluster 6 Work Programme: **EI 1**: sustainable, productive, climate-neutral and resilient farming systems […], while minimising pressure on ecosystems, restoring and enhancing biodiversity, […] generating fair economic returns for farmers. **EI 3**: sustainable, healthy and inclusive food systems delivering co-benefits for climate mitigation and adaptation, […], sustainable healthy nutrition, safe food consumption, food poverty reduction, the inclusion of marginalised people, the empowerment of communities, and flourishing businesses. These pathways are defined by clear, measurable, quantified, and assessable **impact indicators** that fall under **3 impact categories**: **scientific, economic & technological and societal impact.** The results of the project will be used or further up-taken by direct **target groups (TG,** Table 2.1). **POMATO** will generate credible and sustainable will generate credible and sustainable impact pathways in line with the **Key Strategic Orientation 1** '*Green Transition'* of the recently published HEU Strategic Plan for 2025-2027.

Table 2.1. POMATO's Target Groups (TG)

TG1. Conventional and organic potato and tomato growers – farmers, land managers, landowners, national and regional potato and tomato growers' cooperatives *(e.g. Agrico, HZPC, IPM Potato Group, TomatoLand Cooperative, Anecoop, Consorzio Casalasco del Pomodoro)*, EU agricultural associations representing the interest of stakeholders in the EU (*e.g. COPA-COGECA, Europatat, APRODEV*), agricultural holdings, agricultural advisors, extension services

TG2. Retailers, Wholesalers, and Distributors – supermarkets, grocery stores, wholesalers, and distribution companies involved in the buying, storing, and selling of potatoes and tomatoes to end users (*e.g. Carrefour, Tesco, Lidl, METRO Cash & Carry, Eroski*)
TG3. Input related industry: seed producers (*e.g. EUROSEEDS*); bio-based plant health services, manufacturers of bio-based plant protection products (*EBIC*, *IBMA*)
TG4. Research and academic institutions - Institutions involved in research related to plant health and pest control, RTOs focused on applied research and technological development, International research partners involved in potato/tomato pest research

[53] https://open-research-europe.ec.europa.eu/

TG5. Technology and innovation sector – AI and data analytics companies developing predictive models and data analysis tools; Sensor technology companies developing polymeric sensors and other detection technologies; Mobile app developers creating detection and monitoring apps.
TG6. Policy makers and regulatory bodies – EU regulatory bodies involved in creating and enforcing agricultural policies and regulations (*e.g. DG SANTE, DG AGRI, EFSA, EPPO, EU Plant Health Standing Committee*); National plant health agencies responsible for plant health regulation at the national level (*e.g. BMEL – Germany, MIPAAF – Italy, LNV – Netherlands*); International regulatory bodies overseeing international standards and regulations for plant health (*e.g. FAO, WTO- SPS Committee, IPPC, OECD – Agriculture and Fisheries*).
TG7. Environmental and conservation organisations – NGOs and advocacy groups focused on environmental protection and sustainable agriculture (*e.g. IUCN, TNC*); EU, national and international biodiversity and ecosystem service organizations dedicated to preserving biodiversity and ecosystem services (*e.g. EEA, EEB, UNEP, BUND – Germany*).
TG8. End consumers – Consumers of potatoes and tomatoes who benefit from healthy and sustainably grown crops; Consumer advocacy groups representing consumer interests in food quality and safety (*e.g. BEUC, Consumers International (Europe), EPHA, ICRT*).
TG9 Media and general public– Agricultural media (publications, websites, and other media outlets) focused on agriculture and farming; General media: newspapers, television, radio, and online platforms that can help raise public awareness and disseminate information about the project; Science and technology media that focus on scientific advancements and technological innovations.

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SCIENTIFIC IMPACT **Increase the information of the molecular and biological characterization of potato/tomato RGs and the quarantine and priority pests *Cs/Rs*.** *Corresponding to project´s objective O1* *Impact Indicator*: Acquisition of data related to potato/tomato RGs and isolation of bacteria from infected areas. *Target groups*: TG1, TG2, TG3, TG4, TG6.	
Scale and Significance: Currently in EU, there are no effective biological or chemical control methods for the disease caused by *Cs/Rs*, and resistance to the disease in potato stocks has not been found. Control relies upon keeping the pathogens out and on strict "decontamination" procedures as for example incineration of infected material, machinery, planted material being destroyed and prohibitions to plant the crops until fields are found free of the pathogens.[7,8] In the USA, yield losses in individual crops due to diseased plants and tuber due to Cs have been as high as 50%. On the other hand, most commercial tomato plants are susceptible to *Rs* and the few resistant varieties that exist make very small tomatoes.[54] **POMATO** proposes a method for studying potato RGs against *Cs/Rs*, and for tomato against *Rs* based on the respective genetic resources. This will ensure a long-term sustainable and successful strategy for managing increasing *Cs/Rs* disease and cultivating more resistant potatoes and tomatoes. **POMATO** will thus offer a strategy in line with the EU Biodiversity Strategy 2030's primary action to enhance genetic variety on agricultural land, which would benefit EU health and economy. To achieve this impact, the project proposes the following pathways:	
Project Contribution to EO 1: *responses to new and/or emerging plant pests*	**Means of Verification**

Short term: Collection of all the available genetic *Cs/Rs*, and identification and characterisation of potato/tomato RGs against diseases caused by these bacteria. ***Medium term****:* Mapping of potato/tomato RGs against quarantine pests *Cs/Rs* in EU.	*Linked to results R1-R6 of the project:* • 100 isolated bacteria from infected areas with 80% effectiveness • Molecular monitor control: 12 metagenomic analysis performed in 100 isolated bacteria. • 200 potato samples (50 native and 20 wild varieties from Europe; 130 native and landrace varieties from LATAM) analysed for their tolerance level. • 200 tomato samples (50 native, 130 wild and 20 improved varieties from LATAM) analysed for their tolerance level. • Establishment and spread of pathogen: 10 non-supressive consortia identification and ~ 6 genetic, environmental and soil factors.

Project Contribution to EI 3: *delivering co-benefits for climate mitigation and adaptation*	**Means of Verification**
Long term*:* Prevent similar losses in potato/tomato production in the EU to those seen in other countries where these pests already exist.[9,10] Similar	At least 5% (of the above 200 samples of potato/tomato resistance

[54] Planas-Marquès et al. (2020). J. Exp Bot 71, 6, 2157-2171

molecular characterisation strategies could be replicated to other type of crops (pepper, eggplant, etc.).	varieties for targeted diseas and pests will reach the market.

with document Ref. Ares(2025)2863795 08/04/2025

ECONOMIC/TECHNOLOGICAL IMPACT **Long-term sustainable pest diseases integrated management, including prevention, early detection surveillance and natural control strategies**. *Corresponding to project´s objectives O2-O4. Impact Indicator*: Catalogue of innovative solutions to respond to quarantine potato/tomato pests. *Target groups*: TG1, TG3, TG5, TG8, TG9.	
Scale and Significance: **POMATO** is structured around the EU's agricultural research and innovation strategy, notably the set of initiatives aimed at "Priority 2: healthier plants".[55] As the EU strives to reduce the use of chemical pesticides,[56] natural biocontrol products are being more widely used in agricultural practices (e.g., biopesticides CAGR 20% vs synthetic pesticides CAGR 4.7% for the period 2018-2023).[57] **POMATO** will develop sustainable and systemic IPM strategies to address the prevention, monitoring, control, and management of *Cs/Rs*, as an alternative to the use of pesticides or other strict methods, in combination with other innovative agronomic and digital solutions. In this regard, the project is fully in accordance with the actions of the "Agriculture and the Green Deal", as it promotes the adoption of a long-term IPM by farmers to achieve a healthy and sustainable food system for people and the Earth.[58] To achieve this impact, the project proposed the following pathway:	
Project Contribution to EO 2: *to prevent, monitor and (bio)control plant pests*	**Means of Verification**
Short term: Integrate sustainable innovative solutions and validate them as an IPM strategy at TRL 5 in 4 different geographical areas (Poland, Spain, Ecuador and Colombia) based on their respective climate conditions and globalisation factors. ***Medium term***: laying the foundations for future development (TRL 5→7/8) of novel plant diseases strategies (e.g., natural products stimulating the expression of RGs).	*Linked to results R7-R18 of the project* • MAGs and GEMs towards plant's immune response against *Cs/Rs* and identify ~10 metabolites at early stages of infection. • At least 10% of the most relevant genetic signatures of the pathogens are identified related to disease progression. • >1,000 high-resolution multispectral images from healthy and infected crops. • >1,000 analysed infected plant images with AI-based models (80% weighted accuracy). • 10 active suppressive metabolites detected by polymeric sensor. • Decrease of 40% the incidence of *Cs/Rs* applying plant-derived organic substances, biocontrol coatings and chemoperception, and 60% when applying PGPM. • Early detection of diseases by the MDD detection device (90% reliability) and by polymeric sensors (95% accuracy). • Validation of **POMATO**'s IPM strategies in 4 CS at 2 LATAM and 2 EU countries in greenhouses and real infected field trials, including a complete study of the safety hazard assessment and safe by design study and environmental and economic analysis.

Project Contribution to EI 1: *resilient farming systems, minimising pressure on ecosystems, restoring and enhancing biodiversity, fair economic returns for farmers.*	**Means of Verification**
Long term: Exploitation and dissemination of a rigorous, sustainable, and economic IPM to control *Cs/Rs* for potato, tomato and other crops of the Solanaceae family. The project envisaged the collaboration with the JRC to develop reference reports in support of a harmonised IPM in the EU legislative acts and standards.[59]	The dissemination of EU guidelines and/or action plans rooted in **POMATO**'s IPM strategy for preventing potato and tomato diseases stands to significantly benefit crop farms across the EU, which constitute 58.2% of all farms.[1]

SOCIETAL IMPACT

Creation of an interactive innovation model to help the implementation of potato and tomato's IPM plans across different professional backgrounds at local, regional, national and international scale ('Multi-Actor Approach'). *Corresponding to project´s objective O5. Impact Indicator*: Practice abstracts of EIP-AGRI. *Target groups*: All TGs

[55] A strategic approach to EU agricultural research & innovation (2016). Final Paper. Available here. [56] Directive 2009/128/EC. Sustainable use of pesticides. Available here. [57] Frost & Sullivan (2019). Global Crop Protection Market, Forecast to 2023. [58] Integrated Pest Management in the EU Green Deal. Available here.
[59] JRC: Reference & Measurements. Available here.

Associated with document Ref. Ares(2025)2863795 - 08/04/2025

Scale and Significance: The EIP-AGRI policy goal of "speeding up" innovation will be met from the start of the project and in the long run thanks to the active participation of farmers and agri-food businesses through the key associations. The multi-actor participatory platform will include the scientific & technical community, companies (SMEs, large industries, agricultural/potato associations, and cooperatives), and citizens/consumers across the potato/tomato value chain to share knowledge and cooperate one with each other to maximize our results and outcomes. This platform will be critical for enhancing rural community engagement at all scales (local, national, and international) and for disseminating the project's proposed IPM strategies. **POMATO**'s impact will be amplified by close engagement with current EIP Operational Groups (OPs) and sister projects. To achieve this impact, the project proposed the following pathway:

Project Contribution to EO3: *Support to relevant EU and Associated Countries' plant health policies*	**Means of Verification**
Short term: Share knowledge, experience, and background due to the complementary disciplines of the multiple actors. ***Medium term***: acceptance and adoption of the **POMATO** solutions by farmers and other actors of the potato value chain production.	*Linked to results R19-R22 of the project:* Multi-actor participatory platform (4 years + 5 additional years after projects end). **25 "practice abstracts"** for EU CAP Network; 4 one-on-one workshops with policymakers; participation in government consultations, and 4 Thematic "Open Days"; 1 Policy Recommendation Report (M40) presented to the EC in a clustering event;1 **POMATO** IPM Plan (translated in at least 3 languages)

Project Contribution to EI 1: *resilient farming systems, minimising pressure on ecosystems, restoring and enhancing biodiversity, fair economic returns for farmers.*	**Means of Verification**
Long term: Implementation of potato/tomato IPM across all the actors and main EU producers will ensure economic sustainability of EU potato/tomato production, increase farm competitiveness and improve key actor's position in the value chain.	Farmers, potato/tomato producers and traders will be aware of the potential threats, and they are also well informed and trained in the strategies to carry in case of potential impact in their lands and production.

2.1.1 POMATO's Contribution to other EC priorities

POMATO is built up with a strong commitment towards the key policies objectives of the **new Common Agricultural Policy (CAP)**, **Farm-to-Fork strategy** and **SDG (2, 8, 13, 15, 17)** for the period 2023-2027, briefly: **1)** "Fostering knowledge & Innovations" → involving farmers to our research activities and outcomes**; 2)** "Halting & reversing biodiversity loss" → preserving more resistance potato/tomato varieties, **3)** "Efficient natural resource management" → developing a sustainable, effective and long-term management strategies to tackle outbreaks of pests; **4)** "Contributing to climate change mitigation" → boosting sustainable and bio-based circular solutions for plant health and reduce the use of pesticides; **5)** "Increasing competitiveness" → involving all relevant actors in the potato/tomato value chain. The project also will support the **Plant Health Law 2016/2031** and all the associated implementing acts related to pests, commodities and plant health requirements[11].

2.1.2 Requirements and potential barriers

The technological barriers facing **POMATO** lie in the potential for low demand of various natural and bio-control solutions among end-users in the potato/tomato industry, as well as within the bio-based agrochemical industry. This challenge has the potential to hinder the further development of the solution as it progresses to higher TRLs. On the

citizen side, a lack of engagement and adoption of IPM plan could undermine the application of **POMATO**'s outcomes. Potential economic externalities such as the fluctuation in demand and prices of crops may also impact the economic viability for farmers to adopt new pest management practices. To address these challenges, **POMATO**'s strategy is two-fold: First, project partners will collaborate with other funded EU projects, initiatives, and OPs of the EIP-AGRI to establish dedicated stakeholder forums. These forums will aim to mitigate potential barriers by showcasing the benefits of the developed strategies to a wide audience. Second, **POMATO** will provide welldocumented results, dissemination materials, and engagement activities to all stakeholders. This comprehensive approach will provide a solid foundation to garner support for our results. Regulatory standards play a pivotal role in the **POMATO** project. All our endeavours will meticulously adhere to the standards set forth by the EPPO, particularly those pertaining to pests, such as PM1/002(32) EPPO A2 Lists of pests recommended for regulation as quarantine pests (2023),[6] and the diagnostic standards PM 7/59 2 for *Cs* and PM 7/21 (3) for *Rs* species complex.[45] This commitment ensures alignment with established norms and prevents potential conflicts with existing standards. **POMATO** will prioritize compliance with the EU's regulatory framework, which significantly influences the production and commercialization of pest management solutions. Key components of this framework include the Plant Health Law 2016/2031,[11] essential for implementing IPM strategies to safeguard plant crops both during and

beyond the project's duration, and the Regulation 2019/2072[5] on protective measures against plant pests, alongside specific regulations targeting the eradication of *Cs/Rs*,[7,8] which will be closely observed to ensure comprehensive adherence to regulatory requirements.

2.2. Measures to maximise the impact

POMATO's Dissemination, Exploitation and Communication Plan (DEC) is crafted to spotlight the innovative features of the project, encompassing different tools and technologies, along with meticulously developed Integrated Pest Management system, to prevent, monitor and (bio)control plant pests.

2.2.1. Dissemination strategy

The **POMATO** Dissemination Strategy will provide a blueprint to guide partners in sharing knowledge and results with target groups for adoption and reuse. Using the AIDA communication model, dissemination activities will follow a multi-actor, multi-step approach to ensure **POMATO** results are effectively communicated. The table outlines tailored Dissemination Objectives, actions, channels, and KPIs for each target group (TG).

Table 2.2 Dissemination Objectives (DO) with corresponding actions/channels, KPIs and TGs

DO-1. Increase Awareness and Knowledge Sharing Among Growers: Raise awareness and share knowledge about innovative pest management strategies and tools developed by the **POMATO** project
Activity/channel: Engagement with growers in LATAM (**2**, one in Colombia and other one in Ecuador) and in Europe (**4**, Poland, Spain, Portugal and Brussels); Open days related to each pillar of **POMATO** IPM strategies in collaboration with agricultural advisors and extension services (**4**), IPM Guidelines for farmers (**2** translated in at least 3 languages); Practice Abstracts (**25**), Explanatory videos showcasing best practices and **POMATO** technologies (**≥3**); **POMATO** Multi-Actor Participatory Platform for information sharing. **Target groups:** TG1
DO-2. Foster Collaboration and Knowledge Exchange: Facilitate collaboration and knowledge exchange between research and academic institutions involved in plant health and pest management
Activity/channel: Present research findings and innovations at scientific conferences & symposia (**≥20** proceedings/posters) and webinars/online workshops (**≥15**), Publish articles in peer-reviewed journals (**≥10**). Establish a network of researchers and institutions for ongoing collaboration and information sharing (including sister EU projects) (**≥10** members outside of consortium). **Target groups:** TG 4,5,7
DO-3. Promote the adoption of sustainable pest management: Promote the adoption of sustainable pest management practices among retailers, wholesalers, and distributors to ensure the supply chain supports and enhances the quality and safety of potatoes and tomatoes, ultimately benefiting end consumers
Activity/channel: Informational brochures (**2**), video (**1**) and joint workshops/seminars (**4**) to discuss best practices and share success stories; Partner with retailers and wholesalers for pilot programs to test and validate **POMATO**'s innovations in real-world supply chains (**≥4** representatives engaged). **Target groups:** TG 2

[45] EPPO standards for *C. sepedonicus* here. *R. solanacearum* here.

DO-4. Engage Policy Makers and Regulatory Bodies: Support the implementation of effective plant health policies and regulations based on **POMATO**'s findings
Activity/channel: Policy briefs (**2** – translated in at least **3** languages), Position paper (**1** – translated in **3** at least languages) to highlighting key findings and recommendations; Policy roundtables and briefings with relevant EU and national regulatory bodies (**≥2** events); Participation of **POMATO** members in consultations and advisory panels to influence policy development (**≥3** events). **Target groups:** TG6&7
DO-5. Promote POMATO Innovation to Industry: Promote the adoption of natural and bio-control solutions and integrated pest management (IPM) strategies within the agricultural industry, including seed producers and manufacturers of plant protection products
Activity/channel: Collaborate with industry partners to scale up and validate bio-control solutions and IPM strategies (**≥5** industry representatives engaged in **POMATO** multi-actor activities); training sessions and workshops for industry stakeholders on implementing **POMATO** solutions (**≥6**). Case studies and success stories demonstrating the economic and environmental benefits of the proposed **POMATO** strategies (**≥10** publications in specialised agriculture media outlets; over **30** related blogs on the **POMATO** Participatory Platform). **Target groups:** TG3
DO-6. Raise Public Awareness and Consumer Engagement: Raise public awareness and engage consumers about the importance of sustainable pest management practices and the benefits of **POMATO**'s innovations for food safety and quality
Activity/channel: Educational content, including infographics (**≥2** translated in **3** languages), videos (**3**), Collaborate with at least **6** consumer advocacy groups to amplify the message and reach a broader audience**.** **Target groups:** TG 8 & 9

2.2.2. Exploitation strategy

POMATO will put in place an Exploitation Plan, which will include: *i)* assessment and categorisation of the key exploitable results (KERs) *ii)* assessment of exploitation strategies for **POMATO** results (scientific & educational, societal, policy and potential commercial) and roadmap for their implementation; *iii)* the joint and individual

exploitation plans; *iv)* assessment on how different multi-actor governance models have been implemented in the project pilots, and how these models could be utilised for the promotion of **POMATO's** KERs. Preliminary provisional list of KERs and end users are:

Table 2.3. Identification of KERs, Background Knowledge, Future Exploitation Intentions and Target Stakeholders

KERs, Background Knowledge, Future Exploitation Intentions and Target Stakeholders
KER1: AI Predictive Models for disease detection, spread & risk (Background: **FUND**, other partners involved in development: **WUR**, **IDEN**, **AGRO. Exploitation intentions:** (non)commercial (year of commercialisation: 1-2 years after project ends**). Target stakeholders:** TG1
KER2: Multiparametric Disease Detection sensing device (Background: IRIS)**. Exploitation intentions:** commercial (year of commercialisation: within 3 years after project ends)**. Target stakeholders:** TG1
KER3: Mathematical modelling tools (Background: **IDEN). Exploitation intentions:** commercial**. Target stakeholders:** TG5
KER4: Polymeric sensors (Background: **UBU). Exploitation intentions:** commercial**. Target stakeholders:** TG1
KER5: AI-driven Cloud data management platform (Background: **IRIS). Exploitation intentions:** commercial **(**year of commercialisation: within 1 years after project ends)**. Target stakeholders:** TG1
KER6: Natural and bio-control solutions, Exploitation intentions: commercial**, Targeted stakeholders:** TG1 ***KER6.1*** *Plant-derived organic substances* ***(****Background:* ***FUND)*** ***KER6.2*** *Biocontrol coatings* ***(****Background:* ***FUND)*** ***KER6.3*** *Plant growth-promoting microorganisms (PGPMs)* ***(****Background:* ***JSI)*** ***KER6.4*** *Chemoperception disruptors.* ***(****Background:* ***CBGP),*** **KER7: Omics marker of infection (WUR), Exploitation intentions:** commercial (year of commercialisation: within 1 years after project ends), **Targeted stakeholders:** TG1
KER8: POMATO's Integrated Pest Management (IPM) plan, guidelines and policy recommendations (Background: **joint). Exploitation intentions:** non-commercial**. Target stakeholders:** TG1, TG3, TG4, TG6, TG7.
KER9: POMATO Multi-Actor Participatory Platform (INO). Exploitation intentions: non-commercial**. Target stakeholders: TG1-9**

*2.2.2.1. Main **POMATO**'S Exploitation Routes*

Scientific exploitation – Academia will use **POMATO** results to establish new R&I projects building upon the project achievements. The scientists and developers releasing new or updated models under opensource modality will leverage open-source communities, widening their visibility. The **POMATO** results will potentially lead to spinoff research both by project partners and by other stakeholders due to the multiplier effect of the pilot and networking activities. **POMATO** will further high-profile research beyond the project through: (a) training a new generation of researchers in AI predictive and mathematical modelling, polymeric sensors, bio-control solutions and integrated pest management (UBU, FUND, USFQ, CBGP); (b) publishing its innovative pest management methods and scientific results in high-impact journals and conferences; (c) exploring opportunities for using **POMATO** results for educational purposes (e.g. launching new academic programs); (d) liaison with entities including FAO, WTOSPS Committee, IPPC, OECD-Agriculture and Fisheries, and its Flagships and Initiatives (T6.3).

Policy related exploitation – **POMATO** results will be developed to create long lasting impacts on the farm to fork strategy and transition to fair, healthy and environmentally-friendly food systems as well as to support EU Regulation on protective measures against pests of plants. Thus, with the active participation of EUROPATAT, UBU, PIN and JRS, **POMATO** will invest significant efforts to influence the targeted policy processes on multiple levels by supporting regional, national or international (e.g. DG SANTE, DG AGRI, FAO, IPPC) reports, policy instruments such as decrees and norms, and policy implementations in programs and projects. To orchestrate the partners' efforts, the **POMATO** activities related to policy and scientific outreach (T5.5) will be planned and executed to bring together and highlight the knowledge created in the WPs. Different forms of outputs of **POMATO** project results (DO's 2 & 4) are envisioned to lead to policy influence and impact, such as: high-quality scientific writing; highquality scientific presentation (oral/keynote at international scientific and policy conferences, workshops or events); training on project outputs (workshops, hands-on sessions, but also videos and other digital artefacts available in the **POMATO** Participatory Platform); publicly disseminated recommendations, guidelines and best practices; written statements on the (potential) policy relevance of project results (policy briefs).

Societal exploitation - Community outreach and educational programs will be developed to raise awareness and engage the public in pest management practices. As such, **POMATO** outputs will include tools, solutions and methods that allow a wide range of stakeholders to actively engage in supporting implementation Farm to Fork

strategy in general and identifying and managing *Cs/Rs* pests in particular. Combined with outreach campaigns that highlight the importance of sustainable pest management for food security and environmental health, collaboration with agricultural services to implement best practices and enhance community resilience against pest outbreaks. **Commercial exploitation** - KERs perceived as having potential for commercial exploitation at this stage are Multiparametric Disease Detection sensing (KER2), Polymeric Sensors (KER4) and four Natural and bio-control solutions (KER6). The partners owning IPR over these solutions will consider investment in their post-project development to reach full market readiness level at TRL 8-9 or expand to other markets. This commercial initiative is envisioned to be examined on an individual basis in cases where IPR belongs to only one partners or bilaterally during cases of joint results during the project course (clarified in the Consortium Agreement). In the latter case partners will consider creating a pool of resources (e.g., monetary, personnel) to cover the activities needed for market penetration, if clear market opportunity and demand are identified during the project lifetime. This will permit sharing risks and costs related to exploitation of results and concentrate efforts on reaching target markets. Detailed market conditions for the commercial exploitation will be identified during the project development phase in D6.4 (M48).

2.2.3. Communication Strategy

POMATO's communication strategy will effectively employ content marketing, social media, and events to reach out to society and show the impact and benefits of the project. Additionally, the **POMATO** Multi-Actor Participatory Platform, in the scope of the website, will be a collaborative space, enabling stakeholders to share best practices, access, and exchange resources.

Branding & Visual Identity	**TGs**: 1-9	M1 to M48
Establish a uniform **POMATO** brand and tone across all media channels, respecting gender dimension.		
Target Values: Printable brand book **1**; **1** Promo set – available as a e-material (Roll-up, Poster, Stickers, Bags, Notebooks, Folders, Pens, etc.); **1** fact sheet per each pilot site; **3** Brochures on solutions, sustainability commitments, technology benefits. No. of hard copy promo material distributed **≥ 1000**		
Website & Participatory Platform	**TGs:** 1-9	M01 to M48
Website (including **POMATO** Multi-Actor Participatory Platform) allowing easy retrieval of main action data while acting as the core source of information about the **POMATO** objectives & activities. The first version of the website will be online by M01, while the Participatory Platform by M18.		
Target Values: Unique visitors from M12: **>2.000**; Unique visitors from M36: **>10.000**.		
Social media	**TGs**: 1-9	M01 to M48
Online presence in social media channels (LinkedIn, X, YouTube) by spreading tailored content to interact with different target groups.		
Target Values: Size of online community (by M36): **>1.000**; No. of impressions (monthly average): **>500;** No. of all project related videos on Youtube **≥ 20**		
Online publishing	**TGs**: 7, 8, 9	M1 to M48
Leveraging online press release publishing platforms (e.g. Eurekalert, AlphaGalileo), magazines that publishes popular-audience articles by academics (e.g. Agrinnovation magazine, The Conversation), as well as CORDIS Research EU Magazine will enable to reach out to a broader audience. Partners will disclose non-confidential information of the project in their language to local media. Pilot partners will lead the stakeholder's engagement in their countries.		
Target Values: Press releases (**≥4** translated in **3** languages), Media engagements at pilot sites (at least **3 media coverage per pilot site**); Blog articles (**≥40**) for dissemination through various media channels.		
Multimedia	**TGs**: 7, 8, 9	M6 to M48
Producing clear presentation of project scope, results and funding. A set of interviews from project experts and other partners, utilising plenary meetings & events.		
Target Values: Short videos **≥4**; Number of interviews produced **≥10**; Number of views >**2.000**		
Public event management & community relations	**TGs**: 7, 8, 9	M6 to M48
Opening pathways for communication with broader audience through project events and providing the information about **POMATO** innovation and solutions.		
Target Values: (Online) events for broader audience (including events at demo site and events co-organised with local actors): **>10**; Visitors per event: **>50**; Co-organisation of clustering events (**≥3**), Project final conference **1**		

POMATO will regularly assess the effectiveness of communications and adapt the strategies by referencing established benchmarks and KPIs, through Google Analytics and social media dashboards.

Dissemination and communication activities after the end of the project: Post-project dissemination and communication efforts will focus on maintaining the website and **POMATO** Multi-Actor Participatory Platform for four years, archiving newsletters, and periodically updating social media to inform stakeholders, such as farmers and policymakers, about ongoing activities. Partner contact information will remain accessible for continued collaboration and support. We will share practice abstracts, policy recommendations, and audio-visual content on our platform and others, like the Horizon Results Platform, adhering to Open Science practices. Open-source tools and datasets will be available in an open-access repository, with defined usage terms, while collaboration with relevant initiatives will enhance outreach and ensure the sustained impact of the project's contributions.

2.2.4. Feedback to policy measures

Key policies and legislative relevant to the **POMATO** project include the Farm to Fork Strategy, the Regulation on Protective Measures Against Pests of Plants (Regulation (EU) 2016/2031), EU Plant Health Legislation, and the Council Directive 2000/29/EC on protective measures against the introduction and spread of harmful organisms within the Community. Throughout the project and within the framework of T5.5 IPM Guidelines for farmers and Policy Recommendations, **POMATO** will focus on understanding these regulatory landscapes and identifying challenges related to plant pest management. This will be accomplished through a phased approach: **Phase 1** (M0112) will involve mapping out regulatory requirements and safety protocols for combating plant pests, including compliance with existing EU regulations and directives. **Phase 2** (M12-36) will leverage insights from project activities to draft and refine EU policy recommendations, ensuring alignment with the Farm to Fork Strategy and other relevant legislative frameworks. **Phase 3** (M36-48) will involve organizing stakeholder roundtables and workshops to validate and enhance these recommendations. Our partners, affiliated with various policy and industry bodies, will actively participate in policy networks and working groups, such as the Standing Committee on Plants, Animals, Food, and Feed, industry meeting of Potato Value Chain, EAPR's (European Association for Potato Research) Triennial Conference, the EUCAP Network, etc., to ensure that **POMATO** results inform and shape future policy measures effectively. Through these efforts, **POMATO** aims to provide robust, evidence-based recommendations that support the EU's goal of sustainable plant health management.

2.2.5. Intellectual Property (IPR) management

POMATO will manage all IP rights under T6.5 Exploitation and IPR Management, distinguishing between two key types of knowledge: *i)* Background knowledge, where consortium members will grant access to each other's background knowledge based on agreements made prior to signing the Grant Agreement (GA), and *ii)* Foreground knowledge, referring to new knowledge developed during the project, which will be protected, owned, and funded by the partner(s) responsible for its creation. IP-related risks, challenges, and opportunities will be addressed in the Consortium Agreement (CA), following EU recommendations on IP and knowledge transfer. **POMATO** will comply with Horizon Europe's open access mandate, publishing on Open Research Europe and making materials available on its website and OpenAIRE. To exploit project results, access rights will be agreed upon before signing the GA and granted either under fair and reasonable conditions or royalty-free. Confidential information will not be shared with third parties without explicit authorization. Ownership of foreground IP will belong to the generating partner, with joint ownership rules established in the CA. In the event of IP-related conflicts, partners will resolve issues following the CA guidelines with the support of the Coordinator's IP Lawyer. To foster innovation, the consortium has already outlined potential KERs and partner-specific IP, forming the basis for discussions during T6.5. INO will also conduct three IP Rights Workshops for consortium partners during the project's lifespan.

2.3. Impact summary

Specific Needs	Expected Results (at the end of the project)	D & E & C measures	Target Groups
The growing threat of ***Clavibacter sepedonicus*** (Cs) and ***Ralstonia solanacearum*** (*Rs*) outbreaks, classified as **EPPO A1 quarantine pests** and subjected to EU regulations aimed at eradicating and preventing their spread (Regulation 2019/2072), which severely **affect potato and tomato crops**, demands urgent action. Current fossil-based disease management practices are unsustainable, and **the spread of these quarantine pests, exacerbated by climate change**, poses a significant risk to EU agriculture. There is a need for **rapid-response**, and **cuttingedge digital technologies** for pest **early detection**, **surveillance**, **control**, and **eradication**.	**POMATO** develops a **I-matrix tool** for the specific isolation of *Rs* from environmental samples, with at least 80% effectiveness and it fully identifies at molecular level the of **resistance genes (RGs)** from over **200 potato and tomato varieties** (both wild and cultivated) from EU and LATAM's germplasm culture collections. It sequences at least **12 pathogen genomes and 12 transcriptomic analyses** to understand **pathogen diversity and virulence, while** identifying at least **6 genetic**, **environmental** and **soil** factor that determine **pathogen's spread** and **establishment** (**R1-R6**). **POMATOs** delivers **metagenomic** and **AI-driven detection tools** for early diagnosis and disease surveillance, including **polymeric sensors** that detect key infection metabolites with **90% reliability**, and **AI models** trained on **1000+ images** for **80% accurate disease prediction** (**R7-R11**). **POMATO's** natural and bio-control solutions **reduces the pathogen's incidence by 40-60%,** including plant-derived substances, chemoperception disruptive products and PGPMs (**R12-R13**). Finally, POMATO's main goal is to validate the Integrated Pest Management (IPM) Plan in 4 case studies (EU and LATAM) at TRL 5 to control the outbreaks of these bacteria (**R14-R17**), providing the foundations and drivers for future implementation of the IPMs by farmers and other target groups through a dedicated multi-actor participatory platform and the C&D&E measures (**R18-R21**)	**Dissemination & Exploitation Measures**: - >10 peer-reviewed publications and >10 KER (Key Exploitable Results). - Organization of >4 thematic Open Days and 3 workshops with field trials across Europe and LATAM. - Creation of 3 videos, 2 infographics, and >15 newsletters, targeting farmers, researchers, and policymakers. **Communication Measures**: - A multi-actor participatory platform (MAPP) for knowledge sharing, cocreation, and networking with at least 25 practice abstracts contributed to the European CAP Network. - Policy briefs and position papers shared with EU regulatory bodies. - Engagement with >800 stakeholders including farmers, industry, and academia	**TG1-TG9**: **1**) Potato/tomato farmers and growers (organic and conventional), cooperatives, and agricultural advisors across the EU and LATAM. **2**) Retailers and distributors (e.g., Carrefour, Lidl) responsible for delivering sustainable produce. **3**) Seed producers and bio-based plant health service providers. **4**) Research and technology organizations (RTOs), universities, and AI/sensor tech companies. **5**) Policymakers, regulatory bodies (e.g., DG AGRI, EPPO), and national plant health agencies. **6**) NGOs and advocacy groups focused on biodiversity and sustainable agriculture. **7**) End consumers of potatoes/tomatoes, particularly those interested in sustainably grown produce
OUTCOMES		**IMPACTS OF POMATO**	

OC1: New resistance genes identified for *Cs* and *Rs*, improving the genetic defence of potato and tomato crops. **OC2**: Advanced AI-driven early detection and monitoring technologies, including polymeric sensors and disease prediction models with 90%+ accuracy. **OC3**: Natural and bio-control solutions for *Cs/Rs* reduction validated in real-field conditions, reducing pest incidence by 40-60%. **OC4**: Validated IPM strategies in real case studies ready for replication in other Solanaceae crops, ensuring long-term pest control. **OC5**: Engagement and adoption of **POMATO** tools across 4 case studies (LATAM and EU), fostering knowledge exchange and sustainable farming practices	**Scientific Impact**: Establishment of molecular and biological data on potato/tomato resistance genes and quarantine pests, leading to improved disease management practices and enhanced biodiversity. **Economic Impact**: Adoption of long-term IPM strategies, reducing reliance on chemical pesticides and improving farmers' economic returns. **Technological Impact**: Development of innovative AI and sensor technologies for early detection and biocontrol of pests. **Societal Impact**: Increased awareness and knowledge sharing among farmers and other stakeholders, leading to more resilient farming systems, reduced environmental pressure, and better food safety for consumers

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3. Quality and efficiency of the implementation

3.1 Work plan and resources

WP	POMATO Work Plan	Leader	YEAR 1	YEAR 2	YEAR 3	YEAR 4
WP1	Project Management	UBU				
T1.1	Scientific and technical coordination	UBU	D 1.1			
T1.2	Administrative and financial management	UBU				
T1.3	Research data management	UBU	D 1.2a		D 1.2b	D 1.2c
T1.4	Risk management	UBU	D 1.3a		D 1.3b	D 1.3c
WP2	Characterization of pathogens, resistance sources and predisposing factors	AGRO				
T2.1	Pathogen Characterization	AGRO			D 2.1 D2.2	
T2.2	Compilation of resistance sources against Cs/Rs	AGRO				D 2.3
T2.3	Linking pathogen–microbiome–host interactions	JSI				D 2.4
T2.4	Compilation of molecular data into a common platform	IDEN			D 2.4a	D 2.5b
WP3	Early Detection and Surveillance Tools	IRIS				
T3.1	Identification of biotic and abiotic factors affecting bacterial disease establishment and spread	IDEN				D 3.1 D 3.2
T3.2	Advanced AI-based models for robust disease detection and climate-driven risk assessment	FUND				D 3.3
T3.3	Smart polymers for straightforward and early disease detection	UBU				D 3.4
T3.4	Multiparametric Disease Detection (MDD) sensing device	IRIS				D 3.5
T3.5	AI powered cloud data management for disease prediction	IRIS				D 3.6 D 3.7
WP4	Natural and Bio-Control solutions	FUND				
T4.1	Biocontrol coating solutions	FUND				D 4.1
T4.2	Plant derived organic substances	FUND				D 4.2
T4.3	Plant growth-promoting microorganisms	JSI				D 4.3
T4.4	Chemoperception disruptive products	CBGP				D 4.4
T4.5	Scale up process of the solutions at pilot plant (TRL5)	CHEM				D 4.5
WP5	Integrated Pest Management Strategies validated in field trials, SSbD strategies and Policy Recommendations	UBU				
T5.1	Validation field trials in Colombia and Ecuador in potato crops: Early Detection & Bio-control solutions (CS#1)	USFQ				D 5.1
T5.2	Validation field trials in Colombia in tomato crops: Early Detection & Bio-control solutions (CS#2)	AGRO				D 5.2
T5.3	Validation field trials in Europe in potato and tomato crops: Early Detection & Bio-control solutions (CS#3, #4)	F4S				D 5.3 D 5.4
T5.4	SSbD Framework and safety hazard application to natural and bio-control solutions	UBU		D 5.5		D 5.6 D 5.7
T5.5	IPM Guidelines for farmers and policy recommendations	EUPAT			D 5.8a	D 5.8b
WP6	Dissemination, Communication, Exploitation and Multi-Actor Engagement	INO				
T6.1	Dissemination, Exploitation and Communication Plan	INO	D 6.1			
T6.2	Public outreach and case study specific promotional campaigns	F4S				
T6.3	POMATO Multi-Actor Participatory Platform	INO	D 6.2a			D 6.2b
T6.4	Multi-Actor approach framework and activities	F4S				
T6.5	Clustering, synergies, and interlinkages with the European R&D community	UBU	D 6.4a	D 6.3a	D 6.4b	D 6.2b D 6.4a
T6.6	Exploitation and IPR Management	INO		D 6.5a		D 6.5b

Deliverable Milestone

POMATO work plan will be implemented in 48 months and it is divided into 6 WPs aimed to fulfil the general requirements set out by the target HEU's topic and promoting the interaction between all participants (Fig. 3.1). In the first part of **POMATO**, the characterization of the pathogens *Cs/Rs*, RGs of potato and tomate and factors affecting the infection will be explored at molecular level (**WP2**). Then, **POMATO** will develop a baterry of integrated cutting-edge digital technologies that will enable the early detection and surveillance of the pathogens which will be finally integrated in a Mulparametric Disease Detection sensing device (**WP3**). In parallel, the project will develop and scale-up 4 pathogen prevention strategies based on sustainble and effective plant natural control solutions (**WP4**). The outcomes of the previous WPs will be integrated and validated in **WP5** where the 4 Case Studies will be carried out in both greenhouses and real infected fields (at TRL5) in Colombia, Ecuador, Poland and Portugal, anchored by the EU SSbD framework. Dedicated dissemination, communication, multi-actor

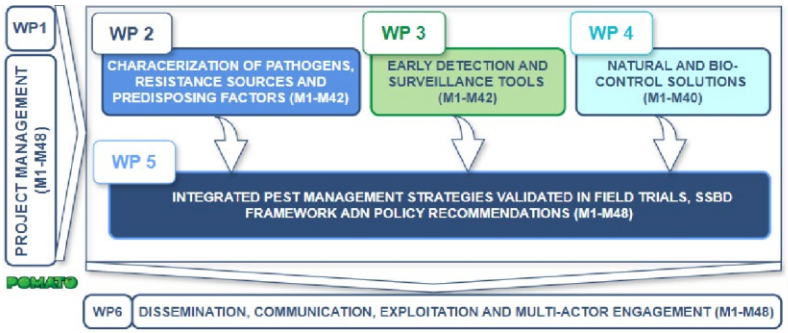

Figure 3.1 POMATO Work Plan

engagement and exploitation activities will secure the maximum impact of the project (**WP6**). The technical WPs will be supported by **WP 1** of project's management.

Table 3.1h: Purchase costs items (travel and subsistence, equipment, and other goods, works and services) *Travel and subsistence: T&S; Equipment: E; Other goods, works and services: OGWS*

3.2 Capacity of participants and consortium as a whole

POMATO consortia is formed by **16 partners** from **8 European Countries** and **2 Latin American** (**AGRO and USFQ**). These partners bring the knowledge and experience for the implementation of **POMATO** bringing together a multidisciplinary with strong competences in the field of plant pest management. In addition, the presence of the

Experts in Cs/Rs and Resistance Genes

IHAR (Poland): IHAR (Plant Breeding and Acclimatization Institute)[61] is one of the largest agricultural

EEAB will provide independent opinions on the research outputs and propose new actions to improve the quality and applicability of project results.

research institutes in Poland with the multi-disciplinary area of crop improvement, biotechnology, germplasm conservation and enhancement. It was founded in 1951 for research in breeding and seed production of major field crops. It currently employs over 300 people (including 1/3 of scientists) and consists of 17 R&D locations, including 6 research centers, 6 experimental stations and 5 plant breeding companies located in different parts of Poland. The group of phytopathology involved in **POMATO** has extensive, specialized research and scientific experience in the field of diagnostics, materials engineering, nano- and biotechnology regarding potato pathogens, including the quarantine ones.

Added value to POMATO: IHAR has quarantine plant pathogens resources as well Potato Gene Bank a source of collection of EU varieties and wild species of potato. Its laboratories allow for the performance of diagnostics needed to prepare for research and identify the pathogens tested under WP2 and WP5. In addition, the team has the appropriate conditions to develop a I-matrix, a tool for the isolation of *Rs* bacteria from environmental samples (WP2, ST 2.1.1). IHAR has the appropriate facilities (equipment and phytotron chambers) to perform tests within WP2 and a quarantine experimental field of 400 m2 allowing for the performance of validation tests with Cs on potatoes in WP5. **Role:** Provide Cs strains and potato varieties for GR studies, developing I-matrix for bacteria isolation **(WP2)** and validation for Cs in potato fields in **WP5 (CS#4 Leader)**.

AGRO (Colombia): AGROSAVIA, the Colombian Agricultural Research Corporation, founded in 1993, focuses on agricultural research, environmental sustainability, and food security. With 23 locations, including 13 research centers, AGROSAVIA operates across Colombia's five natural regions. It employs over 380 researchers and 360 support staff, publishing over 600 research papers annually.

Added value to POMATO: Dr. Mauricio Soto-Suárez's laboratory specializes in plant pathology, studying pathogen and host interactions to develop sustainable disease management strategies. The team, experts in various fields, has published over 85 scientific articles in the last decade. **Role:** AGRO's role includes genomic and transcriptomic characterization of potato and tomato resistance (**WP2 Leader**). Their advanced facilities feature high-throughput sequencing, gene expression analysis, and controlled growth chambers. The Tibaitatá Research Center has a quarantine greenhouse and laboratory for biosafe experiments and field trials with local farmers (**CS#2 Leader, WP5**)

USFQ (Ecuador): USFQ, a leading research university in Quito operates 3 research stations in the Amazon rainforest, Galapagos Islands, and Andean region. For 15 years, their lab has focused on pathogen identification, plant responses to stress, and plant-microbe interactions, working with crops like potato, tomato, banana, broccoli, and maize. The team includes a fulltime professor, a junior researcher, a lab technician, two PhD students, three master's students, and five undergrad students. USFQ have published 38 scientific papers with a total of 5445 citations, showcasing the impact of our research.

[61] Link to IHAR Institute.

Added value to POMATO: collection of native and commercial potato varieties, essential for studying genetic

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resistance to various stresses. USFQ facilities include greenhouses and fields for pathogenicity assays, a molecular lab for pathogen identification, and a microbiology lab for pathogen multiplication and morphological descriptions. **Role**: Provide potato varieties for GR studies **(WP2). CS#1 Leader** in connection with local potato farmers (**WP5**).

CBGP (Spain): The "Phytopathogenic Bacteria"[46] group from CBGP, a center of excellence Severo Ochoa, belongs to Polytechnical University of Madrid and it is a leading research group in plant pathology and biotechnology, with a focus on understanding and combating bacterial diseases in
plants. The group has a long and contrasted experience in molecular biology and phyto-bacteriology techniques and works with different models of phytopathogenic bacteria that affect tomato and potato crops. The group has a highly qualified and multidisciplinary team of researchers and technicians and has published several relevant papers in the

Added value to POMATO: All the tasks carried out by CBGP will be carried out in their own facilities, with the specific requirements for the research objectives. CBGP has P1 and P2 facilities for GMOs and a P3 laboratory for quarantine pathogens. The research group is authorized to use all the CBGP facilities. The research team involved in **POMATO** is made up of highly qualified members in the field of Agrobiotechnology with high experience in the approaches to be developed in **POMATO**: Prof. Emilia López Solanilla, Prof. José Juan Rodríguez Herva, Prof. Gema López Torrejón, Prof. Saray Santamaría Hernando. **Role:** Natural products to disrupt bacterial chemoperception **(T4.4 Leader).**

Experts in Computational analysis, IA predictive models, and digital tools

IRIS (Spain): IRIS Technology Solutions, S.L. is an advanced engineering company specializing in

last 10 years.

real-time process monitoring and Digital Analytical Platforms. It excels in Photonics and AI, offering solutions in NIR technology, Raman spectroscopy, hyperspectral technology, and machine vision, crucial for Industry 4.0 innovation.

Added value to POMATO: IRIS has developed SWIR, hyperspectral imaging, and Raman devices, alongside predictive software (Scadalytics), applicable in sectors such as food, pharmaceuticals, chemicals, plastics, and wood industries. Their solutions enhance industry intelligence, sustainability, efficiency, process improvement, quality, and resource efficiency while reducing waste and preventing off-spec products. **Role: WP 3 Leader,** Developer of the cutting-edge Portable Disease Detection Multiparametric Device

IDEN (Spain): IDENER Research and Development is a RTO composed of a team of

engineers, scientists and software developers exploring the potential of AI for solving challenging science and engineering problems. To do so, IDEN is using an extended interpretation of AI which fuses machine learning, systems modelling, advanced control and mathematical optimization.

[46] Link to the "Phytopathogenic Bacteria" group

Added value to POMATO: Along many different European initiatives, IDEN has developed a strong background in the development and application of computational tools to biotechnology processes, including machine learning, process modelling and assessment and optimization tools. **Role:** Big-data analysis; ML techniques; GEM models for studying microbial interactions in **WP 2** and **3**.

WUR (Netherlands): The Laboratory of Systems and Synthetic Biology (SSB)[47] at WUR is a leading group in developing and applying computational and experimental methods to study microbes and their interactions with the environment and hosts. The group has expertise in metabolic modelling, omics data analysis and integration, synthetic biology, data mining, and FAIR data management.

Added value to POMATO: The group brings to the **POMATO** project the ability to analyse AI predictions with knowledge-based approaches and to associate biomolecular data with disease symptoms. The group consists of 31 members, including four senior researchers who are directly involved in the project: Prof. Dr. María Suarez Diez, Dr. Edoardo Saccenti, Dr. Jasper Koehorst, and Dr. Cristina Furlan.

Role: Computational analysis methods of biological data; AI predictive models of disease progression and interpretation (participation in **T2.4** and **WP 3**)

FUND (Spain): FUND is an RTO with expertise in Agrifood and Digital Transformation. FUND will participate with a team of three researchers with experience in edible coatings, natural and biocontrol solutions, and AI models for image analysis and disease prediction. FUND will coordinate WP4 on natural and bio-control coating solutions and contribute to WP3 on AI-based early detection and risk assessment. FUND has participated in 13 EU projects and published several scientific articles and patents.

Added value to POMATO: FUND coordinates the project PATAFEST related to IPM strategies to control the pest in the potato plant caused by bacteria zebra chip (ZC) and other threats during the postharvest activities. The researchers working in Patafest will be directly involved in the project: Dr. Christian Ghidelli, Dr. Belinda Pérez Bibbins and Álvaro Huertas. **Role:** AI for image analysis, AI models for pest spread in tomato and potato (**T3.X**) and plant-derived organic substances and bio-control coating (**WP 4 Leader)**

UBU (Spain): The University of Burgos features a versatile research team composed of the Polymers Group and the ICCRAM-EST Research Group. This team is skilled in developing innovative strategies for early plant pest detection using polymeric sensory materials and applying the SSbD framework for safety assessments of biocontrol solutions.

Associated with document Ref. Ares(2025)2863795 - 08/04/2025

Added value to POMATO: The Polymers Group specializes in detecting molecules using polymeric sensors, focusing on designing, developing, and characterizing novel sensory materials for environmental applications. Their expertise includes creating smart materials that detect target molecules with visible changes, such as color or fluorescence, and developing accessible detection methods using a colorimetric display and smartphone app. ICCRAM complements these capabilities with its focus on Environmental Biotechnology, Sustainability, and Toxicology. **Role:** UBU, with extensive experience coordinating EU projects, will serve as the **coordinator** of **POMATO**, having led 41 projects and participated in 99 across various EU programs Having participated in 35 EU projects, The Polymers Group is **leader of WP 5**, coordinating all the activities performed in each CSs and is also leading the development of the polymeric sensors (**T3.3**). ICCRAM excels in toxicity assessments and sustainability evaluations. Their work includes life cycle assessments and applying the SSbD framework for material safety and sustainability (**T5.4**)

JSI (Slovenia): Institute Jožef Stefan is the largest public institute in Slovenia performing research in biology, chemistry and physics and is composed of several departments. The Department of Environmental Sciences is one of the largest departments hosting predominantly chemists and molecular biologists and is the department where the Laboratory for Colloid Biology is located and where studies of microbial communities in different environments are carried out. The research group is also dealing with ecological and physiological aspects of prokaryotic microorganisms in relation to other prokaryotes or organisms like plants.

Added value to POMATO: The group for colloid biology specializes in understanding the ecological roles and functioning of bacterial communities. In the project the team will be characterizing the microbiome that show protective properties against potato and tomato pathogens. Additionally, the team specialized in obtaining the established functionally interconnected bacterial consortia from environmental samples, which can be applied in different ways to the seeds, rhizosphere or the soil to enhance plant protection. **Role:** Characterization of natural

[47] Link to the The Laboratory of Systems and Synthetic Biology (SSB)

plant-related microbiota, preparation of formulations of bacterial consortia with plant protective characteristics. Development of artificial consortia to be scaled up and applied in plant expermients in laboratory and field stage.
CHEM (Italy): CHEMIA is an Italian company that specializes in fine chemicals and agro-chemistry products. It has a history of innovation and environmental awareness, with 11 patents on biopesticides and other formulations. Chemia has a farm, an R&D laboratory, and a control quality laboratory, where it can produce, develop, and test its products. Chemia owns all its infrastructures and does not need external
access for the project activities.
Added value to POMATO: Formulation of the solutions and capacity to scale-up the products for the greenhouses and field validations following highest standard of quality, safety and respect for the environment. **Role**: Scale-up process at pre-industrial scale (TRL 5) (Participation in **WP4 & 5**).
FERT (Serbia): FERTICO, an expert in industrial development of crop fertilizers, will be responsible for manufacturing solutions developed at the pre-industrial (pilot plant) scale. Their areas of expertise are microbiology,

Added value to POMATO: Four registered microbiological products for seed treatment available on the market: SPEED for SEED GRAIN, CORN, SOYA, and SUNFLOWER.[48] **Role**: Formulation of the solutions developed by JSI and capacity to scale-up the products (TRL5) for the greenhouses and field validations following highest standard of quality, safety and respect for the environment. (Participation in **WP4 & 5**).

Multi-actor engagement, including direct contact with farmers and policy makers

INO (Serbia): INO is a consultancy specialized in impact maximization and business services

phytopathology, plant protection and chemistry.

INOSENS

for diverse European value chains, such as agriculture and food production, Earth observation, and water management.

Added value to POMATO: Vladimir Mrkajic, with over 10 years of experience in implementing EU projects, will occupy the role of **POMATO** Dissemination & Exploitation Manager. Maja Sremacki (PhD), with over 5 years of experience in communicating scientific research, will occupy the role of **POMATO** Communication Manager**. Role**: INO will lead **WP6** and optimize dissemination activities to ensure the sustainability of project results, as well as deliver a wide array of communication activities across different target groups using appropriate channels. INO will also lead the project exploitation, managing IPR, and ultimately supporting the increased uptake of project results.

PIN (Poland): The Polish Seed Trade Association (PIN) represents 83 affiliates, covering 80% of

Associated with document
Ref. Ares(2025)2863795 - 08/04/2025

PIN

Poland's seed market. Its Potato Section controls over 90% of the country's potato seed market and focuses on breeding, seed production, and phytosanitary safety, particularly combating *Clavibacter*.
PIN has led efforts to eradicate *Clavibacter* in Poland, collaborating with national authorities, European organizations, and the potato sector.

Added value to POMATO: Extensive network and influence within the Polish potato seed industry and policy makers. **Role:** PIN will facilitate effective communication and collaboration among stakeholders in the potato value chain. Its strong ties with national authorities, local organizations, and European bodies will help promote project outcomes, ensuring that key players are engaged and that the project's innovations reach the necessary audiences (**WP6**).

Europatat

EUPAT (Belgium): Europatat is the European Association of Potato Traders (including breeders, distributors, storers, packers, importers and exporters) delivering seed potatoes to farmers, as well as raw material to the food industry, packed potatoes to the retailers and food service sector. In total EUPAT has 69 members from over

[48] Link to the certified products of FERTICO

Added value to POMATO: Recent Eurostat production figures, incorporating the EU 27, UK, Norway, and Switzerland, show that the 19 national members of EUPAT account for over 80% of Europe's total potato output, highlighting the extensive reach the project has to key players in the potato sector. **Role:** facilitate the collection of data and information towards all the members, influence policy decision-makers about the project's outcomes and promote networking and communication activities (**T5.5** and **WP6**).

F4S (Portugal): Associação para a Inovação no Alimento Sustentável (Food4Sustainability, F4S)

20 countries. It represents more than 80% of the European potato production and works with different sectors of the potato value chain, EU institutions and research centres.

focuses on sustainable food production and consumption. Their mission is to achieve carbon neutrality and reduce environmental impact through innovative agricultural technologies, education, and partnerships. F4S specializes in Circular Economy, Sustainable Agriculture, Territory, Functional Nutrition, and Entrepreneurship & Innovation, supported by a team of 30 R&D experts.

Added value to POMATO: F4S offers specialized training, workshops, and webinars on sustainable agriculture. They have organized 50+ webinars and co-hosted the Soil-Microbiology Mini-conference, engaging stakeholders in discussions on soil health and biocontrol. **Role**: In **POMATO**, F4S will be involved in knowledge transfer and multiactor activities (**WP6**), and in remote sensing image collection of healthy/infected tomato/potato fields in Portugal and Poland (**WP3**). F4S will be in close contact with Direção-Geral de Alimentação e Veterinária (DGAV), the Portuguese authority responsible for plant protection, plant health policies, phytosanitary measures, as well as the agriculture notices service of the Portuguese Ministry of Agriculture. **CS#3 Leader (WP5)**.

4. Ethics self-assessment

Ethical dimension of the objectives, methodology and likely impact

HUMAN CELLS/TISSUES:

The SSbD strategy within POMATO includes the application and validation of cutting-edge in vitro assays involving human cell lines for testing the potential hazards and toxicity of the solutions proposed in WP4. The human cell lines used within these assays are widely used within the field of toxicology assessment. These tests are relevant to evaluate potential hazards and risks of the developed solution early in the development phase in order to guide the design towards safer alternatives. All the cell are commercially available.

ANIMALS:

Animal experiments performed in POMATO will ONLY involve aquatic toxicity test Acute Toxicity to Fish (OECD 203) to determine the LC50 for fish and the Honey Bee Toxicity Test (OECD 213) to determine contact and oral toxicity to honey bees. Thus not subjected to request of authorization. However, POMATOS's partners engage in minimizing as much as possible the number of animals used in this project.

PERSONAL DATA:

POMATO will involve human participants through the conduct of interviews, surveys, workshop and other relevant events along the project duration, especially in the activities planned in T6.2 and T6.4 where the multi-actor approach activities will be carried out.

In more details human participant will be participating in:

1) engagement of the farmers and wider society (including youth and general publics) through workshops and four "POMATO Pilot Open Days" in order to raise awareness about environmental and economic problems related to pest outbreaks and inform them about POMATO solutions and cutting-edge technologies to address these challenge; 2) consultations and one-on-one meetings with key industry and policy stakeholders at relevant events;

3) Identified stakeholders will be contact to assess their real-life needs, problems, and opportunities to engage in the co-creation process of POMATO's solution development and evaluation, as well as knowledge sharing throughout the project's lifespan.

NON-EU COUNTRIES: Colombia (AGRO) and Ecuador (USFQ).

AGROSAVIA's role includes genomic and transcriptomic characterization of potato and tomato resistance (WP2 Leader). Their advanced facilities feature high-throughput sequencing, gene expression analysis, and controlled growth chambers. The Tibaitatá Research Centre has a quarantine greenhouse and laboratory for biosafe experiments and field trials with local farmers. AGROSAVIA is leading the case study CS#2 in WP5.
USFQ's will provide the collection of native and commercial potato varieties, essential for studying genetic resistance to various stresses. USFQ facilities include greenhouses and fields for pathogenicity assays, a molecular lab for pathogen identification, and a microbiology lab for pathogen multiplication and morphological descriptions. USFG is involved in potato varieties for GR studies (WP2) and is the leader of the case study CS#1 Leader in connection with local potato farmers (WP5).
Both partners are included in the POMATO strategically as they are currently the same outbreaks that POMATO is dealing with (Clavibacter sepedonicus (Cs) and Ralstonia solanacearum (Rs). Both partners will ensure the exchange of good practices while at the same time providing access to a wide germplasm collection of potato/tomato to carry out the tasks in WP 2 and to infected fields to evaluate the developed strategies of POMATO in two case studies (WP5).
Regarding the importing material from EU countries to this LATAM countries: AGROSAVIA is in charge of the characterization at phenotypic and genotypic level of both bacteria (WP2). For this purpose, AGROSAVIA will use isolated strains from potato/tomato infected and germplasm collections from the partner IHAR (Poland). Potato and tomato samples obtained from the European (IHAR, Poland) culture collections will be used to study their resistance genes (WP2). In order to ensure that there will be no problems with shipping from Poland to Ecuador, IHAR has already been in contact with the Office of Phytosanitary Supervision and International cooperation of our Phytosanitary Inspection to follow the rules applied to this kind of samples.

ARTIFICIAL INTELLIGENCE:

POMATO will include AI-based models in the development of the early detection tools. AI-based predictive models will be developed with a focus on technical and social robustness, ensuring accuracy and reproducibility in various environments. This will involve systematic data acquisition, preprocessing, and model training, using Bayesian hyperparameter optimization. Model performance will be rigorously evaluated with metrics like the Matthews Correlation Coefficient to handle potential inaccuracies. The models Will be tailored to agricultural contexts, ensuring reliability. An AI-powered Cloud platform will integrate sensor data for disease prediction. Trustworthiness will be fostered by maintaining both absolute and relative trust values, supporting informed human trust decisions.

Compliance with ethical principles and legislations

Processing of personal data: according to article 89.1 of the General Regulation on Data Protection, "the processing for archiving purposes in the public interest, scientific or historical research purposes or statistical purposes, shall be subject to appropriate safeguards, in accordance with this Regulation, for the rights and freedoms of the data subject". POMATO will put in place a centralized system, in which only the lead partner (UBU) will have the competencies to authorize access and, by extension, withdraw access to any sort of personal data concerning the project. With this, POMATO wishes to successfully ensure that the rights of the participants in the project will be safeguarded. For the informed consent, POMATO will provide each participant with a consent form that will always have to be signed before participation in the project. Since one of the fundamental principles of the GDPR (article 5) is transparency, it is on best interest of POMATO to provide as much information as possible and to make the information as transparent as possible. The consent form includes the following subjects, according to the EU Guidelines on consent under Regulation 2016/679: i) The controller's identity; ii) The purpose of each of the processing operations for which consent is sought; iii) What (type of) data will be collected and used; iv) The existence of the right to withdraw consent; v) Information about the use of the data for automated decision-making in accordance with Article 22 (2)(c) where relevant; vi) On the possible risks of data transfers due to the absence of an adequacy decision and of appropriate safeguards as described in Article 46.

Ethics Summary Report

Human participants will be involved in research and communication activities such as surveys, workshops, and consultations. Data processing including transfer between EU and non-EU countries will take place. Standard toxicity tests will make use of commercially available human cell lines and fish and honeybees. Export of bacterial pathogens for potatoes and tomatoes is planned and infected experimental sites will be exploited in the project. Finally, the proposal included the development of AI tools for early detection and prediction.

Due to the diversity and the number of ethics issues raised by the planned activities and the large number of partners involved, it is recommended to appoint an ethics mentor to advise the project participants on the ethics issues and to keep a report on the activities on file to be made available promptly upon request.

The Consortium declares that compliance with the ethical principles and the applicable EU, international, and national laws for the ethics issues identified in the Ethics Summary Report and any additional ethics issues that may emerge in the course of the grant, will be ensured.
The Consortium confirms that for any applicable ethics issue, the guidance provided in the European Commission Ethics Self-Assessment Guidelines will be rigorously followed.

ANNEX 2